



ESC Guidelines for the management of acute myocardial infarction in patients presenting with ST-segment elevation

The Task Force on the management of ST-segment elevation acute myocardial infarction of the European Society of Cardiology (ESC)

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Antithrombotic therapy • Secondary prevention

Table of Contents

Abbreviations and Acronyms	2570
1. Preamble	2572
2. Introduction	2573
2.1. Definition of acute myocardial infarction	2573
2.2. Epidemiology of ST-segment elevation myocardial infarction	2573
3. Emergency care	2574
3.1. Initial diagnosis	2574
3.2. Relief of pain, breathlessness and anxiety	2576
3.3. Cardiac arrest	2576
3.4. Pre-hospital logistics of care	2577
3.4.1. Delays	2577
3.4.2. Emergency medical system	2578
3.4.3. Networks	2578
3.4.4. General practitioners	2579
3.4.5. Admission procedures	2579
3.4.6. Logistics	2579
3.5. Reperfusion therapy	2580
3.5.1. Restoring coronary flow and myocardial tissue reperfusion	2580
3.5.2. Selection of a strategy for reperfusion	2581
3.5.3. Primary percutaneous coronary intervention	2582
3.5.4. Fibrinolysis and subsequent interventions	2586
3.5.5. Coronary bypass surgery and multivessel coronary revascularization	2590
3.5.6. Non-reperused patients	2590
3.6. Management of hyperglycaemia in the acute phase of ST-segment elevation myocardial infarction	2592
4. Management during hospitalization and at discharge	2593
4.1. Coronary care unit logistics and monitoring	2593
4.1.1. Coronary care unit	2593
4.1.2. Monitoring	2593
4.1.3. Ambulation	2593
4.1.4. Length of stay	2593
4.2. Risk assessment and imaging	2594
4.2.1. Indications and timing	2594
4.3. Assessment of myocardial viability	2595
4.4. Long-term therapies for ST-segment elevation myocardial infarction	2595
4.4.1. Lifestyle interventions and risk factor control	2595
4.4.2. Antithrombotic therapy	2596
4.4.3. Beta-blockers	2597
4.4.4. Lipid-lowering therapy	2598
4.4.5. Nitrates	2598
4.4.6. Calcium antagonists	2598

4.4.7. Angiotensin-converting enzyme inhibitors and angiotensin receptor blockers	2598
4.4.8. Aldosterone antagonists	2598
4.4.9. Magnesium, glucose–insulin–potassium, lidocaine	2598
5. Complications following ST-segment elevation myocardial infarction	2600
5.1. Haemodynamic disturbances	2600
5.1.1. Heart failure	2600
5.1.2. Management of heart failure following ST-segment elevation myocardial infarction (Table 23)	2601
5.1.3. Arrhythmias and conduction disturbances in the acute phase	2603
5.2. Cardiac complications	2606
5.2.1. Mitral valve regurgitation	2606
5.2.2. Cardiac rupture	2607
5.2.3. Ventricular septal rupture	2607
5.2.4. Right ventricular infarction	2607
5.2.5. Pericarditis	2607
5.2.6. Left ventricular aneurysm	2607
5.2.7. Left ventricular thrombus	2607
6. Gaps in the evidence and areas for future research	2608

Abbreviations and Acronyms

ACE	angiotensin-converting enzyme
ACS	acute coronary syndrome
ADP	adenosine diphosphate
AF	atrial fibrillation
AMI	acute myocardial infarction
AV	atrioventricular
AIDA-4	Abciximab Intracoronary vs. intravenously Drug Application
APACHE II	Acute Physiology And Chronic Health Evaluation II
ATOLL	Acute myocardial infarction Treated with primary angioplasty and intravenous enoxaparin or unfractionated heparin to Lower ischaemic and bleeding events at short- and Long-term follow-up

aPTT	activated partial thromboplastin time	GRACIA	GRupo de Análisis de la Cardiopatía Isquémica Aguda
ARB	angiotensin receptor blocker	GUSTO	Global Utilization of Streptokinase and Tissue plasminogen activator for Occluded coronary arteries
ASSENT 3	ASsessment of the Safety and Efficacy of a New Thrombolytic 3	HbA1c	haemoglobin A1c
ATLAS ACS (etc.)	Anti-Xa Therapy to Lower cardiovascular events in Addition to Standard therapy in subjects with Acute Coronary Syndrome—Thrombolysis In Myocardial Infarction 51	HORIZONS—AMI	Harmonizing Outcomes with Revascularization and Stents in Acute Myocardial Infarction
b.i.d.	bis in die (twice daily)	i.c.	intracoronary
BMI	body mass index	i.v.	intravenous
BMS	bare-metal stent	IABP	intra-aortic balloon pump
BNP	B-type natriuretic peptide	INFUSE—AMI	Intracoronary abciximab infusion and aspiration thrombectomy for anterior ST-segment Elevation Myocardial Infarction
BRAVE-3	Bavarian Reperfusion Alternatives Evaluation-3	IRA	infarct-related artery
CAD	coronary artery disease	ISIS-2	Second International Study of Infarct Survival
CAPITAL-AMI	Combined Angioplasty and Pharmacological Intervention vs. Thrombolytics Alone in Acute Myocardial Infarction	Lab	catheterization laboratory
CHA2DS2-VASc	Cardiac failure, Hypertension, Age ≥ 75 [Doubled], Diabetes, Stroke [Doubled] – VASascular disease, Age 65–74 and Sex category [Female])	LBbB	left bundle branch block
CHADS2	Cardiac failure, Hypertension, Age, Diabetes, Stroke (Doubled)	LDL	low-density lipoprotein
CK-MB	creatinine kinase myocardial band	LV	left ventricular
CLARITY-TIMI 28	CLlopidogrel as Adjunctive Reperfusion Therapy—Thrombolysis in Myocardial Infarction 28	LVAD	left ventricular assist device
COMMIT	Clopidogrel and Metoprolol in Myocardial Infarction Trial	NORDISTEMI	NORwegian study on District treatment of ST-Elevation Myocardial Infarction
CPG	Committee for Practice Guidelines	NRMI	National Registry of Myocardial Infarction
CRISP AMI	Counterpulsation to Reduce Infarct Size Pre-PCI-Acute Myocardial Infarction	NSTE-ACS	non-ST-segment elevation acute coronary syndromes
CRT	cardiac resynchronization therapy	OASIS	Optimal Antiplatelet Strategy for Intervention
CVLPRIT	Complete Versus Lesion-only Primary PCI Trial	OAT	Occluded Artery Trial
CT	computed tomography	ON-TIME 2	ONgoing Tirofiban In Myocardial infarction Evaluation
DAPT	dual antiplatelet therapy	OPTIMAAL	Optimal Therapy In Myocardial infarction with the Angiotensin II Antagonist Losartan
DES	drug-eluting stent	p.o.	per os
DIGAMI	Diabetes, Insulin Glucose Infusion in Acute Myocardial Infarction	PAMI-II	Primary Angioplasty in Myocardial Infarction II
EAPCI	European Association of Percutaneous Cardiovascular Interventions	PET	positron emission tomography
ECG	electrocardiogram	PCI	percutaneous coronary intervention
EMS	emergency medical system	PLATO	PLATelet inhibition and patient Outcomes
EPHESUS	Eplerenone Post-AMI Heart failure Efficacy and SURvival Study	PRAMI	PReventive Angioplasty in Myocardial Infarction trial
ESC	European Society of Cardiology	PRIMARY PCI	primary percutaneous coronary intervention
ExTRACT-TIMI 25	Enoxaparin and Thrombolysis Reperfusion for ACute myocardial infarction Treatment—Thrombolysis In Myocardial Infarction 25	PROVE IT-TIMI 22	PRavastatin Or atorVastatin Evaluation and Infection Therapy—Thrombolysis In Myocardial Infarction 22
FINESSE	Facilitated INTERvention with Enhanced reperfusion Speed to Stop Events	RBBB	right bundle branch block
FMC	first medical contact	r-PA	reteplase
GP	glycoprotein	RIFLE-STEACS	Radial Vs. Femoral randomized investigation in ST elevation Acute Coronary Syndrome
		RIVAL	Radial Vs. femoral access for coronary intervention
		SBP	systolic blood pressure
		SHOCK	SHould we emergently revascularize Occluded coronaries for Cardiogenic shock

STEMI	ST-segment elevation myocardial infarction
STREAM	STrategic Reperfusion Early After Myocardial infarction
t-PA	tissue plasminogen activator
TACTICS	Treat angina with Aggrastat and determine Cost of Therapy with an Invasive or Conservative Strategy
TAPAS	Thrombus Aspiration during Percutaneous coronary intervention in Acute myocardial infarction
TIA	transient ischaemic attack
TNK-tPA	tenecteplase
TRANSFER	Trial of Routine ANgioplasty and Stenting after Fibrinolysis to Enhance Reperfusion in acute myocardial infarction
TRITON—TIMI 38	TRial to assess Improvement in Therapeutic Outcomes by optimizing platelet Inhibition with prasugrel—Thrombolysis in Myocardial Infarction 38
UFH	unfractionated heparin
VALIANT	VALsartan In Acute myocardial iNfarction Trial
VF	ventricular fibrillation
VT	ventricular tachycardia

1. Preamble

Guidelines summarize and evaluate all available evidence—at the time of the writing process—on a particular issue, with the aim of assisting physicians in selecting the best management strategies for an individual patient with a given condition, taking into account the impact on outcome, as well as the risk–benefit ratio of particular diagnostic or therapeutic means. Guidelines are not

substitutes but are complements for textbooks and cover the ESC Core Curriculum topics. Guidelines and recommendations should help physicians to make decisions in their daily practice. However, the final decisions concerning an individual patient must be made by the responsible physician(s).

A great number of guidelines have been issued in recent years by the European Society of Cardiology (ESC), as well as by other societies and organizations. Because of their impact on clinical practice, quality criteria for the development of guidelines have been established, in order to make all decisions transparent to the user. The recommendations for formulating and issuing ESC guidelines can be found on the ESC web site (<http://www.escardio.org/guidelines-surveys/esc-guidelines/about/Pages/rules-writing.aspx>). ESC guidelines represent the official position of the ESC on a given topic and are regularly updated.

Members of this Task Force were selected by the ESC to represent professionals involved with the medical care of patients with this condition. Selected experts in the field undertook a comprehensive review of the published evidence for diagnosis, management and/or prevention of a given condition, according to ESC Committee for Practice Guidelines (CPG) policy. A critical evaluation of diagnostic and therapeutic procedures was performed, including assessment of the risk–benefit ratio. Estimates of expected health outcomes for larger populations were included, where data exist. The levels of evidence and the strengths of recommendation of particular treatment options were weighed and graded according to predefined scales, as outlined in *Tables 1* and *2*.

The experts of the writing and reviewing panels filled in Declaration of Interest forms, in order to identify what might be perceived as real or potential sources of conflicts of interest. These forms were compiled into a single file and can be found on the ESC web site (<http://www.escardio.org/guidelines>). Any changes in declarations of interest that arise during the writing period must be notified to the ESC and updated. The Task Force received

Table 1 Classes of recommendations

Classes of recommendations	Definition	Suggested wording to use
Class I	Evidence and/or general agreement that a given treatment or procedure is beneficial, useful, effective.	Is recommended/is indicated
Class II	Conflicting evidence and/or a divergence of opinion about the usefulness/efficacy of the given treatment or procedure.	
<i>Class IIa</i>	<i>Weight of evidence/opinion is in favour of usefulness/efficacy.</i>	Should be considered
<i>Class IIb</i>	<i>Usefulness/efficacy is less well established by evidence/opinion.</i>	May be considered
Class III	Evidence or general agreement that the given treatment or procedure is not useful/effective, and in some cases may be harmful.	Is not recommended

Table 2 Levels of evidence

Level of evidence A	Data derived from multiple randomized clinical trials or meta-analyses.
Level of evidence B	Data derived from a single randomized clinical trial or large non-randomized studies.
Level of evidence C	Consensus of opinion of the experts and/or small studies, retrospective studies, registries.

its entire financial support from the ESC, without any involvement from the healthcare industry.

The ESC CPG supervises and co-ordinates the preparation of new guidelines produced by task forces, expert groups or consensus panels. The Committee is also responsible for the endorsement process of these Guidelines. The ESC Guidelines undergo extensive review by the CPG and external experts. After appropriate revisions, it is approved by all the experts involved in the Task Force. The finalized document is approved by the CPG for publication in the *European Heart Journal*.

The task of developing ESC Guidelines covers not only the integration of the most recent research, but also the creation of educational tools and implementation programmes for the recommendations. To implement the guidelines, condensed pocket guidelines editions, summary slides, booklets with essential messages, and electronic versions for digital applications (smartphones, etc.) are produced. These versions are abridged and, thus, if needed, one should always refer to the full text version, which is freely available on the ESC web site. The national societies of the ESC are encouraged to endorse, translate and implement the ESC Guidelines. Implementation programmes are needed because it has been shown that the outcome of disease may be favourably influenced by the thorough application of clinical recommendations.

Surveys and registries are needed to verify that real-life daily practice is in keeping with what is recommended in the guidelines, thus completing the loop between clinical research, writing of guidelines, and implementing them into clinical practice.

The guidelines do not, however, override the individual responsibility of health professionals to make appropriate decisions according to the circumstances of individual patient, in consultation with that patient and, where appropriate and necessary, the patient's guardian or carer. It is also the health professional's responsibility to verify the rules and regulations applicable to drugs and devices at the time of prescription.

2. Introduction

2.1 Definition of acute myocardial infarction

The management of acute myocardial infarction continues to undergo major changes. Good practice should be based on sound evidence, derived from well-conducted clinical trials. Because of

Table 3 Universal definition of myocardial infarction^a

Detection of rise and/or fall of cardiac biomarker values (preferably troponin) with at least one value above the 99th percentile of the upper reference limit and with at least one of the following:

- ♦ Symptoms of ischaemia;
- ♦ New or presumably new significant ST-T changes or new LBBB;
- ♦ Development of pathological Q waves in the ECG;
- ♦ Imaging evidence of new loss of viable myocardium, or new regional wall motion abnormality;
- ♦ Identification of an intracoronary thrombus by angiography or autopsy.

Cardiac death with symptoms suggestive of myocardial ischaemia, and presumably new ECG changes or new LBBB, but death occurring before blood cardiac biomarkers values are released or before cardiac biomarker values would be increased.

Stent thrombosis associated with MI when detected by coronary angiography or autopsy in the setting of myocardial ischaemia and with a rise and/or fall of cardiac biomarker values with at least one value above the 99th percentile URL.

ECG = electrocardiogram; LBBB = left bundle branch block.

^aExcluding myocardial infarction associated with revascularization procedures or criteria for prior myocardial infarction.

the great number of trials on new treatments performed in recent years, and in view of new diagnostic tests, the ESC decided that it was opportune to upgrade the previous guidelines and appointed a Task Force. It must be recognized that, even when excellent clinical trials have been undertaken, their results are open to interpretation and that treatment options may be limited by resources. Indeed, cost-effectiveness is becoming an increasingly important issue when deciding upon therapeutic strategies.

Owing to major changes in the biomarkers available for diagnosis, criteria for acute myocardial infarction have been revised. The current international consensus definition states that the term 'acute myocardial infarction' (AMI) should be used when there is evidence of myocardial necrosis in a clinical setting consistent with myocardial ischaemia.² Under these conditions, any one of the criteria described in Table 3 meets the diagnosis for spontaneous myocardial infarction. The present guidelines pertain to patients presenting with ischaemic symptoms and *persistent* ST-segment elevation on the electrocardiogram (ECG). Most of these patients will show a typical rise in biomarkers of myocardial necrosis and progress to Q-wave myocardial infarction. Separate guidelines have recently been developed by another Task Force of the ESC for patients presenting with ischaemic symptoms but *without* persistent ST-segment elevation and for patients undergoing myocardial revascularization in general.^{3,4}

2.2 Epidemiology of ST-segment elevation myocardial infarction

Worldwide, coronary artery disease (CAD) is the single most frequent cause of death. Over seven million people every year die from CAD, accounting for 12.8% of all deaths.⁵ Every sixth man and every seventh woman in Europe will die from myocardial infarction. The incidence of hospital admissions for AMI with ST-segment elevations (STEMI) varies among countries that

belong to the ESC.⁶ The most comprehensive STEMI registry is probably in Sweden, where the incidence is 66 STEMI/100 000/year. Similar figures were also reported in the Czech Republic,⁷ Belgium,⁶ and the USA:⁸ the incidence rates (per 100 000) of STEMI decreased between 1997 and 2005 from 121 to 77, whereas the incidence rates of non-STEMI increased slightly from 126 to 132. Thus, the incidence of STEMI appears to be declining, while there is a concomitant increase in the incidence of non-STEMI.⁹ The mortality of STEMI is influenced by many factors, among them: age, Killip class, time delay to treatment, mode of treatment, history of prior myocardial infarction, diabetes mellitus, renal failure, number of diseased coronary arteries, ejection fraction, and treatment. The in-hospital mortality of unselected STEMI patients in the national registries of the ESC countries varies between 6% and 14%.¹⁰ Several recent studies have highlighted a fall in acute and long-term mortality following STEMI, in parallel with greater use of reperfusion therapy, primary percutaneous coronary intervention (primary PCI), modern antithrombotic therapy and secondary prevention treatments.^{6,8,11,12} Still, mortality remains substantial with approximately 12% of patients dead within 6 months,¹³ but with higher mortality rates in higher-risk patients,¹⁴ which justifies continuous efforts to improve quality of care, adherence to guidelines and research.

3. Emergency care

3.1 Initial diagnosis

Management—including both diagnosis and treatment—of AMI starts at the point of first medical contact (FMC), defined as the point at which the patient is either initially assessed by a paramedic or physician or other medical personnel in the pre-hospital setting, or the patient arrives at the hospital emergency department—and therefore often in the outpatient setting.¹⁵ A working diagnosis of myocardial infarction must first be made. This is usually based on a history of chest pain lasting for 20 min or more, not responding to nitroglycerine. Important clues are a history of CAD and radiation of the pain to the neck, lower jaw or left arm. The pain may not be severe. Some patients present with less-typical symptoms, such as nausea/vomiting, shortness of breath, fatigue, palpitations or syncope. These patients tend to present later, are more likely to be women, diabetic or elderly patients, and less frequently receive reperfusion therapy and other evidence-based therapies than patients with a typical chest pain presentation. Registries show that up to 30% of patients with STEMI present with atypical symptoms.¹⁶ Awareness of these atypical presentations and a liberal access to acute angiography for early diagnosis might improve outcomes in this high-risk group.

Timely diagnosis of STEMI is key to successful management. ECG monitoring should be initiated as soon as possible in all patients with suspected STEMI to detect life-threatening arrhythmias and allow prompt defibrillation if indicated. A 12-lead ECG should be obtained and interpreted as soon as possible at the point of FMC (Table 4).¹⁷ Even at an early stage, the ECG is seldom normal. Typically, ST-segment elevation in acute myocardial infarction, measured at the J point, should be found in two contiguous leads and be ≥ 0.25 mV in men below the age of 40 years,

Table 4 Recommendations for initial diagnosis

Recommendations	Class ^a	Level ^b	Ref ^c
A 12-lead ECG must be obtained as soon as possible at the point of FMC, with a target delay of ≤ 10 min.	I	B	17, 19
ECG monitoring must be initiated as soon as possible in all patients with suspected STEMI.	I	B	20, 21
Blood sampling for serum markers is recommended routinely in the acute phase but one should not wait for the results before initiating reperfusion treatment.	I	C	-
The use of additional posterior chest wall leads (V_7 – V_9 , ≥ 0.05 mV) in patients with high suspicion of inferobasal myocardial infarction (circumflex occlusion) should be considered.	IIa	C	-
Echocardiography may assist in making the diagnosis in uncertain cases but should not delay transfer for angiography.	IIb	C	-

ECG = electrocardiogram; FMC = first medical contact; STEMI = ST-segment elevation myocardial infarction.

^aClass of recommendation.

^bLevel of evidence.

^cReference

≥ 0.2 mV in men over the age of 40 years, or ≥ 0.15 mV in women in leads V_2 – V_3 and/or ≥ 0.1 mV in other leads (in the absence of left ventricular (LV) hypertrophy or left bundle branch block (LBBB)).² In patients with inferior myocardial infarction, it is advisable to record right precordial leads (V_3R and V_4R) seeking ST elevation, in order to identify concomitant right ventricular infarction.^{2,18} Likewise, ST-segment depression in leads V_1 – V_3 suggests myocardial ischaemia, especially when the terminal T-wave is positive (ST-elevation equivalent), and may be confirmed by concomitant ST elevation ≥ 0.1 mV recorded in leads V_7 – V_9 .²

The ECG diagnosis may be more difficult in some cases (Table 5), which nevertheless deserve prompt management. Among these:

- **BBB:** in the presence of LBBB, the ECG diagnosis of acute myocardial infarction is difficult, but often possible if marked ST abnormalities are present. Somewhat complex algorithms have been offered to assist the diagnosis,²² but they do not provide diagnostic certainty.²³ The presence of concordant ST elevation (i.e. in leads with positive QRS deflections) appears to be one of the best indicators of ongoing myocardial infarction with an occluded infarct artery.²⁴ Previous data from thrombolysis trials have shown that reperfusion therapy is beneficial overall in patients with LBBB and suspected myocardial infarction. However, most LBBB patients evaluated in the emergency

department do not have an acute coronary occlusion, nor do they require primary PCI. A previous ECG may be helpful in determining whether the LBBB is new (and, therefore, the suspicion of ongoing myocardial infarction is high). Importantly, in patients with clinical suspicion of ongoing myocardial ischaemia with new or presumed new LBBB, reperfusion therapy should be considered promptly, preferably using emergency coronary angiography with a view to primary PCI or, if unavailable, intravenous (i.v.) thrombolysis. A positive point-of-care troponin test 1–2 h after symptom onset in patients with BBB of uncertain origin may help decide whether to perform emergency angiography with a view to primary PCI. Patients with myocardial infarction and RBBB also have a poor prognosis,²⁵ although RBBB usually will not hamper interpretation of ST-segment elevation. Prompt management should be considered when persistent ischaemic symptoms occur in the presence of RBBB, regardless of whether or not the latter is previously known.

- **Ventricular pacing** may also prevent interpretation of ST-segment changes and may require urgent angiography to confirm diagnosis and initiate therapy. Reprogramming the pacemaker—allowing an evaluation of ECG changes during intrinsic heart rhythm—may be considered in patients known not to be dependent on ventricular pacing, without delaying invasive investigation.
- **Patients without diagnostic ECG:** some patients with acute coronary occlusion may have an initial ECG without ST-segment elevation, sometimes because they are seen very early after symptom onset (in which case, one should look for hyper-acute T waves, which may precede ST-segment elevation). It is important to repeat the ECG or monitor the ST segment. In addition, there is a concern that some patients with genuine acute occlusion of a coronary artery and ongoing myocardial infarction (such as those with an occluded circumflex coronary artery,^{26,27} acute occlusion of a vein graft, or left main disease), may present without ST-segment elevation and be denied reperfusion therapy, resulting in larger infarction and worse outcomes. Extending the standard 12-lead ECG with V₇–V₉ leads, while useful, does not always identify these patients. In any case, ongoing suspicion of myocardial ischaemia—despite medical therapy—is an indication for emergency coronary angiography with a view to revascularization, even in patients without diagnostic ST-segment elevation.³
- **Isolated posterior myocardial infarction:** Acute myocardial infarction of the infero-basal portion of the heart, often corresponding to the left circumflex territory in which isolated ST-depression ≥ 0.05 mV in leads V₁ through V₃ represents the dominant finding, should be treated as a STEMI. The use of additional posterior chest wall leads [V₇–V₉ ≥ 0.05 mV (≥ 0.1 mV in men <40 years old)] is recommended to detect ST elevation consistent with infero-basal myocardial infarction.
- **Left main coronary obstruction—lead aVR ST elevation and infero-lateral ST depression:** The presence of ST-depression >0.1 mV in eight or more surface leads, coupled with ST elevation in aVR and/or V₁ but an otherwise unremarkable ECG, suggests ischaemia due to multivessel or left main coronary artery obstruction, particularly if the patient presents with haemodynamic compromise.²⁸

Table 5 Atypical ECG presentations that deserve prompt management in patients with signs and symptoms of ongoing myocardial ischaemia

• LBBB
• Ventricular paced rhythm
• Patients without diagnostic ST-segment elevation but with persistent ischaemic symptoms
• Isolated posterior myocardial infarction
• ST-segment elevation in lead aVR

ECG = electrocardiogram; LBBB = left bundle branch block.

In patients with a suspicion of myocardial ischaemia and ST-segment elevation or new or presumed new LBBB, reperfusion therapy needs to be initiated as soon as possible. However, the ECG may be equivocal in the early hours and, even in proven infarction, may never show the classical features of ST-segment elevation and new Q waves. If the ECG is equivocal or does not show evidence to support the clinical suspicion of myocardial infarction, ECGs should be repeated and, when possible, the current ECG should be compared with previous tracings. Additional recordings of, for example, lead V₇, V₈ and V₉ may be helpful in making the diagnosis in selected cases.

Blood sampling for serum markers is routinely carried out in the acute phase but one should not wait for the results before initiating reperfusion treatment. Troponin (T or I) is the biomarker of choice, given its high sensitivity and specificity for myocardial necrosis. In patients who have both a clinically low or intermediate likelihood of ongoing myocardial ischaemia and a long prior duration of symptoms, a negative troponin test may help to avoid unnecessary emergency angiography in some patients.

If in doubt regarding the possibility of acute evolving myocardial infarction, emergency imaging (as opposed to waiting for the biomarkers to become elevated) allows the provision of timely reperfusion therapy to these patients. If locally available, emergency coronary angiography is the modality of choice, as it can be followed immediately by primary PCI if the diagnosis is confirmed. In hospitals or settings in which coronary angiography is not immediately available—provided it does not delay transfer—rapid confirmation of segmental wall-motion abnormalities by two-dimensional echocardiography may assist in making a decision for emergency transfer to a PCI centre, since regional wall-motion abnormalities occur within minutes following coronary occlusion, well before necrosis. However, wall-motion abnormalities are not specific to acute myocardial infarction and may be due to other causes such as ischaemia, an old infarction or ventricular conduction defects. Two-dimensional echocardiography is of particular value for the diagnosis of other causes of chest pain, such as pericardial effusion, massive pulmonary embolism or dissection of the ascending aorta (Table 4). The absence of wall-motion abnormalities excludes major myocardial infarction. In the emergency setting, the role of computed tomography (CT) scan should be

confined to differential diagnosis of acute aortic dissection or pulmonary embolism.

Stress-induced (Takotsubo) cardiomyopathy is a recently recognized syndrome, which may be difficult to differentiate from STEMI as symptoms and findings, ranging from slight chest pain to cardiogenic shock, may mimic an acute myocardial infarction but the ECG changes at presentation are usually modest and do not correlate with the severity of ventricular dysfunction. It is often triggered by physical or emotional stress and characterized in its typical form by transient apical or mid-left ventricular dilation and dysfunction. Because there is no specific test to rule out myocardial infarction in this setting, emergency angiography should not be delayed and, in the absence of myocardial infarction, will show neither significant culprit coronary artery stenosis nor intracoronary thrombi. The diagnosis is confirmed by the finding, on imaging, of transient apical- to mid-ventricular ballooning with compensatory basal hyperkinesis, and by disproportionately low plasma levels of cardiac biomarkers with respect to the severity of ventricular dysfunction and, eventually, by recovery of left ventricular function.²⁹

3.2 Relief of pain, breathlessness and anxiety

Relief of pain is of paramount importance, not only for humane reasons but because the pain is associated with sympathetic activation that causes vasoconstriction and increases the workload of the heart. Titrated i.v. opioids (e.g. morphine) are the analgesics most commonly used in this context (Table 6). Intramuscular injections should be avoided. Repeated doses may be necessary. Side-effects include nausea and vomiting, hypotension with bradycardia, and respiratory depression. Anti-emetics may be administered concurrently with opioids to minimize nausea. The hypotension and bradycardia will usually respond to atropine and the respiratory depression to naloxone (0.1–0.2 mg i.v. every 15 min when indicated), which should always be available.

Oxygen (by mask or nasal prongs) should be administered to those who are breathless, hypoxic, or who have heart failure. Whether oxygen should be systematically administered to patients without heart failure or dyspnoea is at best uncertain.³⁰ Non-invasive

monitoring of blood oxygen saturation greatly helps when deciding on the need to administer oxygen or ventilatory support.

Anxiety is a natural response to the pain and the circumstances surrounding a heart attack. Reassurance of patients and those closely associated with them is of great importance. If the patient becomes excessively disturbed, it may be appropriate to administer a tranquilizer, but opioids are frequently all that is required.

3.3 Cardiac arrest

Many deaths occur early during the first few hours after STEMI, due to ventricular fibrillation (VF). Since this arrhythmia occurs most frequently at an early stage, these deaths usually happen out of hospital. Therefore it is crucial that all medical and paramedical personnel caring for suspected myocardial infarction have access to defibrillation equipment and are trained in cardiac life support and that, at the point of FMC, ECG monitoring be immediately implemented in all patients with suspected myocardial infarction (Table 7).

In patients with resuscitated cardiac arrest, whose ECG shows ST-segment elevation, immediate angiography with a view to primary PCI is the strategy of choice, provided that the guidelines-mandated times can be met.^{31–33} Given the high prevalence of coronary occlusions and potential difficulties in interpreting the

Table 6 Recommendations for relief of pain, breathlessness and anxiety

Recommendations	Class ^a	Level ^b
Titrated i.v. opioids are indicated to relieve pain.	I	C
Oxygen is indicated in patients with hypoxia (SaO ₂ <95%), breathlessness, or acute heart failure.	I	C
Tranquillizer may be considered in very anxious patients.	IIa	C

i.v. = intravenous; SaO₂ = saturated oxygen.
^aClass of recommendation.
^bLevel of evidence.

Table 7 Cardiac arrest

Recommendations	Class ^a	Level ^b	Ref ^c
All medical and paramedical personnel caring for a patient with suspected myocardial infarction must have access to defibrillation equipment and be trained in cardiac life support.	I	C	-
It is recommended to initiate ECG monitoring at the point of FMC in all patients with suspected myocardial infarction.	I	C	-
Therapeutic hypothermia is indicated early after resuscitation of cardiac arrest patients who are comatose or in deep sedation.	I	B	34–36
Immediate angiography with a view to primary PCI is recommended in patients with resuscitated cardiac arrest whose ECG shows STEMI.	I	B	31–33
Immediate angiography with a view to primary PCI should be considered in survivors of cardiac arrest without diagnostic ECG ST-segment elevation but with a high suspicion of ongoing infarction.	IIa	B	31, 33

ECG = electrocardiogram; FMC = first medical contact; PCI = percutaneous coronary intervention; STEMI = ST-segment elevation myocardial infarction.
^aClass of recommendation.
^bLevel of evidence.
^cReferences.

ECG in patients after cardiac arrest, immediate angiography should be considered in survivors of cardiac arrest having a high index of suspicion of ongoing infarction (such as the presence of chest pain before arrest, history of established CAD, and abnormal or uncertain ECG results).^{31,33} Additionally, there is evidence that survivors of out-of-hospital cardiac arrest who are comatose have improved neurological outcomes when cooling is provided early after resuscitation. Therefore, these patients should rapidly receive therapeutic hypothermia.^{34–36} The optimal sequence of cooling and primary PCI in these patients is unclear.

The implementation of local/regional protocols to optimally manage out-of-hospital cardiac arrest is pivotal to providing prompt cardiopulmonary resuscitation, early defibrillation (if needed), and effective advanced cardiac life support. Availability of automated external defibrillators is a key factor in increasing survival. Prevention and improved treatment of out-of-hospital cardiac arrest is key to reductions in mortality related to CAD. For a more detailed discussion of these issues, refer to the recent European Resuscitation Council Guidelines for Resuscitation.³⁷

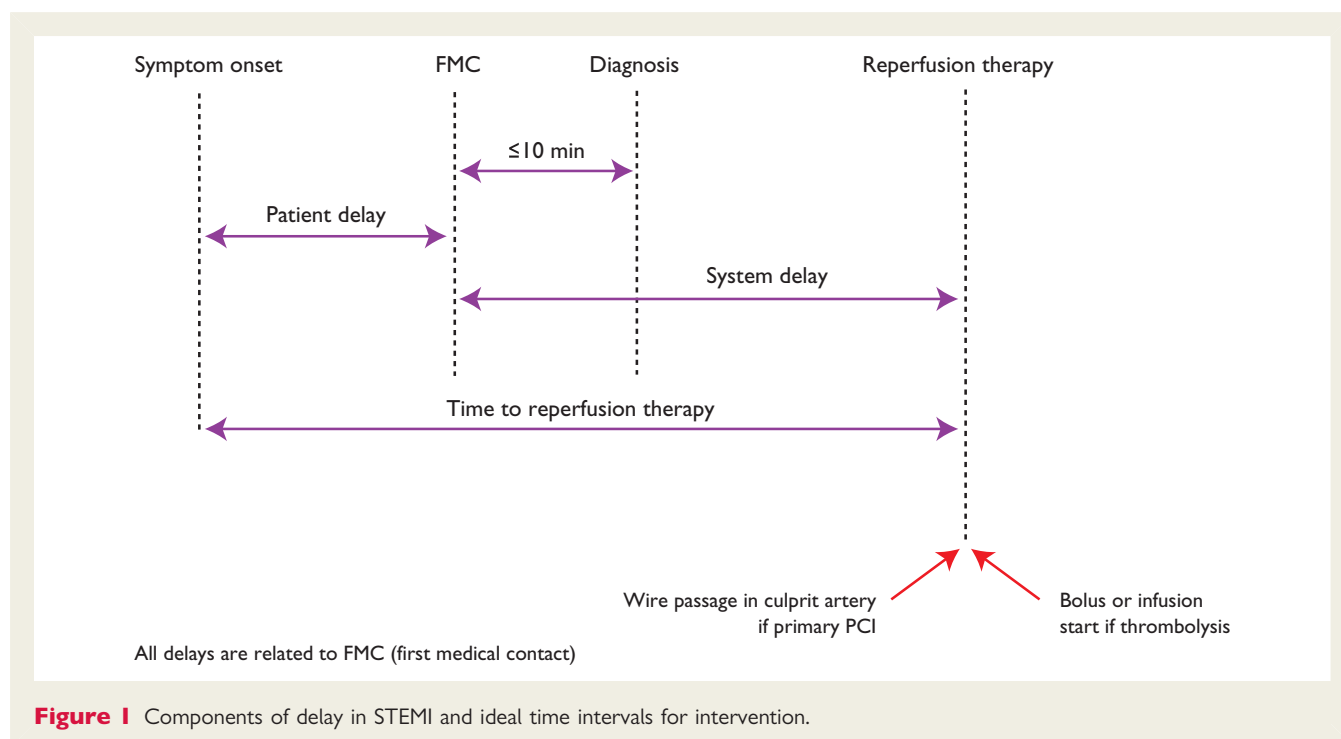
3.4 Pre-hospital logistics of care

3.4.1 Delays

Prevention of delays is critical in STEMI for two reasons: first, the most critical time of an acute myocardial infarction is the very early phase, during which the patient is often in severe pain and liable to cardiac arrest. A defibrillator must be made available to the patient with suspected acute myocardial infarction as soon as possible, for immediate defibrillation if needed. In addition, early provision of therapy, particularly reperfusion therapy, is critical to its benefit.³⁸ Thus, minimizing delays is associated with improved outcomes. In addition, delays to treatment are the most readily available, measurable index of quality of care in STEMI; they should be

recorded in every hospital providing care to STEMI patients and be monitored regularly, to ensure that simple quality-of-care indicators are met and maintained over time. Although still debated, public reporting of delays may be a useful way of stimulating improvement in STEMI care. If targets are not met, then interventions are needed to improve performance. There are several components of delay in STEMI and several ways to record and report them. For simplicity, it is advised to describe and report as shown in Figure 1.

- *Patient delay*: that is, the delay between symptom onset and FMC. To minimize patient delay, the public should be made aware of how to recognize common symptoms of acute myocardial infarction and to call the emergency services, but the effectiveness from public campaigns has not yet been clearly established.³⁸ Patients with a history of CAD, and their families, should receive education on recognition of symptoms due to acute myocardial infarction and the practical steps to take, should a suspected acute coronary syndrome (ACS) occur. It may be wise to provide stable CAD patients with a copy of their routine baseline ECG for comparison purposes by medical personnel.
- *Delay between FMC and diagnosis*: a good index of the quality of care is the time taken to record the first ECG. In hospitals and emergency medical systems (EMSs) participating in the care of STEMI patients, the goal should be to reduce this delay to 10 min or less.
- *Delay between FMC and reperfusion therapy*: This is the 'system delay'. It is more readily modifiable by organizational measures than patient delay. It is an indicator of quality of care and a predictor of outcomes.³⁹ If the reperfusion therapy is primary PCI, the goal should be a delay (FMC to wire passage into the culprit artery) of ≤ 90 min (and, in high-risk cases with large anterior infarcts and early presenters within 2 h, it should be



≤ 60 min).^{40,41} If the reperfusion therapy is fibrinolysis, the goal is to reduce this delay (FMC to needle) to ≤ 30 min.

- In PCI-capable hospitals, the goal should be to achieve a 'door-to-balloon' delay ≤ 60 min between presentation in the hospital and primary PCI (defined as wire passage into the culprit artery). This delay reflects the organization and performance of the PCI-capable hospital.
- From the patient's perspective, the delay between symptom onset and provision of reperfusion therapy (either starting fibrinolysis or passing a wire through the culprit vessel) is possibly the most important, since it reflects total ischaemic time. It should be reduced as much as possible.

3.4.2 Emergency medical system

An EMS with an easily remembered and well publicized unique telephone number for medical emergencies is important in order to avoid transportation delays. A teleconsultation between the EMS and a reference cardiology centre is ideal, but is only available in a limited number of countries. Therefore, a well-trained EMS and an updated and shared, written STEMI management protocol are critically important. Although the use of an EMS decreases the delay and is the preferred mode of initial care for patients with suspected STEMI, it is under-utilized in many countries and, not infrequently, patients self-present to the emergency department. The ambulance service has a critical role in the management of acute myocardial infarction and should be considered not only a mode of transport but also a place for initial diagnosis, triage and treatment. Pre-hospital diagnosis, triage and initial emergency treatment in the ambulance has been shown to be associated with greater use of reperfusion therapies, reduced delays and improved clinical outcomes.^{39,42} In addition, EMS transportation allows for the diagnosis and treatment of cardiac arrest. The quality of the care given depends on the training of the staff concerned. All ambulance personnel should be trained to recognize the symptoms of an AMI, administer oxygen, relieve pain and provide basic life support (Table 8). All emergency ambulances should be equipped with ECG recorders, defibrillators, and at least one person on board trained in advanced life support. There is evidence that properly trained paramedical personnel can effectively identify AMI and provide timely reperfusion, and that physician-manned ambulances—which are available in only a few countries—are not necessary for effective pre-hospital management of AMI.⁴³ Paramedics trained to administer thrombolytics do so safely and effectively. Since pre-hospital thrombolysis is an attractive therapeutic option in patients presenting early after symptom onset, especially when transfer time is prolonged,^{40,44,45} ongoing training of paramedics to undertake these functions is recommended, even in the era of primary PCI. In specific regions, air ambulance systems further reduce transportation delays and improve outcomes.⁴⁶ Ambulance staff should be able to record an ECG for diagnostic purposes and either interpret it or transmit it, so that it can be reviewed by experienced staff in a coronary care unit or elsewhere. The recording, interpretation and, sometimes, teletransmission of an ECG before hospital admission can greatly accelerate in-hospital management and increase the probability of timely reperfusion therapy.

3.4.3 Networks

Optimal treatment of STEMI should be based on the implementation of networks between hospitals with various levels of technology, connected by an efficient ambulance service. The goal of these networks is to provide optimal care while minimizing delays, in order to improve clinical outcomes. Cardiologists should actively collaborate with all stakeholders, particularly emergency physicians, in establishing such networks. The main features of such a network are:

- Clear definition of geographical areas of responsibility
- Shared protocols, based on risk stratification and transportation by trained paramedic staff in appropriately equipped ambulances or helicopters
- Pre-hospital triage of STEMI patients to the appropriate institutions, bypassing non-PCI hospitals whenever primary PCI can be implemented within the recommended time limits
- On arrival at the appropriate hospital, the patient should immediately be taken to the catheterization laboratory, bypassing the emergency department
- Patients presenting to a non-PCI-capable hospital and awaiting transportation for primary or rescue PCI must be attended in an appropriately monitored and staffed area
- If the diagnosis of STEMI has not been made by the ambulance crew, and the ambulance arrives at a non-PCI-capable hospital, the ambulance should await the diagnosis and, if STEMI is confirmed, should continue to a PCI-capable hospital.

To maximize staff experience, primary PCI centres should perform the procedure systematically on a twenty-four hours, seven days a week (24/7) basis for all STEMI patients. Other models, although not ideal, may include weekly or daily rotation of primary PCI centres or multiple primary PCI centres in the same region. Hospitals that cannot offer a 24/7 service for primary PCI should be allowed to perform primary PCI in patients already admitted for another reason, who develop STEMI during their hospital stay. These hospitals should, however, be discouraged from initiating a service limited to daytime- or within-hours primary PCI, since this generates confusion with the EMS operators and is unlikely to match the door-to-balloon time and quality of intervention of focussed 24/7 true-primary PCI centres. The current catchment population for network systems in European countries that offer primary PCI to the majority of their population is 0.3–1.0 million.⁶ In small service areas the experience may be suboptimal, due to an insufficient number of STEMI patients. However, the optimal size of the catchment area is not clear. Geographical areas where the expected transfer time to the primary PCI centre makes it impossible to achieve the maximal allowable delays indicated in the recommendations below (see section 3.4.6) should develop systems for rapid thrombolysis, preferably in-ambulance/out-of-hospital, with subsequent immediate transfer to primary PCI centres.

Such networks reduce treatment delays and increase the proportion of patients receiving reperfusion.^{47–49} In each network, the quality of care, time delays and patient outcomes should be measured and compared at regular intervals and appropriate measures taken to bring about improvement. In a large survey in the USA, several strategies were associated with shorter delays

before primary PCI, including the ability to activate the catheterization laboratory by a single call, preferably while the patient is en route to hospital, expecting laboratory staff to arrive in the catheterization laboratory within 20 min of being paged, having a cardiologist on site, and using real-time data feedback between the upstream care and the catheterization laboratory.⁵⁰ The most effective strategies for increasing the proportion of patients receiving effective reperfusion and reduce delays to primary PCI may differ in other healthcare systems. In order to address the issue of access to primary PCI and effective implementation of networks across Europe,⁶ the ESC working group on acute cardiac care, the European Association of Percutaneous Cardiovascular Interventions (EAPCI), and EuroPCR, have joined forces in the *Stent for Life* initiative, to improve access to timely, effective primary PCI through focussed implementation programmes, tailored to each specific national healthcare setting and attempting to learn from success.⁵¹ Experience acquired through this initiative, across various European systems of care, is published regularly and provides tips and resources to increase and improve the implementation of primary PCI (www.stentforlife.com).⁵²

3.4.4 General practitioners

In some countries, general practitioners play a major role in the early care of acute myocardial infarction and are often the first to be contacted by patients. If general practitioners respond quickly they can be very effective, since they usually know the patient and can perform and interpret the ECG. Their first task

after the ECG diagnosis should be to alert the EMS. But they are also able to administer opioids and antithrombotic drugs (including fibrinolytics if that is the management strategy), and can undertake defibrillation if needed. In most settings, however, consultation with a general practitioner—instead of a direct call to the EMS—increases pre-hospital delay. Therefore, in general, the public should be educated to call the EMS, rather than the primary care physician, for symptoms suggestive of myocardial infarction.

3.4.5 Admission procedures

The processing of patients once they arrive in hospital must be speedy, particularly with regard to the diagnosis and administration of fibrinolytic agents or the performance of primary PCI, if indicated. Candidates for primary PCI should, as often as possible, be admitted directly to the catheterization laboratory, bypassing the emergency department and/or intensive coronary care unit, while patient candidates for fibrinolysis must be treated directly in the pre-hospital setting, in the emergency department or in the coronary care unit.^{53,54}

3.4.6 Logistics

In the optimal situation (Figure 2), the patient calls a central EMS number for help as soon as possible after the onset of chest pain. The EMS dispatches a fully equipped ambulance with personnel trained to perform and interpret a 12-lead ECG. Once the ECG reveals ST-segment elevation or new (or presumed new)

Table 8 Logistics of pre-hospital care

Recommendations	Class ^a	Level ^b	Ref ^c
Ambulance teams must be trained and equipped to identify STEMI (with use of ECG recorders and telemetry as necessary) and administer initial therapy, including thrombolysis where applicable.	I	B	43
The prehospital management of STEMI patients must be based on regional networks designed to deliver reperfusion therapy expeditiously and effectively, with efforts made to make primary PCI available to as many patients as possible.	I	B	47
Primary PCI-capable centres must deliver a 24/7 service and be able to start primary PCI as soon as possible but always within 60 min from the initial call.	I	B	6, 52, 55
All hospitals and EMSs participating in the care of patients with STEMI must record and monitor delay times and work to achieve and maintain the following quality targets: • first medical contact to first ECG ≤10 min; • first medical contact to reperfusion therapy; • for fibrinolysis ≤30 min; • for primary PCI ≤90 min (≤60 min if the patient presents within 120 min of symptom onset or directly to a PCI-capable hospital).	I	B	56, 57
All EMSs, emergency departments, and coronary care units must have a written updated STEMI management protocol, preferably shared within geographic networks.	I	C	
Patients presenting to a non-PCI-capable hospital and awaiting transportation for primary or rescue PCI must be attended in an appropriately monitored area.	I	C	
Patients transferred to a PCI-capable centre for primary PCI should bypass the emergency department and be transferred directly to the catheterization laboratory.	Ila	B	41, 50, 58

ECG = electrocardiogram; EMS = emergency medical system; PCI = percutaneous coronary intervention; 24/7 = 24 hours a day, seven days a week; STEMI = ST-segment elevation myocardial infarction.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

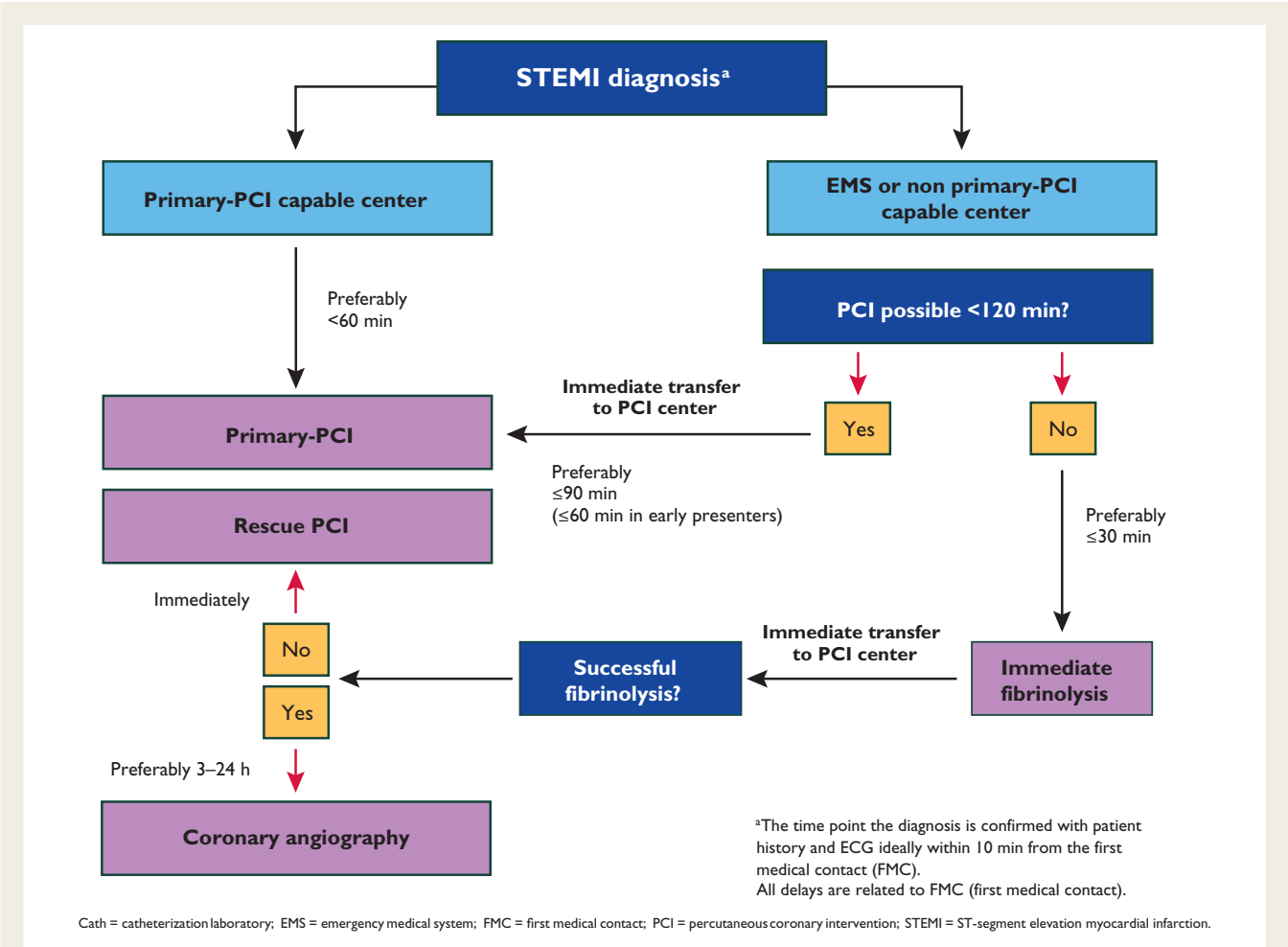


Figure 2 Prehospital and in-hospital management, and reperfusion strategies within 24 h of FMC (adapted from Wijns et al.).⁴

LBBB, the nearest PCI hospital is informed of the expected time of patient arrival. During the ambulance transfer, the catheterization laboratory is prepared and staff summoned, if necessary, allowing direct transfer of the patient to the catheterization laboratory table (bypassing the emergency department and coronary care unit). In cases where the diagnostic ECG has been done elsewhere (e.g. in a non-PCI hospital, at a physician's office, etc.), the EMS is called for transfer and the above chain followed. This scenario is best accomplished in a regional network with one high-volume PCI centre, several surrounding non-PCI hospitals and a single regional EMS. Such regional networks should have predefined management protocols for STEMI patients.

3.5 Reperfusion therapy

3.5.1 Restoring coronary flow and myocardial tissue reperfusion

For patients with the clinical presentation of STEMI within 12 h of symptom onset and with persistent ST-segment elevation or new or presumed new LBBB, early mechanical (PCI) or pharmacological reperfusion should be performed as early as possible (Table 9).

There is general agreement that reperfusion therapy should be considered if there is clinical and/or electrocardiographic evidence of ongoing ischaemia, even if, according to the patient, symptoms started >12 h before as the exact onset of symptoms is often unclear, or when the pain and ECG changes have been stuttering.⁵⁹

There is, however, no consensus as to whether PCI is also beneficial in patients presenting >12 h from symptom onset in the absence of clinical and/or electrocardiographic evidence of ongoing ischaemia. In such asymptomatic late-comers, a small (n = 347) randomized study has shown myocardial salvage and improved 4-year survival resulting from primary PCI, compared with conservative treatment alone, in patients without persistent symptoms 12–48 h after symptom onset.^{60,61} However, in stable patients with persistent occlusion of the infarct-related artery, the large (n = 2166) Occluded Artery Trial (OAT) revealed no clinical benefit from routine coronary intervention with medical management,^{62,63} beyond that from medical management alone, when the occlusion was identified 3–28 days after acute myocardial infarction, including in the subgroup of 331 patients randomized between 24 and 72 h after onset of infarction.⁶⁴ A meta-analysis of trials, testing whether late re-canalization of an occluded infarct artery is beneficial, provided results consistent with those from OAT.⁵¹

Table 9 Recommendations for reperfusion therapy

Recommendations	Class ^a	Level ^b	Ref ^c
Reperfusion therapy is indicated in all patients with symptoms of <12 h duration and persistent ST-segment elevation or (presumed) new LBBB.	I	A	65,66
Reperfusion therapy (preferably primary PCI) is indicated if there is evidence of ongoing ischaemia, even if symptoms may have started >12 h beforehand or if pain and ECG changes have been stuttering.	I	C	67
Reperfusion therapy with primary PCI may be considered in stable patients presenting 12–24 h after symptom onset.	IIb	B	60,61
Routine PCI of a totally occluded artery >24 h after symptom onset in stable patients without signs of ischaemia (regardless of whether fibrinolysis was given or not) is not recommended.	III	A	62–64

ECG = electrocardiogram; i.v. = intravenous; LBBB = left bundle branch block; PCI = percutaneous coronary intervention.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

3.5.2 Selection of a strategy for reperfusion

Primary PCI—defined as an emergent percutaneous catheter intervention in the setting of STEMI, without previous fibrinolytic treatment—is the preferred reperfusion strategy in patients with STEMI, provided it can be performed expeditiously (i.e. within guideline-mandated times), by an experienced team, and regardless of whether the patient presents to a PCI-capable hospital (Figure 1). If FMC is via an EMS or at a non-PCI-capable centre, transfer via the EMS to the catheterization laboratory for PCI should be implemented immediately. An experienced team includes not only interventional cardiologists, but also skilled support staff. This means that only hospitals with an established interventional cardiology programme (available 24/7) should use primary PCI as a routine treatment. Lower mortality rates among patients undergoing primary PCI are observed in centres with a high volume of PCI procedures. Primary PCI is effective in securing and maintaining coronary artery patency and avoids some of the bleeding risks of fibrinolysis. Randomized clinical trials comparing timely primary PCI with in-hospital fibrinolytic therapy in high-volume, experienced centres have repeatedly shown that primary PCI is superior to hospital fibrinolysis.^{68–71} (In these trials there was no routine follow-up rescue PCI or angiography.) In settings where primary PCI cannot be performed within 120 min of FMC by an experienced team, fibrinolysis

Table 10 A summary of important delays and treatment goals in the management of acute ST-segment elevation myocardial infarction

Delay	Target
Preferred for FMC to ECG and diagnosis	≤10 min
Preferred for FMC to fibrinolysis ('FMC to needle')	≤30 min
Preferred for FMC to primary PCI ('door to balloon') in primary PCI hospitals	≤60 min
Preferred for FMC to primary PCI	≤90 min (≤60 min if early presenter with large area at risk)
Acceptable for primary PCI rather than fibrinolysis	≤120 min (≤90 min if early presenter with large area at risk) if this target cannot be met, consider fibrinolysis.
Preferred for successful fibrinolysis to angiography	3–24 h

FMC = first medical contact; PCI = percutaneous coronary intervention.

should be considered, particularly if it can be given pre-hospital (e.g. in the ambulance)^{45,72,73} and within the first 120 min of symptom onset (Figure 2).^{40,74} It should be followed by consideration of rescue PCI or routine angiography.

Both randomized studies and registries have indicated that long delays to primary PCI are associated with worse clinical outcomes. Time delay to reperfusion is defined in section 3.4.1, above. The 'PCI-related delay' is the theoretical difference between the time of FMC to balloon inflation, minus the time from FMC to start of fibrinolytic therapy (i.e. 'door-to-balloon' minus 'door-to-needle'). The extent to which the PCI-related delay diminishes the advantages of PCI over fibrinolysis has been the subject of many analyses and debates. Because no specifically designed study has addressed this issue, caution is needed when interpreting the results of these *post-hoc* analyses. From randomized trials, it was calculated that the PCI-related delay that may mitigate the benefit of mechanical intervention varies between 60 and 110 min. In another analysis of these trials, a benefit of primary PCI over fibrinolytic therapy was calculated, up to a PCI-related delay of 120 min.⁶⁶ In 192 509 patients included in the US National Registry of Myocardial Infarction (NRM1) 2–4 registry,⁴¹ the mean PCI-related time delay, where mortality rates of the two reperfusion strategies were comparable, was calculated at 114 min. This study also indicated that this delay varied considerably according to age, symptom duration and infarct location: from <1 h for an anterior infarction in a patient <65 years of age presenting <2 h after symptom onset, to almost 3 h for a non-anterior infarction in a patient >65 years of age presenting >2 h after symptom onset. Although these results were derived from a *post-hoc* analysis of a registry and reported delays are sometimes inaccurate, this study suggests that an individualized, rather than a uniform, approach for selecting the optimal reperfusion modality could be more appropriate when PCI cannot be performed

expeditiously. Taking into account the studies and registries mentioned above, a target for quality assessment is that primary PCI (wire passage) should be performed within 90 min after FMC in all cases. In patients presenting early, with a large amount of myocardium at risk, the delay should be shorter (<60 min). In patients presenting directly in a PCI-capable hospital, the goal should also be to achieve primary PCI within 60 min of FMC. Although no specific studies have been performed, a maximum delay of only 90 min after FMC seems a reasonable goal in these patients. Note that these target delays for implementation of primary PCI are quality indicators and that they differ from the maximal PCI-related delay of 120 min, which is useful in selecting primary PCI over immediate thrombolysis as the preferred mode of reperfusion (Table 10).

3.5.3 Primary percutaneous coronary intervention

3.5.3.1 Procedural aspects of primary percutaneous coronary intervention (Table 11)

Approximately 50% of STEMI patients have significant multivessel disease. Only the infarct-related artery should be treated during the initial intervention. There is no current evidence to support emergency intervention in non-infarct-related lesions.^{75,76} The only exceptions, when multivessel PCI during acute STEMI is justified, are in patients with cardiogenic shock in the presence of multiple, truly critical ($\geq 90\%$ diameter) stenoses or highly unstable lesions (angiographic signs of possible thrombus or lesion disruption), and if there is persistent ischaemia after PCI of the supposed culprit lesion. However, in patients with multivessel disease and cardiogenic shock, non-culprit lesions without critical stenoses should not routinely be stented.⁷⁷ See also section 3.5.4.9.

Because of the need for potent antithrombotic and antiplatelet agents, bleeding is more frequent when PCI is performed during ACS (and STEMI in particular) when compared with bleeding occurring during an elective procedure. Use of drugs with a more potent antithrombotic effect is often accompanied by an increase in the risk of bleeding, mostly related to the arterial puncture site. The radial approach has been shown to reduce the incidence of acute bleeding events, especially in ACS; in the Radial vs. femoral (RIVAL) access for coronary intervention trial, using radial rather than femoral access actually reduced mortality in the subset of STEMI patients.⁷⁸ Similar findings were also observed in the RIFLE STEACS trial.⁷⁹ In RIVAL there was, however, an interaction between benefit of the radial access route and operator experience, suggesting that the benefit of radial access over femoral depends upon the radial expertise of operators.

In primary PCI, drug-eluting stents (DES) reduce the risk of repeated target vessel revascularization, compared with bare-metal stents (BMS).⁸⁰ There have been concerns about increased risks of very late stent thrombosis and reinfarction with DES, compared with BMS.⁸⁰ However, use of DES has not been associated with an increased risk of death, myocardial infarction or stent thrombosis on long-term follow up.⁸² An issue with the routine use of DES in this setting is that it is often difficult to determine reliably the ability of patients to comply with or tolerate the protracted use of dual antiplatelet therapy (DAPT). Whether newer generations of DES provide improved clinical outcomes—compared with older generation DES or BMS—following primary PCI is currently being tested.

Table 11 Primary PCI: indications and procedural aspects

Recommendations	Class ^a	Level ^b	Ref ^c
Indications for primary PCI			
Primary PCI is the recommended reperfusion therapy over fibrinolysis if performed by an experienced team within 120 min of FMC.	I	A	69, 99
Primary PCI is indicated for patients with severe acute heart failure or cardiogenic shock, unless the expected PCI related delay is excessive and the patient presents early after symptom onset.	I	B	100
Procedural aspects of primary PCI			
Stenting is recommended (over balloon angioplasty alone) for primary PCI.	I	A	101, 102
Primary PCI should be limited to the culprit vessel with the exception of cardiogenic shock and persistent ischaemia after PCI of the supposed culprit lesion.	IIa	B	75, 103–105
If performed by an experienced radial operator, radial access should be preferred over femoral access.	IIa	B	78, 79
If the patient has no contraindications to prolonged DAPT (indication for oral anticoagulation, or estimated high long-term bleeding risk) and is likely to be compliant, DES should be preferred over BMS.	IIa	A	80, 82, 106, 107
Routine thrombus aspiration should be considered.	IIa	B	83–85
Routine use of distal protection devices is not recommended.	III	C	86, 108
Routine use of IABP (in patients without shock) is not recommended.	III	A	97, 98

BMS = bare-metal stent; DAPT = dual antiplatelet therapy; DES = drug-eluting stent; IABP = intra-aortic balloon pump; PCI = percutaneous coronary intervention.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

One single-centre randomized trial, the Thrombus Aspiration during Percutaneous coronary intervention in Acute myocardial infarction (TAPAS) trial,⁸³ showed improvement in indices of myocardial reperfusion (ST-segment resolution and myocardial blush) from routine use of manual thrombus aspiration before a balloon or a stent is introduced into the coronary artery. One-year follow-up from that trial found a reduction in mortality with thrombus aspiration as a secondary endpoint.⁸⁴ A meta-analysis of TAPAS and several smaller trials found similar results.⁸⁵ Mechanical thrombectomy or embolic protection devices have not been found to provide similar benefits. However, the difference in clinical impact between the various models is still unclear.⁸⁶ In the recent INTracoronary abciximab inFusion and aSpiration thrombEctomy in patients undergoing percutaneous coronary intervention for Anterior ST segment elevation Myocardial Infarction (INFUSE-AMI) randomized trial, thrombus aspiration did not affect infarct size.⁸⁷ Several large, randomized trials have been initiated to attempt to confirm the results of TAPAS.^{88,89}

Operators performing primary PCIs in STEMI should be aware of the importance of selecting an appropriate stent size. Most patients with STEMI have some degree of coronary spasm and, thus, intracoronary administration of nitrates is recommended before starting the coronary angiographic sequence used for stent size selection. The presence of thrombus can also lead to stent under-sizing (or otherwise suboptimal deployment), which is a frequent cause of re-stenosis or stent thrombosis in real-life practice.

Preliminary clinical studies have explored the value of myocardial pre- and post-conditioning to improve myocardial salvage. A small, randomized trial tested the value of remote conditioning using intermittent arm ischaemia through four cycles of 5 min inflations and deflation of a blood pressure cuff.⁹⁰ This was associated with improvement in surrogate markers of myocardial salvage, measured by myocardial perfusion imaging at 30 days. It is unknown whether this is associated with clinical benefits. The role of post-conditioning has been explored by small trials, using either repeated balloon inflations or cyclosporine infusions. The results are conflicting.^{91–95} Given the preliminary nature of these findings and the small size of the trials, confirmation of a clinical benefit of myocardial pre- and post-conditioning by ongoing randomized trials is warranted before these procedures can be recommended in routine clinical practice.

The *Counterpulsation to Reduce Infarct Size Pre-PCI-Acute Myocardial Infarction* (CRISP AMI) trial showed no benefit from a routine intra-aortic balloon pump (IABP) in anterior myocardial infarction without shock,⁹⁷ and did show increased bleeding, which is consistent with data available regarding the role of IABPs in patients with acute myocardial infarction without cardiogenic shock.⁽⁹⁸⁾

3.5.3.2 Periprocedural pharmacotherapy (Table 12)

Patients undergoing primary PCI should receive a combination of DAPT with aspirin and an adenosine diphosphate (ADP) receptor blocker, as early as possible before angiography, and a parenteral anticoagulant. No trials to date have evaluated the commencement

of DAPT prior to hospital admission, rather than in hospital, nor its use before, rather than during, angiography in the setting of STEMI, but this is common practice in Europe and is consistent with the pharmacokinetic data for oral antithrombotic agents, suggesting that the earliest administration would be preferable to achieve early efficacy.

Aspirin should preferably be given orally (preferably 150–300 mg) including chewing, to ensure complete inhibition of TXA₂-dependent platelet aggregation, but may be given intravenously in patients who are unable to swallow. There is little clinical data on the optimal i.v. dosage, but pharmacological data suggest that a lower dose range than orally may avoid inhibition of prostacyclin and therefore a bolus dose range of 80–150 mg should be preferred for i.v. aspirin.

The preferred ADP-receptor blockers are prasugrel [60 mg per os (p.o.) loading dose, 10 mg maintenance dose] or ticagrelor [180 mg p.o. loading dose, 90 mg maintenance dose *bis in die* (b.i.d)]; these drugs have a more rapid onset of action and greater potency and have proved superior to clopidogrel in large outcome trials.^{109,110} In the TRial to assess Improvement in Therapeutic Outcomes by optimizing platelet inhibition–Thrombolysis In Myocardial Infarction 38 (TRITON–TIMI 38), prasugrel reduced the composite primary endpoint (cardiovascular death, non-fatal MI, or stroke) in clopidogrel-naïve patients undergoing PCI, either primary or secondary PCI for STEMI, or moderate-to-high-risk non-ST-segment elevation acute coronary syndromes (NSTEMI-ACS) once coronary angiography had been performed.¹⁰⁹ In the whole cohort, there was a significant increase in the rate of non-CABG-related TIMI major bleeding. In the subset of patients with STEMI undergoing primary or secondary PCI, the benefit was consistent, without significant increase in non-CABG-related bleeding risk.¹¹¹ Prasugrel is contraindicated in patients with prior stroke/transient ischaemic attack (TIA). Its use is generally not recommended in patients aged ≥75 years or in patients with lower body weight (<60 kg) as it was not associated with net clinical benefit in these subsets. The European label indicates that, if used in these patients, a similar loading dose but a reduced maintenance dose of 5 mg should be considered, but no outcome data are available with this dose and there are alternative ADP receptor blockers in this setting.¹¹² In the PLATelet inhibition and patient Outcomes (PLATO) trial, ticagrelor reduced the composite primary endpoint (cardiovascular death, non-fatal MI, or stroke) and also reduced cardiovascular mortality in clopidogrel naïve or pretreated patients with either STEMI (planned for primary PCI) or moderate-to-high risk NSTEMI-ACS (planned for either conservative or invasive management).^{109,110} Although there was no significant difference in overall PLATO-defined major bleeding rates between the clopidogrel and ticagrelor groups, PLATO-defined and TIMI-defined major bleeding that was unrelated to CABG surgery was increased with ticagrelor. In the subset of patients with STEMI, the benefit was consistent.¹¹³ Ticagrelor may cause transient dyspnoea at the onset of therapy, which is not associated with morphological or functional lung abnormalities, and which rarely leads to discontinuation.¹¹⁴ In PLATO, patients experiencing dyspnoea had a mortality benefit

Table 12 Periprocedural antithrombotic medication in primary percutaneous coronary intervention

Recommendations	Class ^a	Level ^b	Ref ^c
Antiplatelet therapy			
Aspirin oral or i.v. (if unable to swallow) is recommended	I	B	133, 134
An ADP-receptor blocker is recommended in addition to aspirin. Options are:	I	A	135, 136
• Prasugrel in clopidogrel-naïve patients, if no history of prior stroke/TIA, age <75 years.	I	B	109
• Ticagrelor.	I	B	110
• Clopidogrel, preferably when prasugrel or ticagrelor are either not available or contraindicated.	I	C	-
GP IIb/IIIa inhibitors should be considered for bailout therapy if there is angiographic evidence of massive thrombus, slow or no-reflow or a thrombotic complication.	IIa	C	-
Routine use of a GP IIb/IIIa inhibitor as an adjunct to primary PCI performed with unfractionated heparin may be considered in patients without contraindications.	IIb	B	137–141
Upstream use of a GP IIb/IIIa inhibitor (vs. in-lab use) may be considered in high-risk patients undergoing transfer for primary PCI.	IIb	B	127, 128, 137, 142
Options for GP IIb/IIIa inhibitors are (with LoE for each agent):			
• Abciximab		A	137
• Eptifibatide (with double bolus)		B	138, 139
• Tirofiban (with a high bolus dose)		B	140, 141
Anticoagulants			
An injectable anticoagulant must be used in primary PCI.	I	C	-
Bivalirudin (with use of GP IIb/IIIa blocker restricted to bailout) is recommended over unfractionated heparin and a GP IIb/IIIa blocker.	I	B	124
Enoxaparin (with or without routine GP IIb/IIIa blocker) may be preferred over unfractionated heparin.	IIb	B	122
Unfractionated heparin with or without routine GP IIb/IIIa blocker must be used in patients not receiving bivalirudin or enoxaparin.	I	C	I
Fondaparinux is not recommended for primary PCI.	III	B	118
The use of fibrinolysis before planned primary PCI is not recommended.	III	A	127, 143

ADP = adenosine diphosphate; GP = glycoprotein; i.v. = intravenous; lab = catheterization laboratory; PCI = percutaneous coronary intervention; TIA = transient ischaemic attack; UFH = unfractionated heparin.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

of ticagrelor consistent with the overall trial population. Ticagrelor may also be associated with asymptomatic bradycardia in the first week of therapy. None of the more potent agents (prasugrel or ticagrelor) should be used in patients with a previous haemorrhagic stroke or in patients with a moderate-to-severe liver disease. When neither of these agents is available (or if they are contraindicated), clopidogrel 600 mg p.o. should be given instead.¹¹⁵ Clopidogrel has not been evaluated against placebo in any large-outcome study in the setting of primary PCI, but a higher regimen of 600 mg loading dose/150 mg maintenance dose in the first week was superior to the 300/75 mg regimen in the subset of patients undergoing PCI in the *Optimal Antiplatelet Strategy for Interventions* (OASIS) 7 trial,¹¹⁵ and use of high clopidogrel loading doses has been demonstrated to achieve more rapid inhibition of the ADP receptor. This is consistent with the

pharmacokinetics of clopidogrel, a pro-drug, which requires extensive metabolism before being active and therefore should be given in higher doses and as early as possible for it to exert its action in the emergency setting of primary PCI. Furthermore, pre-treatment with high dose clopidogrel was superior to in-laboratory treatment in observational studies.^{116,117} All ADP receptor blockers should be used with caution in patients at high risk of bleeding or with significant anaemia.

Anticoagulant options for primary PCI include unfractionated heparin (UFH), enoxaparin and bivalirudin. Use of fondaparinux in the context of primary PCI was associated with potential harm in the OASIS 6 trial and is therefore not recommended.¹¹⁸ There have been no placebo-controlled trials evaluating UFH in primary PCI but there is a large body of experience with this agent. Dosage should follow standard recommendations for PCI

[i.e. initial bolus 70–100 U/kg when no glycoprotein (GP) IIb/IIIa inhibitor is planned or 50–60 U/kg when the use of GP IIb/IIIa inhibitors is expected]. There are no solid data recommending the use of activated clotting time to tailor dose or monitor UFH and, if activated clotting time is used, it should not delay recanalization of the infarct-related artery. Enoxaparin (0.5 mg/kg i.v. followed by s.c. treatment) was suggested by several non-randomized studies to provide benefit over UFH in primary PCI.^{119–121} It was compared with UFH in one randomized open label trial, the Acute myocardial infarction Treated with primary angioplasty and intravenous enoxaparin or unfractionated heparin to Lower ischaemic and bleeding events at short- and Long-term follow-up (ATOLL) trial. The primary composite endpoint of 30-day death, complication of myocardial infarction, procedural failure and major bleeding was not significantly reduced (17% reduction, $P = 0.063$), but there were reductions in the composite main secondary endpoint of death, recurrent myocardial infarction or ACS or urgent revascularization, and in other secondary composite endpoints such as death, or resuscitated cardiac arrest and death, or complication of myocardial infarction. Importantly, there was no indication of increased bleeding from use of enoxaparin over UFH.¹²² Based on these considerations and on the considerable clinical experience with enoxaparin in other PCI settings,^{109–111} enoxaparin may be preferred over UFH.

One large open-label trial demonstrated the superiority of bivalirudin over the combination of UFH + GP IIb/IIIa inhibitor,¹²³ a benefit driven by a marked reduction in bleeding, associated with an initial increase in stent thrombosis, which disappeared after 30 days.¹²⁴ Importantly, that study reported a reduction in all-cause and cardiovascular mortality at 30 days, which was maintained up to 3 years.⁸² A large fraction of patients in the Harmonizing Outcomes with Revascularization and Stents in Acute Myocardial (HORIZONS-AMI) trial, received prerandomization UFH and approximately 10% bailout GP IIb/IIIa blockers. This is noteworthy because the interpretation of the trial result is slightly confounded by an interaction between prerandomization use of UFH, use of a 600mg loading dose of clopidogrel and reduced risk of stent thrombosis.¹²⁵

Several trials, performed before the routine use of DAPT, mostly using abciximab, had documented clinical benefits of GP IIb/IIIa inhibitors as adjuncts to primary PCI performed with UFH.¹²⁶ The Facilitated Intervention with Enhanced reperfusion Speed to Stop Events (FINESSE) trial¹²⁷ found that routine upstream use of abciximab before primary PCI did not yield clinical benefit but increased bleeding risk, compared with routine use in the catheterization laboratory, suggesting that, for patients going on to primary PCI, there does not appear to be any appreciable benefit and only harm in starting GP IIb/IIIa inhibitors in the pre-hospital setting. A *post-hoc* subset analysis of the FINESSE trial, focussing on patients presenting within 4 h of symptom onset to non-PCI hospitals and requiring transfer, suggested they might derive a survival benefit from use of abciximab.¹²⁸ More recently, the ONgoing Tirofiban in Myocardial infarction Evaluation 2 (ON-TIME 2) trial¹²⁹ found an improvement in surrogate

markers of reperfusion from the use of tirofiban started during the pre-hospital phase, upstream of primary PCI, and continued for up to 18 h after the procedure (compared to only provisional use (i.e. not systematic use) in the catheterization laboratory). There was also a reduction in the composite secondary endpoint of death in recurrent myocardial infarction in urgent target vessel revascularization and thrombotic bailout. Finally, in the large HORIZONS-AMI trial,¹²⁴ there was no clear benefit from using a combination of GP IIb/IIIa inhibitor + UFH, compared to bivalirudin (with a substantial fraction of patients receiving UFH before randomization) and the Bavarian Reperfusion Alternatives Evaluation-3 (BRAVE-3) trial did not find evidence of a reduction in infarct size from treatment with abciximab in primary PCI patients treated with 600 mg of clopidogrel.¹³⁰ Therefore, there is no definitive answer regarding the current role of routine use of GP IIb/IIIa inhibitors in primary PCI in the era of potent DAPT, particularly when prasugrel or ticagrelor is used, and the value of starting upstream of the catheterization laboratory is, at best, uncertain. Using GP IIb/IIIa inhibitors as bailout therapy in the event of angiographic evidence of large thrombus, slow or no-reflow and other thrombotic complications is reasonable, although it has not been tested in a randomized trial. In conclusion, the existing data suggest that, if bivalirudin is chosen as the anti-coagulant, there is no benefit of routine addition of GP IIb/IIIa blockers and a strategy of bivalirudin alone (with provisional bailout use of GP IIb/IIIa blockers) leads to lower bleeding rates and reduced mortality. If UFH or enoxaparin is chosen as the anti-coagulant, the role of routine—as opposed to ‘bailout’—use of GP IIb/IIIa blockers remains debatable.

Intracoronary (i.c.) rather than i.v. administration of GP IIb/IIIa inhibitors has been tested in several small studies and is associated with some benefits.¹³¹ The Intracoronary abciximab infUSion and aspiration thrombectomy for anterior ST-segment Elevation Myocardial Infarction (INFUSE-AMI) trial⁸⁷ randomized 452 patients undergoing percutaneous coronary intervention with bivalirudin to local delivery of abciximab vs. no abciximab. Intracoronary abciximab reduced the 30-day infarct size, evaluated by magnetic resonance imaging, but did not improve abnormal wall motion score, ST-segment resolution, post-PCI coronary flow or myocardial perfusion. The large Abciximab Intracoronary vs. intravenously Drug Application 4 (AIDA-4) randomized trial, which enrolled 2065 patients (i.e. more than all previous studies combined) found no clinical benefit (but also no harm) in this route of administration in terms of the composite of death, reinfarction and heart failure, and found a borderline reduction in the secondary endpoint of heart failure.¹³² Therefore, the i.c. route may be considered but the i.v. route should remain the standard of care for administration of GP IIb/IIIa inhibitors.

Routine post-procedural anticoagulant therapy is not indicated after primary PCI, except when there is a separate indication for either full-dose anticoagulation (due, for instance, to atrial fibrillation, mechanical valves or LV thrombus) or prophylactic doses for prevention of venous thromboembolism in patients requiring prolonged bed rest.

3.5.3.3 Prevention and treatment of microvascular obstruction and no-reflow

Inadequate myocardial perfusion after successful mechanical opening of the infarct-related artery is often referred to as 'no-reflow'. The diagnosis of no-reflow is usually made when post-procedural thrombolysis in myocardial infarction (TIMI) flow is <3, or in the case of a TIMI flow of 3 when myocardial blush grade is 0 or 1, or when ST resolution within 4 h of the procedure is <70%.¹⁴⁴ Other non-invasive techniques are contrast echocardiography, single-photon emission tomography, positron emission tomography (PET), and contrast-enhanced magnetic resonance imaging (MRI).

There have been many attempts to treat no-reflow using intra-coronary vasodilators, i.v. infusion of adenosine or abciximab, but there is no definitive proof that these therapies affect clinical outcomes. Likewise, although it is widely used in clinical practice, there is no firm evidence that manual thrombus aspiration reduces distal embolization.^{83–86,145}

3.5.4 Fibrinolysis and subsequent interventions

3.5.4.1 Benefit of fibrinolysis

Fibrinolysis is an important reperfusion strategy, particularly in those settings where primary PCI cannot be offered to STEMI patients within the recommended timelines. The benefit of fibrinolytic therapy in patients with STEMI is well established:¹⁴⁶ compared with placebo, approximately 30 early deaths are prevented per 1000 patients treated within 6 h after symptom onset. Overall, the largest absolute benefit is seen among patients with the highest risk, even though the proportional benefit may be similar. The benefit is also seen in the elderly: in a subgroup of 3300 patients over the age of 75 years presenting within 12 h of symptom onset and with either ST-segment elevation or bundle-branch block, mortality rates were reduced significantly by fibrinolytic therapy.¹⁴⁷

3.5.4.2 Time to treatment

An analysis of studies in which >6000 patients were randomized to pre-hospital or in-hospital thrombolysis, showed a significant reduction (17%) in early mortality with pre-hospital treatment.⁷² In a meta-analysis of 22 trials,⁶⁵ a much larger mortality reduction was found in patients treated within the first 2 h than in those treated later. These data support pre-hospital initiation of fibrinolytic treatment if this reperfusion strategy is indicated. More recent *post-hoc* analyses of several randomized trials and data from registries have confirmed the clinical usefulness of pre-hospital fibrinolysis.^{40,44,47,143} Most of these studies reported outcome data similar to those of primary PCI, provided that early angiography and PCI were performed in those needing intervention (especially those who appear to have failed lysis). However, whether pre-hospital fibrinolysis is associated with a similar or better clinical outcome than primary PCI in early-presenting patients has not been studied prospectively in an adequately sized, randomized fashion. The ongoing STRategic Reperfusion Early After Myocardial infarction (STREAM) study is addressing this issue.¹⁴⁸

3.5.4.3 Hazards of fibrinolysis

Fibrinolytic therapy is associated with a small but significant excess of strokes,¹⁴⁶ with all of the excess hazard appearing on the first day after treatment. These early strokes are largely attributable to cerebral haemorrhage; later strokes are more frequently thrombotic or embolic. Advanced age, lower weight, female gender, prior cerebrovascular disease, and systolic and diastolic hypertension on admission are significant predictors of intracranial haemorrhage.¹⁴⁹ In the latest trials, intracranial bleeding occurred in 0.9–1.0% of the total population studied.^{150,151} Major non-cerebral bleeds (bleeding complications requiring blood transfusion or that are life-threatening) occur in 4–13% of the patients treated.^{150–152} Administration of streptokinase may be associated with hypotension, but severe allergic reactions are rare. Re-administration of streptokinase should be avoided because of antibodies, which can impair its activity, and because of the risk of allergic reactions.

3.5.4.4 Comparison of fibrinolytic agents

In the Global Utilization of Streptokinase and Tissue plasminogen activator for Occluded coronary arteries (GUSTO) trial,¹⁵³ tissue plasminogen activator (tPA; alteplase) with concomitant activated partial thromboplastin time (aPTT)-adjusted i.v. UFH resulted in 10 fewer deaths per 1000 patients treated, when compared with streptokinase, at the cost of three additional strokes, only one of which led to a residual neurological deficit. Several variants of tPA have been studied. Double-bolus r-PA (reteplase) does not offer any advantage over accelerated tPA, except for its ease of administration.¹⁵¹ Single-bolus weight-adjusted TNK-tPA (tenecteplase) is equivalent to accelerated tPA for 30-day mortality and is associated with a significantly lower rate of non-cerebral bleedings and less need for blood transfusion.¹⁵⁰ Bolus fibrinolytic therapy is easier to use in the pre-hospital setting.

3.5.4.5 Contraindications to fibrinolytic therapy

Absolute and relative contraindications to fibrinolytic therapy are listed in Table 13. Successful resuscitation does not contraindicate fibrinolytic therapy. However, lytic therapy is not effective and increases bleeding, and is not indicated in patients who are refractory to resuscitation. Prolonged, or traumatic but successful, resuscitation increases bleeding risk and is a relative contraindication to fibrinolysis.¹⁵⁴

Fibrinolytic therapy is recommended within 12 h of symptom onset if primary PCI cannot be performed within 90 min of being able to administer fibrinolysis and within 120 min from FMC (see section 3.4.6 and Figure 1) and there are no contraindications (Table 14). The later the patient presents (particularly after 6 h), the more consideration should be given to transfer for primary PCI (in preference to fibrinolytic therapy) as the efficacy and clinical benefit of fibrinolysis decrease over time, which, in later presentations, has the effect of increasing the acceptable time delay before transfer for primary PCI.⁷⁴

Where appropriate facilities exist, with trained medical or paramedical staff able to analyse on-site or to transmit the ECG to the hospital for supervision, fibrinolytic therapy should be initiated in

Table 13 Contraindications to fibrinolytic therapy

Absolute
Previous intracranial haemorrhage or stroke of unknown origin at any time
Ischaemic stroke in the preceding 6 months
Central nervous system damage or neoplasms or atrioventricular malformation
Recent major trauma/surgery/head injury (within the preceding 3 weeks)
Gastrointestinal bleeding within the past month
Known bleeding disorder (excluding menses)
Aortic dissection
Non-compressible punctures in the past 24 h (e.g. liver biopsy, lumbar puncture)
Relative
Transient ischaemic attack in the preceding 6 months
Oral anticoagulant therapy
Pregnancy or within 1 week postpartum
Refractory hypertension (systolic blood pressure >180 mmHg and/or diastolic blood pressure >110 mmHg)
Advanced liver disease
Infective endocarditis
Active peptic ulcer
Prolonged or traumatic resuscitation

the pre-hospital setting. The aim is to start fibrinolytic therapy within 30 min of FMC. For patients arriving at the hospital, a realistic aim is to initiate fibrinolysis within 30 min (door-to-needle time). A fibrin-specific agent should be preferred. The doses of fibrinolytic agents are shown in Table 15.

3.5.4.6 Adjunctive antiplatelet and anticoagulant therapies

The doses of antiplatelet and antithrombin co-therapies are given in Table 16.

Convincing evidence of the effectiveness of aspirin in addition to fibrinolysis was demonstrated by the Second International Study of Infarct Survival (ISIS-2), in which the benefits of aspirin and streptokinase were seen to be additive.¹³³ The first dose of 150–300 mg should be chewed or given intravenously (though at a lower dose range) and a lower dose (75–100 mg) given orally daily thereafter. In the CLOpidogrel as Adjunctive Reperfusion Therapy–Thrombolysis In Myocardial Infarction 28 (CLARITY-TIMI 28) trial, clopidogrel added to aspirin reduced the risk of cardiovascular events in patients ≤75 years of age who had been treated with fibrinolysis, and in the Clopidogrel and Metoprolol in Myocardial Infarction Trial (COMMIT), clopidogrel reduced overall mortality in such patients.^{156,157} Accordingly, there is a good case for the routine use of clopidogrel added to aspirin as an adjunct to lytic therapy. Prasugrel and ticagrelor have not been studied as adjuncts to fibrinolysis and should not be given.

The role of GP IIb/IIIa inhibitors used in conjunction with early routine post-thrombolysis PCI is unclear. In the GRUpo de Análisis de la Cardiopatía Isquémica Aguda (GRACIA-3) trial,¹⁷³ 436 patients with STEMI, treated with tenecteplase, enoxaparin and

aspirin, were randomly assigned to receive tirofiban or no tirofiban. There was no evidence that administration of tirofiban improved epicardial or myocardial perfusion.

Parenteral anticoagulation has been used extensively during and after fibrinolysis and should preferably be given until revascularization (if performed). Otherwise it should be given for at least 48 h or for the duration of hospital stay, up to 8 days. UFH was found to improve coronary patency after alteplase but not after streptokinase.^{174,175} Careful dosing and close monitoring of i.v. UFH therapy is mandatory; aPTT values >70 s are associated with a higher likelihood of bleeding, reinfarction and death. In spite of an increased risk of major bleeding, the net clinical benefit favoured enoxaparin over UFH in more recent studies: in the ASsessment of the Safety and Efficacy of a New Thrombolytic 3 (ASSENT 3) trial (n = 6095), a standard dose of enoxaparin given in association with tenecteplase for a maximum of 7 days reduced the risk of in-hospital reinfarction or in-hospital refractory ischaemia when compared with UFH.¹⁵⁸ However, in the ASSENT-3 PLUS trial (n = 1639),¹⁵⁹ pre-hospital administration of the same dose of enoxaparin resulted in a significant increase in the intracranial haemorrhage rate in elderly patients. In the large Enoxaparin and Thrombolysis Reperfusion for ACute myocardial infarction Treatment–Thrombolysis In Myocardial Infarction 25 (ExTRACT–TIMI 25) trial (n = 20 506), a lower dose of enoxaparin was given to patients >75 years of age and to those with impaired renal function (estimated creatinine clearance < 30 mL/min). Enoxaparin was associated with a reduction in the risk of death and reinfarction at 30 days when compared with a weight-adjusted UFH dose, but at the cost of a significant increase

Table 14 Fibrinolytic therapy

Recommendations	Class ^a	Level ^b	Ref ^c
Fibrinolytic therapy is recommended within 12 h of symptom onset in patients without contraindications if primary PCI cannot be performed by an experienced team within 120 min of FMC.	I	A	1, 41
In patients presenting early (<2 h after symptom onset) with a large infarct and low bleeding risk, fibrinolysis should be considered if time from FMC to balloon inflation is >90 min.	IIa	B	40, 41, 73
If possible, fibrinolysis should start in the prehospital setting.	IIa	A	72, 73, 155
A fibrin-specific agent (tenecteplase, alteplase, reteplase) is recommended (over non-fibrin specific agents).	I	B	150, 153
Oral or i.v. aspirin must be administered.	I	B	133
Clopidogrel is indicated in addition to aspirin.	I	A	156, 157
Antithrombin co-therapy with fibrinolysis			
Anticoagulation is recommended in STEMI patients treated with lytics until revascularization (if performed) or for the duration of hospital stay up to 8 days. The anticoagulant can be:	I	A	118, 153, 158–164
• Enoxaparin i.v. followed by s.c. (using the regimen described below) (preferred over UFH).	I	A	158–163
• UFH given as a weight-adjusted i.v. bolus and infusion.	I	C	153
In patients treated with streptokinase, fondaparinux i.v. bolus followed by s.c. dose 24 h later.	IIa	B	118, 164
Transfer to a PCI-capable centre following fibrinolysis			
Is indicated in all patients after fibrinolysis.	I	A	165–167, 168–171
Interventions following fibrinolysis			
Rescue PCI is indicated immediately when fibrinolysis has failed (<50% ST-segment resolution at 60 min).	I	A	165, 166
Emergency PCI is indicated in the case of recurrent ischaemia or evidence of reocclusion after initial successful fibrinolysis.	I	B	165
Emergency angiography with a view to revascularization is indicated in heart failure/shock patients.	I	A	167
Angiography with a view to revascularization (of the infarct-related artery) is indicated after successful fibrinolysis.	I	A	168–171
Optimal timing of angiography for stable patients after successful lysis: 3–24 h.	IIa	A	172

aPTT = activated partial thromboplastin time; FMC = first medical contact; i.v. = intravenous; s.c. = subcutaneous; UFH = unfractionated heparin.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

Table 15 Doses of fibrinolytic agents

	Initial treatment	Specific contraindications
Streptokinase (SK)	1.5 million units over 30–60 min i.v.	Prior SK or anistreplase
Alteplase (tPA)	15 mg i.v. bolus 0.75 mg/kg over 30 min (up to 50 mg) then 0.5 mg/kg over 60 min i.v. (up to 35 mg)	
Reteplase (r-PA)	10 units + 10 units i.v. bolus given 30 min apart	
Tenecteplase (TNK-tPA)	Single i.v. bolus: 30 mg if <60 kg 35 mg if 60 to <70 kg 40 mg if 70 to <80 kg 45 mg if 80 to <90 kg 50 mg if ≥90 kg	

i.v. = intravenous.

in non-cerebral bleeding complications. The net clinical benefit (absence of death, non-fatal infarction and intracranial haemorrhage) favoured enoxaparin.^{160,161} Finally, fondaparinux was shown in the large OASIS-6 trial to be superior to placebo or UFH in preventing death and reinfarction,^{118,164} especially in patients who received streptokinase.

In a large trial with streptokinase,¹⁷⁶ no mortality reduction was observed at 30 days, but significantly fewer reinfarctions were seen with bivalirudin (a direct antithrombin, given for 48 h), compared with UFH, though at the cost of a modest and non-significant increase in non-cerebral bleeding complications. Bivalirudin has not been studied with fibrin-specific agents. Thus there is no evidence in support of direct thrombin inhibitors as an adjunct to fibrinolysis.

Tenecteplase, aspirin, enoxaparin and clopidogrel comprise the antithrombotic combination that has been most extensively studied as part of a pharmacoinvasive strategy, viz. Trial of Routine Angioplasty and Stenting after Fibrinolysis to Enhance Reperfusion in acute myocardial infarction (TRANSFER),¹⁶⁸ NORwegian study on District treatment of ST-Elevation Myocardial Infarction (NORDISTEMI),¹⁷⁰ GRACIA-2,¹⁷⁷ and GRACIA-3.¹⁷³

Table 16 Doses of antiplatelet and antithrombin co-therapies

Doses of antiplatelet co-therapies	
With primary PCI	
Aspirin	Loading dose of 150–300 mg orally or of 80–150 mg i.v. if oral ingestion is not possible, followed by a maintenance dose of 75–100 mg/day.
Clopidogrel	Loading dose of 600 mg orally, followed by a maintenance dose of 75 mg/day.
Prasugrel	Loading dose of 60 mg orally, followed by a maintenance dose of 10 mg/day. In patients with body weight <60 kg, a maintenance dose of 5 mg is recommended. In patients >75 years, prasugrel is generally not recommended, but a dose of 5 mg should be used if treatment is deemed necessary.
Ticagrelor	Loading dose of 180 mg orally, followed by a maintenance dose of 90 mg b.i.d.
Abciximab	Bolus of 0.25 mg/kg i.v. and 0.125 µg/kg/min infusion (maximum 10 µg/min) for 12 h.
Eptifibatide	Double bolus of 180 µg/kg i.v. (given at a 10-min interval) followed by an infusion of 2.0 µg/kg/min for 18 h.
Tirofiban	25 µg/kg over 3 min i.v., followed by a maintenance infusion of 0.15 µg/kg/min for 18 h.
With fibrinolytic therapy	
Aspirin	Starting dose 150–500 mg orally or i.v. dose of 250 mg if oral ingestion is not possible.
Clopidogrel	Loading dose of 300 mg orally if aged ≤75 years, followed by a maintenance dose of 75 mg/day.
Without reperfusion therapy	
Aspirin	Starting dose 150–500 mg orally.
Clopidogrel	75 mg/day orally.
Doses of antithrombin co-therapies	
With primary PCI	
Unfractionated heparin	70–100 U/kg i.v. bolus when no GP IIb/IIIa inhibitor is planned. 50–60 U/kg i.v. bolus with GP IIb/IIIa inhibitors.
Enoxaparin	0.5 mg/kg i.v. bolus.
Bivalirudin	0.75 mg/kg i.v. bolus followed by i.v. infusion of 1.75 mg/kg/h for up to 4 h after the procedure as clinically warranted. After cessation of the 1.75 mg/kg/h infusion, a reduced infusion dose of 0.25 mg/kg/h may be continued for 4–12 h as clinically necessary.
With fibrinolytic therapy	
Unfractionated heparin	60 U/kg i.v. bolus with a maximum of 4000 U followed by an i.v. infusion of 12 U/kg with a maximum of 1000 U/h for 24–48 h. Target aPTT: 50–70 s or 1.5 to 2.0 times that of control to be monitored at 3, 6, 12 and 24 h.
Enoxaparin	In patients <75 years of age: 30 mg i.v. bolus followed 15 min later by 1 mg/kg s.c. every 12 h until hospital discharge for a maximum of 8 days. The first two doses should not exceed 100 mg. In patients >75 years of age: no i.v. bolus; start with first s.c. dose of 0.75 mg/kg with a maximum of 75 mg for the first two s.c. doses. In patients with creatinine clearance of <30 mL/min, regardless of age, the s.c. doses are given once every 24 h.
Fondaparinux	2.5 mg i.v. bolus followed by a s.c. dose of 2.5 mg once daily up to 8 days or hospital discharge.
Without reperfusion therapy	
Unfractionated heparin	Same dose as with fibrinolytic therapy.
Enoxaparin	Same dose as with fibrinolytic therapy.
Fondaparinux	Same dose as with fibrinolytic therapy.

aPTT = activated partial thromboplastin time; b.i.d. = twice a day; GP = glycoprotein; i.v. = intravenous; PCI = percutaneous coronary intervention; s.c. = subcutaneous; UFH = unfractionated heparin.

3.5.4.7 Angiography after fibrinolysis

Following initiation of lytic therapy, patients should be transferred to a PCI centre (see section 3.4.6). In cases of failed fibrinolysis, or if there is evidence of re-occlusion or reinfarction with recurrence of ST-segment elevation, the patient should undergo immediate angiography and rescue PCI.¹⁶⁵ Re-administration of fibrinolysis has not been shown to be beneficial. Even if it is likely that fibrinolysis will be successful (ST-segment resolution >50% at 60–90 min; typical reperfusion arrhythmia; disappearance of chest pain), a

strategy of routine early angiography is recommended if there are no contraindications. Several randomized trials^{168–171,178,179} and three contemporary meta-analyses^{172,180} have shown that early routine post-thrombolysis angiography with subsequent PCI (if required) reduced the rates of reinfarction and recurrent ischaemia compared with a ‘watchful waiting’ strategy, in which angiography and revascularization were indicated only in patients with spontaneous or induced severe ischaemia or LV dysfunction. The benefits of early routine PCI after thrombolysis were seen in the

absence of increased risk of adverse events (stroke or major bleeding). Thus, early referral for angiography with subsequent PCI (if indicated) should be the standard of care after thrombolysis: the so-called 'pharmacoinvasive' strategy. A crucial issue is the optimal delay between lysis and PCI: there was a wide variation in delay in trials, from a median of 1.3 h in the Combined Angioplasty and Pharmacological Intervention versus Thrombolytics ALone in Acute Myocardial Infarction (CAPITAL-AMI) trial to 16.7 h in the GRACIA-1 trial.^{171,179} Based on the three most recent trials, all of which had a median delay between start of lysis and angiography of 2–3 h, a time window of 3–24 h after successful lysis is recommended.^{168–170} The ongoing STREAM¹⁴⁸ and GRACIA-4 trials are exploring whether lysis performed with modern adjunctive therapies, and followed by subsequent PCI, can achieve similar or better outcomes compared with primary PCI.

3.5.4.8 Adjunctive antithrombotic therapy for delayed percutaneous coronary intervention after lysis

For patients undergoing PCI several hours or days after fibrinolysis, PCI should be supported by DAPT (aspirin and an ADP antagonist) and antithrombin therapy, in doses similar to those used for primary PCI.

3.5.4.9 Revascularization strategy for ST-segment elevation myocardial infarction with multivessel disease

Apart from patients in cardiogenic shock, and in patients with continuous ischaemia after opening the supposed culprit lesion, performing PCI of non-culprit vessels in the acute setting is generally discouraged. The best strategy for STEMI patients with multivessel disease, who underwent primary PCI of the infarct-related artery in the acute phase with remaining multivessel disease, is still not well established. Among the possible strategies, two that are frequently used are either a **conservative** approach—which uses medical therapy after primary PCI, and revascularization of other arteries only if there are symptoms or evidence of ischaemia in provocative tests—or a **staged revascularization approach**, using PCI or coronary bypass surgery of non-infarct arteries several days or weeks after primary PCI, often after confirmation of the stenosis severity with measurements of fractional flow reserve. A multidisciplinary approach is often needed, including a heart team and appropriate informed consent of the patient.

In STEMI patients with multivessel disease initially treated with primary or post-thrombolysis culprit-artery PCI and confirmed presence of ischaemia in non-infarcted territories, staged revascularization may be performed before discharge or in the days to weeks after initial PCI.¹⁸¹ A comparison of in-hospital complete revascularization [infarct-related artery (IRA) and non-IRA] vs. conservative approach (IRA only) is being undertaken in the Complete Vs. Lesion-only PRimary PCI Trial (CVLPRIT) and also in the Preventive Angioplasty in Myocardial Infarction (PRAMI) trial. Both assess the benefit/risk of treating non-infarct-related lesions. Likewise, the DANish study of optimal acute treatment of patients with ST-elevation Myocardial Infarction 3 (DANAMI-3) trial is currently testing whether or not to treat non-culprit lesions in patients treated previously with primary PCI.

3.5.5 Coronary bypass surgery and multivessel coronary revascularization

The number of patients who require CABG surgery in the acute phase of STEMI is small, but CABG may be indicated in patients with anatomy unsuitable for PCI but who have a patent infarct-related artery, since patency of this artery provides time for transfer to the surgical team. It may also be indicated in patients in cardiogenic shock if the coronary anatomy is not amenable to PCI, or at the time of repair for patients with mechanical complications. CABG is rarely used and its benefits are uncertain in patients with failed PCI, coronary occlusion not amenable to PCI, and in the presence of refractory symptoms after PCI since, in most of these cases, time for implementation of surgical reperfusion will be long and the risks associated with surgery are maximal in this setting.

3.5.5.1 Withholding adenosine diphosphate inhibitors for surgery

The risk of bleeding related to surgery must be balanced against the risk of recurrent ischaemic events related to discontinuation of therapy, bearing in mind the nature of the surgery, the ischaemic risk and extent of CAD, the time since the acute episode, the time since PCI and the risk of stent thrombosis. Clopidogrel is associated with an increased risk of bleeding if discontinued less than 5 days before surgery. Prasugrel is also associated with a marked increase in bleeding risk.¹⁰⁹ With respect to ticagrelor, data from the PLATO trial,¹¹⁰ suggest that ticagrelor, discontinued 3–5 days before CABG surgery, yielded similar CABG-related major bleeding and transfusions for clopidogrel and ticagrelor. Although non-fatal myocardial infarction and stroke rates in the two groups were not significantly different in this cohort, there was a halving of mortality in the ticagrelor group. In stabilized patients, it is reasonable to stop clopidogrel at least 5 days before surgery and to stop prasugrel 7 days before surgery. Given the PLATO data, ticagrelor may be discontinued 3 to 5 days before surgery.

Whether ADP receptor antagonists should be restarted after CABG surgery has not been addressed in any specific trial and the optimal timing of such restarting remains uncertain. However, given the reduction of the primary endpoint and mortality with ticagrelor in the PLATO trial and the continued risk for ischaemic events in patients post-CABG it is reasonable to restart DAPT as soon as considered safe in relation to bleeding risk.

In very-high-risk patients in whom cessation of antiplatelet therapy before surgery seems to carry a high risk (e.g. within the first weeks after stent implantation), it has been suggested to switch, before surgery, to a short half-life and reversible antiplatelet agent, e.g. the GP IIb/IIIa receptor inhibitors tirofiban or eptifibatide,¹⁸² but there is no clinical evidence to support this approach based solely on pharmacokinetic or pharmacodynamic studies. In the future, the use of cangrelor, an i.v. reversible ADP receptor antagonist, may permit platelet inhibition to be maintained up to surgery in patients discontinuing oral antiplatelet therapy.¹⁸³

3.5.6 Non-reperfused patients

3.5.6.1 Antithrombotic use

In patients presenting within 12 h of symptom onset, and in whom reperfusion therapy was not given, or in patients presenting beyond 12 h, aspirin, clopidogrel and an antithrombin agent (UFH, enoxaparin or fondaparinux) should be given as soon as

possible (see section 3.4.6).^{1,156,184} In OASIS-6, fondaparinux was superior to UFH in a subgroup of 1641 such patients and might be the preferred antithrombin for this indication.¹⁸⁵ If PCI is needed in a patient receiving fondaparinux, i.v. UFH should be administered during the procedure, using the same doses as for primary PCI, to minimize the risk of catheter thrombosis.¹⁸⁶ Recommended doses are given in Table 16. None of the oral agents have been studied in this particular subset of patients, but the benefit of clopidogrel over placebo was consistent in ACS patients, independent of revascularization strategy, in the Clopidogrel in Unstable angina to prevent Recurrent Events (CURE) trial.¹⁸⁷ Ticagrelor was superior to clopidogrel in ACS patients who were randomized for an early non-invasive strategy, with a similar trend also in those who were not revascularized during the index hospitalization.¹⁸⁸

3.5.6.2 Invasive evaluation and revascularization

Patients sometimes seek medical attention too late and either do not receive reperfusion therapy, or undergo unsuccessful reperfusion therapy. It has been suggested that achieving late coronary patency in either of these situations may still have a beneficial effect by preventing adverse LV remodelling, improving LV function, increasing electrical stability and inducing collateral vessels to other coronary beds for protection against future events (the 'open artery' hypothesis). Several trials have evaluated this hypothesis, of which the largest by far was the OAT trial (see above),⁶² in which 20% of the patients received fibrinolytic therapy for the index event. PCI did not reduce the occurrence of death, reinfarction or heart failure, compared to medical therapy alone. Furthermore, there was a trend towards excess reinfarction during four years of follow-up in the invasive therapy group, compared with the medical therapy group. A meta-analysis of all trials in this setting provided similar results.⁶³ These studies demonstrate that late PCI of an occluded infarct-related artery after myocardial infarction in stable patients has no incremental benefit over optimal medical therapy. Thus, in patients presenting days after the acute event with a completed myocardial infarction, only those with recurrent angina or documented residual ischaemia, and proven viability on non-invasive imaging in a large myocardial territory, may be considered for revascularization when the infarct artery is occluded.⁴

Special patient subsets

Several specific patient subsets deserve particular consideration (Table 17):

- **Women** tend to present later and may have somewhat atypical symptoms more frequently than men.¹⁹¹ Yet myocardial infarction remains the leading cause of death in women and it is therefore important to maintain a high degree of awareness for myocardial infarction in women with potential symptoms of ischaemia. In addition, several observational studies have shown that women tend to undergo fewer interventions than men and that they also less frequently receive reperfusion therapy;¹⁹² also that this may not be fully accounted for by the age difference, i.e. women experiencing myocardial infarction at a later age than men.^{193,194} When women are given effective reperfusion therapy, such as primary PCI, they

Table 17 Special subsets

Recommendations	Class ^a	Level ^b	Ref ^c
Both genders must be managed in a similar fashion.	I	C	-
A high index of suspicion for myocardial infarction must be maintained in women, diabetics, and elderly patients with atypical symptoms.	I	B	189
Special attention must be given to proper dosing of antithrombotics in elderly and renal failure patients.	I	B	190

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

experience the same risk of death as men.¹⁹⁵ It is therefore crucial to provide reperfusion therapy as effectively in women as in men. Women generally have lower body weight and are more susceptible to bleeding, which is why antithrombotic therapies and their doses should be used with close attention to bleeding risk.

- **Elderly patients** often present with atypical or mild symptoms, which may result in delayed or missed diagnoses of myocardial infarction (MI).¹⁸⁹ The elderly are at particular risk of bleeding and other complications from acute therapies because bleeding risk increases with age, because renal function tends to decrease and because the prevalence of comorbidities is high. In addition, observational studies have shown frequent overdosing of antithrombotic therapies.¹⁹⁰ It is therefore key to maintain a high index of suspicion for myocardial infarction in elderly patients who present with atypical complaints and to pay specific attention to proper dosing of antithrombotic therapies, particularly in relation with renal function.
- **Renal dysfunction** is present in approximately 30–40% of patients with ACS and is associated with a worse prognosis and increased bleeding risk.¹⁹⁶ Decisions on reperfusion in patients with STEMI have to be made before any assessment of renal function is available, but it is important to estimate the glomerular filtration rate as soon as possible after admission. ACS patients with chronic kidney disease are frequently overdosed with antithrombotics, leading to increased bleeding risk.¹⁹⁰ The benefit of ticagrelor was consistent or enhanced in patients with renal dysfunction: GFR < 60 mL/min in the PLATO trial.¹⁹⁷ In patients with known or anticipated reduction of renal function, several antithrombotic agents should be either withheld or their doses reduced appropriately (Table 18). Ensuring proper hydration during and after primary PCI, and limiting the dose of contrast agents, are important in minimizing the risk of contrast-induced nephropathy.⁴
- **Diabetic patients** are at higher risk of death and complications, but selection of antithrombotic therapies and reperfusion therapy is the same as in non-diabetics. The benefits of the potent oral

Table 18 Initial dosing of antithrombotic agents in patients with chronic kidney disease (estimated creatinine clearance <60 mL/min)

	Recommendation
Aspirin	No dose adjustment.
Clopidogrel	No dose adjustment.
Prasugrel	No dose adjustment. No experience with end-stage renal disease/dialysis.
Ticagrelor	No dose adjustment. No experience with end-stage renal disease/dialysis.
Enoxaparin	No adjustment of bolus dose. Following thrombolysis, in patients with creatinine clearance <30 mL/min, the s.c. doses are given once every 24 h.
Unfractionated heparin	No adjustment of bolus dose.
Fondaparinux	No dose adjustment. No experience in patients with end-stage renal disease or dialysis patients.
Bivalirudin	- In patients with moderate renal insufficiency (GFR 30–59 mL/min) a lower initial infusion rate of 1.4 mg/kg/h should be given. The bolus dose should not be changed. - In patients with severe renal insufficiency (GFR <30 mL/min) and in dialysis-dependent patients bivalirudin is contraindicated.
Abciximab	No specific recommendation. Careful consideration of bleeding risk.
Eptifibatide	- In patients with moderate renal insufficiency (GFR ≥30 to <50 mL/min), an i.v. bolus of 180 µg should be administered followed by a continuous infusion dose of 1.0 µg/kg/min for the duration of therapy. - In patients with severe renal insufficiency (GFR <30 mL/min) eptifibatide is contraindicated.
Tirofiban	In patients with severe renal insufficiency (GFR <30 mL/min) the infusion dose should be reduced to 50%.

GFR = glomerular filtration rate; i.v. = intravenous; s.c. = subcutaneous.

P2Y₁₂ receptor inhibitors (prasugrel or ticagrelor) vs. clopidogrel are consistent or enhanced in patients with diabetes.^{198,199}

3.6 Management of hyperglycaemia in the acute phase of ST-segment elevation myocardial infarction

Hyperglycaemia on admission is common in patients with an ACS and is a powerful predictor of mortality and in-hospital complications. These elevated glucose concentrations have been associated with an adverse prognosis, both in diabetic and non-diabetic patients. However, elevated glucose concentrations may also be a sign of disturbed long-term glucose metabolism, because of undiagnosed diabetes or impaired glucose tolerance.²⁰⁰ It was recently shown, in STEMI patients without known diabetes, that hyperglycaemia and elevated haemoglobin A1c (HbA_{1c}) are associated with a poor prognosis through different mechanisms, with hyperglycaemia especially predicting short-term prognosis in

Table 19 Management of hyperglycaemia in ST-segment elevation myocardial infarction

Recommendations	Class ^a	Level ^b	Ref ^c
Measurement of glycaemia is indicated at initial evaluation in all patients, and should be repeated in patients with known diabetes or hyperglycaemia.	I	C	-
Plans for optimal outpatient glucose control and secondary prevention must be determined in patients with diabetes before discharge.	I	C	-
The goals of glucose control in the acute phase should be to maintain glucose concentrations ≤11.0 mmol/L (200 mg/dL) while avoiding fall of glycaemia <5 mmol/L (<90 mg/dL). In some patients, this may require a dose-adjusted insulin infusion with monitoring of glucose, as long as hypoglycaemia is avoided.	IIa	B	202, 204, 207
A measurement of fasting glucose and HbA _{1c} and, in some cases, a post-discharge oral glucose tolerance test should be considered in patients with hyperglycaemia but without a history of diabetes.	IIa	B	208
Routine glucose-insulin-potassium infusion is not indicated.	III	A	118, 203

HbA_{1c} = haemoglobin A1c.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

association with a larger infarct size, whereas elevated HbA_{1c} was associated with long-term effects on outcome through a higher baseline risk.²⁰¹

Although correction of hyperglycaemia by insulin may be of benefit, clinical trials evaluating the effect of metabolic intervention in patients with STEMI showed conflicting results.²⁰² In particular, the benefits of tight glucose control through i.v. insulin shown in the *Diabetes, Insulin Glucose infusion in Acute Myocardial Infarction* (DIGAMI) trial was not confirmed in the subsequent DIGAMI-2 trial. Glucose–insulin–potassium infusions were found to be of no value and potentially harmful in a combined analysis of two large randomized trials.²⁰³ Additionally, in critically ill patients, there is a high risk of hypoglycaemia-related events when using intensive insulin therapy.²⁰⁴ The definitive answer with regard to glucose management in patients with STEMI, including treatment thresholds and glucose targets, is lacking and therefore a strategy of ‘strict, but not too strict’ glucose control in STEMI seems to be a practical approach. In the acute phase, it is reasonable to manage hyperglycaemia (i.e. maintain a blood glucose

concentration ≤ 11.0 mmol/L) but absolutely avoid hypoglycaemia.^{205,206} This may require a dose-adjusted insulin infusion with monitoring of glycaemia in some patients.

Given the frequency of unrecognised diabetes and impaired glucose metabolism in STEMI patients, it is reasonable to measure HbA_{1c} and fasting blood glucose in all patients without known diabetes, who developed hyperglycaemia during the acute phase (Table 19). If equivocal, an oral glucose tolerance test may be needed after discharge. This should preferably be measured 4 days after the acute phase. The best therapeutic strategy to specifically lower elevated HbA_{1c}-associated mortality risk remains uncertain, apart from strategies of secondary prevention (antiplatelet therapy, aggressive lipid control, blood pressure control, lifestyle modification, and cardiac rehabilitation), which should be implemented in all survivors of acute myocardial infarction. Whether the results of more intensive, early glycaemic therapy with oral agents provides cardiovascular protection is not known and warrants further study.²⁰⁷

4. Management during hospitalization and at discharge

4.1 Coronary care unit logistics and monitoring

4.1.1 Coronary care unit

STEMI patients should be admitted to an intensive cardiac care or coronary care unit (Table 20), or equivalent monitored unit, following reperfusion treatment. A coronary care unit is an intensive care unit designed to provide specialized care to patients with cardiovascular disease requiring constant monitoring. The staff should be thoroughly familiar with the management of ACS, arrhythmias, heart failure, mechanical circulatory support, and complex invasive

and non-invasive haemodynamic monitoring (arterial and pulmonary artery pressures), respiratory monitoring (continuous positive airway pressure and biphasic positive airway pressure), and support, as well as body cooling techniques. The unit should be able to manage patients with serious renal and pulmonary disease. The desirable organization, structure and criteria of the coronary care unit have been described in an ESC position paper.²⁰⁹

4.1.2 Monitoring

ECG monitoring for arrhythmias and ST-segment deviations should be continued for at least 24 h after symptom onset in all STEMI patients. Further monitoring for arrhythmia depends upon perceived risk and equipment available. When a patient leaves the coronary care unit, monitoring may be continued by telemetry.

4.1.3 Ambulation

Patients with significant LV damage should initially rest in bed before a first assessment of infarct extent and severity is possible for detection of early heart failure and arrhythmias. In uncomplicated cases, the patient can usually sit out of bed on the first day, be allowed to use a commode and undertake self-care and self-feeding. Ambulation can often start early (particularly in patients treated via the radial access). Patients who have experienced complications should be kept in bed for longer and their physical activity resumed as a function of symptoms and extent of myocardial damage.

4.1.4 Length of stay

The optimal length of stay in the coronary care unit and hospital should be determined on an individual basis, considering the patient's particular medical and social situation, including pre-morbid health. Over the years, there has been a progressive reduction

Table 20 Logistical issues for hospital stay

Recommendations	Class ^a	Level ^b	Ref ^c
All hospitals participating in the care of STEMI patients should have a coronary care unit equipped to provide all aspects of care for STEMI patients, including treatment of ischaemia, severe heart failure, arrhythmias and common comorbidities.	I	C	-
Length of stay in the coronary care unit			
Patients undergoing uncomplicated successful reperfusion therapy should be kept in the coronary care unit for a minimum of 24 h, after which they may be moved to a step-down monitored bed for another 24–48 h.	I	C	-
Transfer back to a referring non-PCI hospital			
Early transfer (same day) may be considered in selected, low-risk patients after successful primary PCI without observed arrhythmia.	IIb	C	-
Hospital discharge			
Early discharge (after approximately 72 h) is reasonable in selected low-risk patients, if early rehabilitation and adequate follow-up are arranged.	IIb	B	212, 215, 216

PCI = percutaneous coronary intervention; STEMI = ST-segment elevation myocardial infarction.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

in length of stay after myocardial infarction—especially following successful primary revascularization—without any increase in subsequent mortality, suggesting that earlier discharge is not associated with late mortality.^{210,211} Moreover, the *Primary Angioplasty in Myocardial Infarction II* (PAMI-II) trial showed that low-risk patients with successful primary PCI could safely be discharged from hospital at day 3 without non-invasive testing.²¹² Overall, early discharge of low-risk patients (within 72 h) is both feasible and safe in patients with uncomplicated STEMI and successful primary PCI.^{211–213} To identify these low-risk patients, schemes such as the PAMI-II criteria or the Zwolle primary PCI Index can be helpful.^{212,213} The PAMI II criteria designate as low risk patients aged <70 years, with a left ventricular ejection fraction >45%, one- or two-vessel disease, successful PTCA and no persistent arrhythmias. Nevertheless, a short hospital stay implies limited time for proper patient education and up-titration of secondary prevention treatments. Consequently, these patients should be offered early post-discharge consultations with a cardiologist or primary care physician and the option of a formal rehabilitation program, either in-hospital or on an outpatient basis.

Current practice may also include early transfer to a local hospital following successful primary PCI. In selected low-risk patients—identified as being asymptomatic without any arrhythmia, haemodynamically stable, not requiring vasoactive or mechanical support and not scheduled for further revascularization—early transfer (same day) under adequate monitoring and supervision appears safe and feasible.²¹⁴

4.2 Risk assessment and imaging

4.2.1 Indications and timing (Table 21)

After reperfusion treatment, it is important to identify patients at high risk of further events such as reinfarction or death, and hopefully to intervene in order to prevent these events. Because the

risk of events decreases with time, early risk assessment is indicated. Assessment of infarct size and resting LV function, usually by echocardiography, should be undertaken before discharge. The timing of further investigations will depend on local facilities and whether angiography and PCI have been performed successfully. With the increasing use of primary PCI, risk assessment for ischaemia before discharge has become less important, since it can be assumed that the infarct-related coronary lesion has been treated and stabilized and the presence or absence of significant lesions in other arteries has been assessed. Several risk scores have been developed, based on readily identifiable parameters in the acute phase before reperfusion.^{217–219} Clinical indicators of high risk in the acute phase include older age, fast heart rate, hypotension, Killip class >I, anterior infarction, previous infarction, elevated initial serum creatinine and history of heart failure. Malignant arrhythmias, persistent chest pain and early angina on minimal exertion are also associated with worse outcome.

If, in spite of the angiography performed in the acute phase, there are concerns about inducible ischaemia, an outpatient exercise-testing or stress-imaging test (using scintigraphy, echocardiography or magnetic resonance imaging) within 4–6 weeks is appropriate (Table 9). Because of high availability and low cost, an exercise ECG is commonly used. However, in patients with previous myocardial infarction, its accuracy is limited. Stress imaging tests are more accurate and allow localization of the ischaemia. The most-validated tests are perfusion scintigraphy and stress echocardiography. In post-myocardial infarction patients, the detection of residual ischaemia is challenging, due to existing wall-motion abnormalities. Computed tomography angiography is a sensitive technique to detect coronary lesions but, as an anatomical test, it does not assess ischaemia, which remains essential for therapeutic decisions. If the main concern is arrhythmia, additional electrophysiological testing may be needed before discharge, and

Table 21 Summary of indications for imaging and stress testing

Recommendations	Class ^a	Level ^b	Ref ^c
At presentation			
In the acute phase, when diagnosis is uncertain, emergency echocardiography may be useful. However, if inconclusive or unavailable and persistent doubt, emergency angiography should be considered.	I	C	-
After the acute phase			
All patients should have an echocardiography for assessment of infarct size and resting LV function,	I	B	220, 221
If echocardiography is not feasible, MRI may be used as an alternative.	IIb	C	-
Before or after discharge			
For patients with multivessel disease, or in whom revascularization of other vessels is considered, stress testing or imaging (e.g. using stress myocardial perfusion scintigraphy, stress echocardiography, positron emission tomography or MRI) for ischaemia and viability is indicated.	I	A	4, 220, 222
Computed tomography angiography has no role in the routine management of STEMI patients.	III	C	-

Echocardiography = transthoracic echocardiography, or transoesophageal if required; LV = left ventricular; MRI = magnetic resonance imaging; STEMI = ST-segment elevation myocardial infarction.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

repeated assessment of LV ejection fraction after discharge may be important for selecting candidates for implantation of a cardioverter-defibrillator as primary prevention (see *below*).

All patients should have their metabolic risk markers measured during the index admission, including total cholesterol, low-density lipoprotein (LDL) cholesterol, high-density lipoprotein cholesterol, fasting triglycerides and plasma glucose, as well as renal function. Since LDL levels tend to decrease during the first days after myocardial infarction, they are best measured as soon as possible after admission.

4.3 Assessment of myocardial viability

LV dysfunction after acute myocardial infarction may be due to necrosis, to stunning of viable myocardium remaining in the infarct territory, to hibernation of viable myocardium, or to a combination of all three. Simple stunning should recover within 2 weeks of the acute ischaemic insult if ischaemia does not persist but, if it does, then recurrent stunning may become hibernation and require revascularization for recovery of function. These concepts are of most relevance in the patient with severely impaired LV function after infarction when the need for revascularization to improve function is considered (e.g. after successful fibrinolysis).

Multiple imaging techniques, including PET, single-photon emission CT, and dobutamine stress echocardiography have been evaluated extensively for assessment of viability and prediction of clinical outcome after myocardial revascularization. In general, nuclear imaging techniques have a high sensitivity, whereas techniques evaluating contractile reserve have a somewhat lower sensitivity but higher specificity. MRI has a high diagnostic accuracy for assessing transmural extent of myocardial scar tissue, but its ability to detect viability and predict recovery of wall motion is not superior to other imaging techniques.²²³ The differences in performance of the various imaging techniques are small, and experience and availability commonly determine which technique is used. Current evidence is mostly based on observational studies or meta-analyses, with the exception of two randomized clinical trials, both relating to PET imaging.²²² Patients with a substantial amount of dysfunctional but viable myocardium are likely to benefit from myocardial revascularization and may show improvements in regional and global contractile function, symptoms, exercise capacity and long-term prognosis.²²⁰

4.4 Long-term therapies for ST-segment elevation myocardial infarction

Coronary heart disease is a chronic condition and patients who have recovered from a STEMI are at high risk for new events and premature death. In fact, in long-term cohorts, most patients with STEMI who die do so after discharge from the index event.¹⁴ Several evidence-based interventions can improve prognosis. Even though long-term management of this large group of patients will be the responsibility of the general practitioner, these interventions will have a higher chance of being implemented if initiated during the hospital stay. In addition, lifestyle changes should be explained and proposed to the patient before discharge. However, habits of a lifetime are not easily changed and the implementation and follow-up of these changes are a long-term

undertaking. In this regard, a close collaboration between the cardiologist and the primary care physician is critically important. With ever-shorter length of hospital stay in patients with STEMI, there is no longer a clear distinction between acute and chronic therapies in STEMI. This chapter summarizes both lifestyle interventions and drug therapies that need to be considered and implemented before hospital discharge (*Table 22*).

4.4.1 Lifestyle interventions and risk factor control

Key lifestyle interventions include cessation of smoking and tight blood pressure control, advice regarding diet and weight control, and the encouragement of physical activity. Detailed recommendations are available from the ESC guidelines on prevention.²²⁴ Even though long-term management of this large group of patients will be the responsibility of the primary care physician, these interventions will have a higher chance of being implemented if initiated during the hospital stay. In addition, the benefits and importance of lifestyle changes should be explained and proposed to the patient—who is the key player—before discharge. However, habits of a lifetime are not easily changed, and the implementation and follow-up of these changes are a long-term undertaking. In this regard, a close collaboration between the cardiologist and the general practitioner, specialist rehabilitation nurses, pharmacists, dieticians, physiotherapists is critically important.

4.4.1.1 Smoking cessation

Unselected ACS patients who are smokers are twice as likely to present with a STEMI, compared with non-smokers, indicating a strong prothrombotic effect of smoking. Observational studies show that patients who stop smoking reduce their mortality in the succeeding years compared with continued smokers. Stopping smoking is potentially the most effective of all secondary prevention measures,²²⁵ and much effort should be devoted to this end. Patients do not smoke during the acute phase of a STEMI and the convalescent period is ideal for health professionals to help smokers to quit. However, resumption of smoking is common after discharge, and continued support and advice are needed during rehabilitation. Nicotine replacement, bupropione and antidepressants may be useful. Nicotine patches have been demonstrated to be safe in ACS patients.²²⁶ A randomized study has also demonstrated the effectiveness of a nurse-directed programme.²²⁷ A smoking cessation protocol should be adopted by each hospital.

4.4.1.2 Diet and weight control

Current guidelines on prevention recommend: ²²⁴ (i) eating a wide variety of foods; (ii) adjustment of calorie intake to avoid obesity; (iii) increased consumption of fruit and vegetables, along with wholegrain cereals and bread, fish (especially oily varieties), lean meat and low-fat dairy products; (iv) replacing saturated and trans fats with monounsaturated and polyunsaturated fats from vegetable and marine sources, and to reduce total fats (of which less than one-third should be saturated) to <30% of total calorie intake, and (v) to reduce salt intake if blood pressure is raised. Many processed and prepared foods are high in salt and in fat of doubtful quality. There is no evidence for the use of

antioxidant supplements, low glycaemic index diets or homocysteine-lowering therapies following STEMI.

Obesity is an increasing problem in patients with STEMI. Current ESC Guidelines define a body mass index (BMI) $<25 \text{ kg/m}^2$ as optimal, and recommend weight reduction when the BMI is 30 kg/m^2 or more, and when waist circumference is $>102 \text{ cm}$ in men or $>88 \text{ cm}$ in women, because weight loss can improve many obesity-related risk factors. However, it has not been established that weight reduction *per se* reduces mortality.

4.4.1.3 Physical activity

Exercise therapy has long been used for rehabilitation purposes following STEMI and the benefit of regular physical exercise in stable CAD patients is also well established. It can reduce the anxiety associated with the life-threatening illness and improve patient self-confidence. Four mechanisms are considered to be important mediators of a reduced cardiac event rate: (i) improvement of endothelial function; (ii) reduced progression of coronary lesions; (iii) reduced thrombotic risk and (iv) improved collateralization. In a large meta-analysis, exercise training as part of coronary rehabilitation programmes was associated with a 26% reduction in cardiac mortality rate in patients with CAD.²²⁸ It should be appreciated that, apart from its influence on mortality, exercise rehabilitation can have other beneficial effects. Exercise capacity, cardiorespiratory fitness, and perception of well-being have also been reported to improve, at least during the actual training period, even in elderly patients. Thirty minutes of moderate intensity aerobic exercise at least five times per week is recommended.²²⁴ Each step of increase in peak exercise capacity is associated with a reduction in all-cause mortality risk in the range of 8–14%.²²⁹

4.4.1.4 Blood pressure control

In hypertensive patients with STEMI, blood pressure should be well controlled. Data from a retrospective analysis of the PRavastatin Or atorVastatin Evaluation and Infection Therapy–Thrombolysis In Myocardial Infarction 22 (PROVE IT–TIMI 22) trial suggest that, after acute coronary syndromes, the blood pressure goal should be a systolic $<140 \text{ mmHg}$ but not $<110 \text{ mmHg}$.²³⁰ The pharmacotherapy (beta-blockers, ACE inhibitors or ARBs) recommended after STEMI, in addition to lifestyle modifications (reduced salt intake, increased physical activity and weight loss) usually helps achieve these goals. Additional drug therapy may be needed.

4.4.1.5 Psychosocial factor interventions

There is evidence that stress management is useful in this setting: in a recent trial 362 patients, aged 75 years or younger, with acute myocardial infarction, PCI or CABG within the past 12 months, were randomized to receive traditional care or traditional care *plus* a cognitive behavioural therapy programme focussed on stress management. During a mean of 94 months of follow-up, the intervention group had a 41% lower rate of fatal and non-fatal first recurrent cardiovascular disease events (45% fewer recurrent acute myocardial infarctions) and a non-significant 28% lower all-cause mortality than the reference group after adjustment for other outcome-affecting variables. There was also a strong

dose-response effect between attendance rate in the group sessions and outcome rate.²³¹

4.4.1.6 Exercise-based rehabilitation programme

Exercise-based rehabilitation has been shown to be effective at reducing all-cause mortality and the risk of reinfarction, as well as improving risk factors, exercise-based capacity and health-related quality of life after myocardial infarction.^{232,233} Yet these benefits were established in the era preceding modern treatment of STEMI and a recent British randomized trial failed to demonstrate benefits of a rehabilitation programme on clinical outcomes or quality of life.²³⁴ In another larger randomized study, a long-term multifactorial, educational and behavioural intervention was proven to be feasible and sustainable over a long period after myocardial infarction, and reduced some clinical outcomes—particularly re-infarction—and global cardiovascular risk.²³⁵ An additional benefit of rehabilitation programmes is to help ensure proper titration and monitoring of key, evidence-based therapies after STEMI. Nowadays, in patients with an uncomplicated course, rehabilitation can often be performed on an outpatient basis with an efficacy similar to that of centre-based cardiac rehabilitation.²³⁶

4.4.1.7 Resumption of activities

No generalizable recommendations can be made regarding the delay to resumption of daily activities. Decisions should be individualized, based on left ventricular function, completeness of revascularization and rhythm control. Extended sick leave is usually negative and light-to-moderate physical activity after discharge should be encouraged. Sexual activity can be resumed early if adjusted to physical ability. Long distance air travel should be avoided for 4–6 weeks if residual ischaemia or left ventricular dysfunction is present.

4.4.2 Antithrombotic therapy

4.4.2.2 Aspirin

Given its established benefits in secondary prevention,²³⁷ aspirin should be used indefinitely in all patients with STEMI. The dosage of aspirin is debated. In respect of the first few days of treatment, the Clopidogrel and aspirin Optimal Dose usage to reduce recurrent events—Seventh organization to assess strategies in ischaemic syndromes (CURRENT/OASIS 7) large, randomized trial failed to demonstrate a difference in hard clinical outcomes when comparing low doses (75–100 mg/day) or relatively high doses of 300–325 mg/day.¹¹⁵ There were, however, fewer gastro-intestinal bleeds with lower doses. For the long term, low doses (70–100 mg) are generally used. Platelet aggregation data suggest that rapid turnover of platelets in diabetic patients may require higher doses or more frequent dosing of aspirin to achieve platelet inhibition,^{238,239} but there is no proof of clinical benefit of such a strategy. Patients with a history of hypersensitivity to aspirin can undergo desensitization and continue therapy indefinitely.^{240–242} Patients who are truly intolerant to aspirin can instead receive clopidogrel (75 mg/day) as long-term secondary prevention.²⁴³

4.4.2.2 Duration of dual antiplatelet therapy and antithrombotic combination therapies after ST-segment elevation myocardial infarction DAPT, combining aspirin and an ADP-receptor blocker (clopidogrel, prasugrel or ticagrelor), is recommended in patients with STEMI who are undergoing primary PCI (for up to 12 months), fibrinolysis (for up to 12 months, although the data available pertain only to one month of DAPT) and in those patients who have not undergone reperfusion therapy (for at least 1 month and up to 12 months). The choice of ADP-receptor blocker has been discussed previously. While there are no trial data to support extended DAPT, treatment for 12 months after stenting and for 9–12 months following STEMI has traditionally been recommended by consensus in prior guidelines, regardless of whether a stent (BMS or DES) was used.^{1,4,244} Some studies have suggested that there is no benefit in extended durations of DAPT beyond 6 or 12 months after placement of a DES to prevent ischaemic events and stent thrombosis,^{245–247} but these studies, even when pooled, include a relatively small number of STEMI patients. Several ongoing large trials, including the Dual Antiplatelet Therapy (DAPT) study,²⁴⁸ are testing whether longer durations of dual antiplatelet therapy following stenting are of clinical benefit. Clearly, after stenting for ACS, particularly STEMI, extended DAPT reduces the risk of stent thrombosis, reinfarction and cardiovascular mortality,²⁴⁹ and more potent DAPT is associated with greater clinical benefits post-ACS of any type.^{109,110,188} Pending the results of ongoing trials, a 9–12 months duration of DAPT is recommended, with a strict minimum of one month for patients who have received a BMS and six months for those who received a DES. It is important to inform patients and their physicians about the need to avoid premature discontinuation of DAPT.

In patients with STEMI and with atrial fibrillation and the need for permanent anticoagulation after primary PCI [based on Cardiac failure, Hypertension, Age, Diabetes, Stroke (Doubled) (CHADS₂) or Cardiac failure, Hypertension, Age ≥ 75 (Doubled), Diabetes, Stroke (Doubled) – VASc disease, Age 65–74 and Sex category (Female) (CHA₂DS₂-VASc) scores of ≥ 2],^{250,251} ‘triple therapy’, combining aspirin, an ADP receptor antagonist and an oral anticoagulant, is recommended to reduce the burden of thromboembolic complications associated with atrial fibrillation and minimize the risk of stent thrombosis.⁴ However, it is also associated with an increase in bleeding complications and should therefore be used for the shortest possible duration.^{252,253} This is an area of controversy, with missing evidence, and several consensus documents have tried to offer algorithms for decision-making.^{253–255} Moreover, in STEMI patients with an indication for anticoagulation, and in whom stents are needed, selection of BMS over DES would appear to minimize the duration of triple therapy and therefore the risk of bleeding. These benefits should be weighed against the benefits of DES in preventing restenosis.^{4,253}

Gastric protection, preferably with a proton pump inhibitor, should be considered for patients with a history of gastrointestinal bleeding and is appropriate for patients with multiple risk factors for bleeding, such as advanced age, concurrent use of anticoagulants, steroids or non-steroidal anti-inflammatory drugs including high dose aspirin, and *Helicobacter pylori* infection.²⁵⁶ There is no

pharmacokinetic interaction between proton pump inhibitors and the new potent P2Y₁₂ receptor inhibitors and no clear evidence that the pharmacokinetic interaction of clopidogrel with some proton pump inhibitors has meaningful clinical consequences.^{257–261} In any case, the benefits of avoiding or minimizing bleeding in patients at high risk outweigh the concerns raised by this pharmacokinetic interaction.

The recent Anti-Xa Therapy to Lower cardiovascular events in Addition to Standard therapy in subjects with Acute Coronary Syndrome–Thrombolysis In Myocardial Infarction 51 (ATLAS ACS 2–TIMI 51) trial tested the addition of rivaroxaban, a factor Xa antagonist to aspirin and clopidogrel following ACS.²⁶² In that trial, a low dose of rivaroxaban (2.5 mg twice daily) reduced the composite primary endpoint of cardiovascular death, myocardial infarction and stroke, but also all-cause mortality. Interestingly, stent thrombosis was reduced by one third. This was associated with threefold increases in non-CABG-related major bleeding, and intracranial haemorrhage. Importantly, the high dose of rivaroxaban (5 mg twice daily) was not associated with similar benefits and was associated with a major increase in the risk of bleeding. The ATLAS ACS 2–TIMI 51 trial did not test a combination of rivaroxaban with prasugrel or ticagrelor, which might be associated with even more bleeding. This trial suggests that, in selected patients at low bleeding risk, the 2.5 mg dose of rivaroxaban may be considered in patients who receive aspirin and clopidogrel after STEMI. However, a phase III trial of another factor Xa antagonist (apixaban), the Apixaban for Prevention of Acute Ischemic and Safety Events (APPRAISE-2) trial,²⁶³ failed to find similar benefits of adding a high dose of apixaban to single or DAPT in a very-high-risk ACS population. Finally, darexaban and dabigatran were both tested in phase-II dose-ranging trials in post ACS patients,^{264,265} with, in both cases, dose-dependent increases in major bleeding but no signal of added efficacy when adding anticoagulant therapy to antiplatelet therapy in this setting. In conclusion, the role of novel anticoagulants in combination with DAPT in secondary prevention of STEMI remains under discussion. The substantial mortality benefit seen with a low dose of rivaroxaban combined with aspirin and clopidogrel is intriguing but interpretation of the totality of evidence for the class is difficult.

4.4.3 Beta-blockers

The benefit of long-term treatment with beta-blockers after STEMI is well established, although mostly from trials pre-dating the advent of modern reperfusion therapy and pharmacotherapy. The role of routine early i.v. administration, on the other hand, is less firmly established. Oral administration of beta-blockers appears to be associated with benefit, but high, early i.v. dosage was associated with an early hazard and increased mortality in the large COMMIT trial.²⁶⁶ Thus, early i.v. use of beta-blockers is contraindicated in patients with clinical signs of hypotension or congestive heart failure. Early use may be associated with a modest benefit in low-risk, haemodynamically stable patients. In most patients, however, it is prudent to wait for the patient to stabilize before starting a beta-blocker and to use oral, rather than i.v., administration. In contemporary trials utilizing primary PCI, beta-blockers have not been investigated, although it is not unreasonable to extrapolate their benefit to this setting.

4.4.4 Lipid-lowering therapy

The benefits of statins in secondary prevention have been unequivocally demonstrated,²⁶⁷ and specific trials have demonstrated the benefit of early and intensive statin therapy.^{268,269} The recent meta-analysis of trials comparing more- vs. less-intensive LDL-cholesterol lowering with statins indicated that, compared with less-intensive regimes, more-intensive statin therapy produced reductions in the risks of cardiovascular death, non-fatal myocardial infarction, ischaemic stroke and coronary revascularization. For every 1.0 mmol/L reduction in LDL cholesterol, these further reductions in risk were similar to the proportional reductions in the trials of statin vs. control. Therefore, statins should be given to all patients with acute myocardial infarction, irrespective of cholesterol concentration. This treatment should be started early during admission, as this increases patient adherence after discharge, and given at high doses, as this is associated with early and sustained clinical benefits.²⁷⁰ The treatment goal is an LDL-cholesterol concentration of <1.8 mmol/L (<70 mg/dL). The use of lower-intensity statin therapy should be considered in patients at increased risk of side-effects from statins (e.g. the elderly, patients with hepatic or renal impairment, with previous side-effects of statins or the potential for interaction with essential concomitant therapy).²⁷⁰ Lipids should be re-evaluated 4–6 weeks after the ACS, to determine whether the target levels have been reached and regarding safety issues; the statin dose can then be adjusted accordingly. Given trial results with high doses of atorvastatin and simvastatin and the risks associated with high-dose simvastatin,²⁷¹ the strongest trial data available so far favour atorvastatin at a dose of 80 mg daily, unless a high dose of statin was poorly tolerated previously in that patient. In patients known to be intolerant of any dose of statin, treatment with ezetimibe should be considered.

The consumption of n-3 polyunsaturated fatty acids reduced mortality in survivors of myocardial infarction in one study,²⁷² but failed to affect clinical outcomes in two more recent trials using modern evidence-based prevention therapies and therefore cannot be recommended in routine practice.^{273,274}

4.4.5 Nitrates

The routine use of nitrates in STEMI has not been shown to be of value and is not therefore recommended. Intravenous nitrates may be useful during the acute phase in patients with hypertension or heart failure, provided there is no hypotension, right ventricular infarction or use of phosphodiesterase type 5 inhibitors in the previous 48 h. In the acute and stable phase, nitrates remain valuable agents to control anginal symptoms.

4.4.6 Calcium antagonists

A meta-analysis of trials involving calcium antagonists early in the course of a STEMI showed a trend towards harm.²⁷⁵ There is no case for using calcium antagonists for prophylactic purposes in the acute phase. In the chronic phase, verapamil may be helpful to prevent reinfarction and death.^{276,277} Thus, in patients with contraindications to beta-blockers, particularly in the presence of obstructive airway disease, calcium antagonists are a reasonable option for patients without heart failure, although caution has to be exercised in patients with impaired LV function. Routine use

of dihydropyridines, on the other hand, have failed to show benefit after STEMI and they should therefore only be prescribed for clear indications such as hypertension or angina.²⁷⁸

4.4.7 Angiotensin-converting enzyme inhibitors and angiotensin receptor blockers

It is well established that angiotensin-converting enzyme (ACE) inhibitors should be given to patients with an impaired ejection fraction ($<40\%$) or who have experienced heart failure in the early phase. A systematic overview of trials of ACE inhibition early in STEMI indicated that this therapy is safe, well tolerated and associated with a small but significant reduction in 30-day mortality, with most of the benefit observed in the first week.²⁷⁹ Opinions still differ as to whether to give ACE inhibitors to all patients or to high-risk patients only. Patients who do not tolerate an ACE inhibitor should be given an angiotensin receptor blocker (ARB).²⁸⁰ Use of ACE inhibitors should be considered in all patients with atherosclerosis, but, given their relatively modest effect, their long-term use cannot be considered mandatory in post-STEMI patients who are normotensive, without heart failure, or have neither LV systolic dysfunction nor diabetes. Two trials have evaluated ARBs, in the context of STEMI, as alternatives to ACE inhibitors: the Optimal Trial In Myocardial infarction with the Angiotensin II Antagonist Losartan (OPTIMAAL) trial with losartan (50 mg) failed to show either superiority or non-inferiority when compared with captopril (50 mg three times daily).²⁸⁰ Conversely, the VALsartan In Acute myocardial infarction Trial compared valsartan alone (160 mg twice daily), full-dose captopril (50 mg three times daily), or both (80 mg twice daily and 50 mg three times daily).²⁸¹ Mortality was similar in the three groups but discontinuations were more frequent in the groups receiving captopril. Therefore valsartan, in the dosages used in the trial, represents an alternative to ACE inhibitors in patients who have clinical signs of heart failure and/or an ejection fraction $\leq 40\%$, particularly in patients who do not tolerate ACE inhibitors.

4.4.8 Aldosterone antagonists

The Eplerenone Post-AMI Heart failure Efficacy and SURvival Study (EPHESUS) trial randomized 6642 post-STEMI patients with LV dysfunction (ejection fraction $<40\%$) and heart failure or diabetes to eplerenone, a selective aldosterone blocker, or placebo.²⁸² After a mean follow-up of 16 months, there was a 15% relative reduction in total mortality and a 13% reduction in the composite of death and hospitalization for cardiovascular events. Serious hyperkalaemia was more frequent in the group receiving eplerenone. The results suggest that aldosterone blockade may be considered for post-STEMI patients with an ejection fraction $\leq 40\%$ and heart failure or diabetes, provided that the creatinine concentration is <221 $\mu\text{mol/L}$ (2.5 mg/dL) in men and <177 $\mu\text{mol/L}$ (2.0 mg/dL) in women, and potassium is <5.0 mEq/L. Routine monitoring of serum potassium is warranted.

4.4.9 Magnesium, glucose–insulin–potassium, lidocaine

There is no benefit in the routine administration of magnesium, glucose–insulin–potassium, or lidocaine in patients with STEMI.

Table 22 Routine therapies in the acute, subacute and long term phase of ST-segment elevation myocardial infarction

Recommendations	Class ^a	Level ^b	Ref ^c
Active smokers with STEMI must receive counselling and be referred to a smoking cessation programme.	I	B	225
Each hospital participating in the care of STEMI patients must have a smoking cessation protocol.	I	C	-
Exercise-based rehabilitation is recommended.	I	B	232, 233
Antiplatelet therapy with low dose aspirin (75–100 mg) is indicated indefinitely after STEMI.	I	A	237
In patients who are intolerant to aspirin, clopidogrel is indicated as an alternative to aspirin.	I	B	243
DAPT with a combination of aspirin and prasugrel or aspirin and ticagrelor is recommended (over aspirin and clopidogrel) in patients treated with PCI.	I	A	109, 110
DAPT with aspirin and an oral ADP receptor antagonist must be continued for up to 12 months after STEMI, with a strict minimum of:	I	C	245–247, 283
• 1 month for patients receiving BMS	I	C	
• 6 months for patients receiving DES	IIb	B	
In patients with left ventricular thrombus, anticoagulation should be instituted for a minimum of 3 months.	IIa	B	344–346
In patients with a clear indication for oral anticoagulation (e.g. atrial fibrillation with CHA ₂ DS ₂ -VASC Score ≥ 2 or mechanical valve prosthesis), oral anticoagulation must be implemented in addition to antiplatelet therapy.	I	C	-
If patients require triple antithrombotic therapy, combining DAPT and OAC, e.g. because of stent placement and an obligatory indication for OAC, the duration of dual antiplatelet therapy should be minimized to reduce bleeding risk.	I	C	-
In selected patients who receive aspirin and clopidogrel, low-dose rivaroxaban (2.5 mg twice daily) may be considered if the patient is at low bleeding risk.	IIb	B	262
DAPT should be used up to 1 year in patients with STEMI who did not receive a stent.	IIa	C	-
Gastric protection with a proton pump inhibitor should be considered for the duration of DAPT therapy in patients at high risk of bleeding.	IIa	C	256
Oral treatment with beta-blockers should be considered during hospital stay and continued thereafter in all STEMI patients without contraindications.	IIa	B	I, 266
Oral treatment with beta-blockers is indicated in patients with heart failure or LV dysfunction.	I	A	284–288
Intravenous beta-blockers must be avoided in patients with hypotension or heart failure.	III	B	266
Intravenous beta-blockers should be considered at the time of presentation in patients without contraindications, with high blood pressure, tachycardia and no signs of heart failure.	IIa	B	266
A fasting lipid profile must be obtained in all STEMI patients, as soon as possible after presentation.	I	C	-
It is recommended to initiate or continue high dose statins early after admission in all STEMI patients without contraindication or history of intolerance, regardless of initial cholesterol values.	I	A	267
Reassessment of LDL-cholesterol should be considered after 4–6 weeks to ensure that a target value of ≤ 1.8 mmol/L (70 mg/dL) has been reached.	IIa	C	270
Verapamil may be considered for secondary prevention in patients with absolute contraindications to beta-blockers and no heart failure.	IIb	B	276
ACE inhibitors are indicated starting within the first 24 h of STEMI in patients with evidence of heart failure, LV systolic dysfunction, diabetes or an anterior infarct.	I	A	279
An ARB, preferably valsartan, is an alternative to ACE inhibitors in patients with heart failure or LV systolic dysfunction, particularly those who are intolerant to ACE inhibitors.	I	B	280, 281
ACE inhibitors should be considered in all patients in the absence of contraindications.	IIa	A	289, 290
Aldosterone antagonists, e.g. eplerenone, are indicated in patients with an ejection fraction $\leq 40\%$ and heart failure or diabetes, provided no renal failure or hyperkalaemia.	I	B	282

ACE = angiotensin-converting enzyme; ACS = acute coronary syndrome; ARB = angiotensin receptor blocker; BMS = bare metal stent; DAPT = dual antiplatelet therapy; DES = drug-eluting stent; LDL = low-density lipoprotein; LV = left ventricular; STEMI = ST-segment elevation myocardial infarction.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

5. Complications following ST-segment elevation myocardial infarction

5.1 Haemodynamic disturbances

5.1.1 Heart failure

Myocardial dysfunction frequently occurs during the acute and sub-acute phases following STEMI. Rapid improvement in ventricular function is usually seen following successful early revascularization of the infarct-related artery by PCI or thrombolysis. However, if the STEMI results in transmural injury and/or microvascular obstruction, especially of the anterior wall, pump failure with pathological remodelling—and the clinical symptoms and signs of heart failure—may complicate the acute phase and result in chronic heart failure. Heart failure may also be the consequence of sustained arrhythmias or mechanical complications of STEMI.

The diagnosis of clinical heart failure during the acute and sub-acute phases of STEMI is based on typical symptoms such as dyspnoea, signs such as sinus tachycardia, a third heart sound or pulmonary rales, and some objective evidence of cardiac dysfunction, such as LV dilatation and reduced ejection fraction. Natriuretic peptides [B-type natriuretic peptide (BNP) and N-terminal pro-BNP] rise in response to increased myocardial wall stress and have been shown to be useful biomarkers in the management of patients with chronic heart failure. Evidence has established their role in diagnosing, staging, making admission/discharge decisions and identifying patients at risk for adverse clinical events. Normal levels have strong negative predictive value. Their value in acute heart failure following MI is less well established, due to the abrupt changes in LV systolic and diastolic function that follow MI and the relatively long half-lives of these peptides. Importantly, conditions such as LV hypertrophy, tachycardia, ischaemia, renal dysfunction, advanced age, obesity and treatment may influence levels. There are no definitive cut-off values in patients with signs and symptoms of heart failure following acute MI, and levels should be interpreted in conjunction with the patient's clinical condition.²⁸¹

LV dysfunction is the single strongest predictor of mortality following STEMI. The mechanisms responsible for LV dysfunction in the acute phase include myocardial loss and remodelling due to infarction, ischaemic dysfunction (stunning), atrial and ventricular arrhythmias and valvular dysfunction (pre-existing or new). There is frequently evidence of both systolic and diastolic dysfunction. Co-morbidities such as infection, pulmonary disease, renal dysfunction, diabetes or anaemia often contribute to the clinical picture. The degree of heart failure following myocardial infarction may be categorized according to the Killip classification: Class I, no rales or third heart sound; Class II, pulmonary congestion with rales over <50% of the lung fields, sinus tachycardia or third heart sound; Class III, pulmonary oedema with rales over 50% of the lung fields and Class IV, cardiogenic shock.

Haemodynamic assessment should be based on thorough physical examination, continuous ECG telemetry of heart and rhythm, oxygen saturation, blood pressure monitoring and hourly urinary output. Patients suspected of having heart failure should be

evaluated early by transthoracic echocardiography/Doppler. Echocardiography is the key diagnostic tool and should be performed to assess LV function and volumes, valvular function, extent of myocardial damage, and to detect mechanical complications. Doppler evaluation permits assessment of flow, gradients, diastolic function and filling pressures. Chest X-ray will assess the extent of pulmonary congestion and detect other important conditions such as pulmonary infection, chronic lung disease and pleural effusion.

Unexpected deterioration of the patient's clinical status, with evidence of haemodynamic compromise, should trigger a re-evaluation with a repeat echocardiographic examination, specifically searching for evidence of progressive LV dysfunction or mechanical complications.

In selected patients who do not respond adequately to conventional measures—and who have evidence of ongoing ischaemia, persistent ST elevation or new LBBB—the need for further revascularization should be considered.

Patients with extensive myocardial injury during the acute phase may develop the symptoms and signs of chronic heart failure. This diagnosis requires management according to guidelines for the treatment of chronic heart failure.²⁸⁴ Selected patients with symptomatic, chronic heart failure and a reduced ejection fraction or electrical dyssynchrony, as evidenced by QRS prolongation, may satisfy criteria for implantation of a cardioverter-defibrillator, cardiac resynchronization therapy (CRT), or a cardiac resynchronization therapy defibrillator. These criteria are presented in a recent guideline focusing on device therapy.²⁹¹

5.1.1.1 Hypotension

Hypotension is defined as persistent systolic blood pressure <90 mmHg. It may be due to heart failure but also to correctable hypovolaemia, treatable rhythm disturbance or mechanical complications. If prolonged, hypotension may cause renal dysfunction, acute tubular necrosis and reduced urinary output.

5.1.1.2 Pulmonary congestion

Pulmonary congestion is characterized by dyspnoea with basal pulmonary rales, reduced arterial oxygen saturation, pulmonary congestion on chest X-ray and clinical response to diuretic and/or vasodilator therapy.

5.1.1.3 Low output states

Low output states combine signs of poor peripheral perfusion and hypotension, renal dysfunction and reduced urinary output. Echocardiography may reveal poor left ventricular function, a mechanical complication or right ventricular infarction.

5.1.1.4 Cardiogenic shock

Cardiogenic shock complicates 6–10% of all cases of STEMI and remains a leading cause of death, with hospital mortality rates approaching 50%.²⁹² Although shock often develops early after the onset of acute myocardial infarction, it is typically not diagnosed on hospital presentation.²⁹² In the SHould we emergently revascularize Occluded coronaries for Cardiogenic shock (SHOCK) trial registry,²⁹³ of the patients who eventually developed shock during hospitalization, this occurred within 6 h in about 50% and within 24 h in 75%. There is a wide spectrum of

clinical symptoms, signs and haemodynamic findings that define the presence and severity of cardiogenic shock and are directly related to short-term outcome.^{294–296} Patients typically present with hypotension, evidence of low cardiac output (resting tachycardia, altered mental status, oliguria, cool peripheries) and pulmonary congestion. The haemodynamic criteria for cardiogenic shock are a cardiac index of <2.2 L/min/m² and an increased wedge pressure of >18 mmHg. Additionally, diuresis is usually <20 mL/h. Shock is also considered present if i.v. inotropes and/or an IABP is needed to maintain a systolic blood pressure >90 mmHg. It is usually associated with extensive LV damage, but may occur in right ventricular infarction. Both short- and long-term mortality appear to be associated with initial LV systolic dysfunction and the severity of mitral regurgitation.²⁹⁵ The presence of right ventricular dysfunction on early echocardiography is also an important predictor of an adverse prognosis, especially in the case of combined left- and right ventricular dysfunction.²⁹⁶ Baseline and follow-up stroke volume index and follow-up stroke work index appear to be the most powerful haemodynamic predictors of 30-day mortality in patients in cardiogenic shock and are more useful than traditional haemodynamic variables.²⁹⁷ Therefore, cardiogenic shock characterization and management do not necessarily need invasive measurement of LV filling pressure and cardiac output through a pulmonary catheter but LV ejection fraction and associated mechanical complications should be evaluated urgently by two-dimensional Doppler echocardiography.^{295–298}

Management of cardiogenic shock complicating acute myocardial infarction includes haemodynamic stability, achieved with medical therapy or mechanical circulatory support, and emergent revascularization by means of PCI or CABG surgery. Drug treatment of cardiogenic shock complicating STEMI includes antithrombotics, fluids, vasopressors and inotropes. Antithrombotics should be given as routinely indicated in STEMI patients, although clopidogrel, prasugrel or ticagrelor should be deferred until angiography, because immediate CABG surgery may be necessary. Fluid administration is often used on a pathophysiological basis, although it has not been analysed in randomized trials. In other forms of shock, however, early fluid support improves survival. Similarly, vasopressors and inotropes are used due to their favourable haemodynamic effects, but none have produced consistent symptomatic improvement and many induced a reduction in survival that may be associated with the deleterious cellular effects of these drugs.²⁹⁹ A recent randomized trial compared norepinephrine with dopamine in 1679 patients with shock, including 280 with cardiogenic shock. Dopamine was associated with higher mortality in the cardiogenic shock subgroup and more adverse events—mainly arrhythmic events—for the overall cohort.³⁰⁰ Therefore, when blood pressure is low, norepinephrine should be the first choice. It should be used at the lowest possible dose and titrated until the systolic arterial pressure rises to at least 80 mmHg. Subsequently—and because its beta-2-adrenergic effect—dobutamine can be given simultaneously to improve contractility.

5.1.2 Management of heart failure following ST-segment elevation myocardial infarction (Table 23)

General measures include: taking a thorough history, including previous medical therapy, and a physical examination with assessment

of the patient's haemodynamic status. It is essential to detect and manage atrial and ventricular dysrhythmias, valvular dysfunction, post-infarction ischaemia and hypertension. Co-morbidities, such as infection, pulmonary disease, renal dysfunction, diabetes, anaemia or other laboratory abnormalities often contribute to the clinical picture. Patients with heart failure usually require oxygen therapy and monitoring of oxygen saturation by oximeter with a target $>95\%$ (90% in chronic obstructive pulmonary disease patients) and periodic blood gas assessment. Care should be taken, in patients with serious obstructive airways disease, to avoid hypercapnia. In hypotensive patients, volume loading should be attempted in patients without evidence of volume overload or congestion. Most patients require diuretic therapy, and an improvement in dyspnoea supports the diagnosis.

In **mild heart failure (Killip Class II)**, i.v. loop diuretics and/or i.v. nitrates are usually effective in achieving preload reduction and alleviating congestion and dyspnoea. Hypertension, if present, should be treated promptly to prevent further decompensation. ACE inhibitors/ARBs and aldosterone antagonists improve dyspnoea, attenuate the remodelling process, improve survival and may be initiated early in the absence of hypotension, hypovolaemia or renal dysfunction.

In **moderate heart failure with pulmonary oedema (Killip Class III)**, i.v. morphine reduces dyspnoea and relieves anxiety. Intravenous loop diuretics and/or i.v. vasodilators are indicated for dyspnoea in patients without hypotension (blood pressure >90 mmHg). In patients who tolerate the apparatus, non-invasive ventilation with continuous positive airway pressure therapy is effective in treating pulmonary oedema. Endotracheal intubation and ventilatory support may be required in patients unable to achieve adequate oxygenation, or with evidence of hypercapnia due to respiratory exhaustion. Systolic arterial blood pressure (SBP) should determine the choice of inotropic or vasopressor agent. In hypotensive patients with signs and symptoms of heart failure and poor organ perfusion (SBP <90 mmHg), dopamine (inotropic/vasopressor) should be considered. In patients with signs and symptoms of heart failure and adequate blood pressure (BP >90 mmHg) dobutamine (inotropic) or levosimendan (inotropic/vasodilator) may be preferable. Noradrenaline (vasopressor) may be preferable in patients with hypotension and signs of cardiogenic shock or septicaemia. The inotropic effect of levosimendan is independent of beta-adrenergic stimulation and represents an alternative for patients on chronic beta-blocker therapy. In patients with an SBP <100 mmHg, initiation of therapy without a bolus is recommended.²⁸⁴ Ultrafiltration to reduce fluid overload refractory to diuretics may be useful, especially in patients with hyponatraemia.

In **severe heart failure with cardiogenic shock (Killip Class IV)** it is essential to detect alternative causes of hypotension such as hypovolaemia, drug-induced hypotension, arrhythmias, tamponade, mechanical complications or right ventricular infarction. Intravenous inotropes/vasopressors are usually required to maintain an SBP >90 mmHg and adequate cardiac output and renal perfusion.

Invasive haemodynamic assessment with a pulmonary artery catheter may permit careful adjustment of filling pressures and assessment of cardiac output. In selected patients who are not

Table 23 Treatment of heart failure and left ventricular dysfunction

Recommendations	Class ^a	Level ^b	Ref ^c
Treatment of mild heart failure (Killip class II)			
Oxygen is indicated to maintain a saturation >95%.	I	C	-
Loop diuretics, e.g. furosemide: 20–40 mg i.v., is recommended and should be repeated at 1–4 h intervals if necessary.	I	C	-
i.v. nitrates or sodium nitroprusside should be considered in patients with elevated systolic blood pressure.	IIa	C	-
An ACE inhibitor is indicated in all patients with signs or symptoms of heart failure and/or evidence of LV dysfunction in the absence of hypotension, hypovolaemia, or renal failure.	I	A	309–312
An ARB (valsartan) is an alternative to ACE inhibitors particularly if ACE inhibitors are not tolerated.	I	B	281
An aldosterone antagonist (eplerenone) is recommended in all patients with signs or symptoms of heart failure and/or evidence of LV dysfunction provided no renal failure or hyperkalaemia.	I	B	282
Hydralazine and isosorbide dinitrate should be considered if the patient is intolerant to both ACE inhibitors and ARBs.	IIa	C	313
Treatment of moderate heart failure (Killip class III)			
Oxygen is indicated.	I	C	-
Ventilatory support should be instituted according to blood gasses.	I	C	-
Loop diuretics, e.g. furosemide: 20–40 mg i.v., are recommended and should be repeated at 1–4 h intervals if necessary.	I	C	-
Morphine is recommended. Respiration should be monitored. Nausea is common and an antiemetic may be required. Frequent low-dose therapy is advisable.	I	C	-
Nitrates are recommended if there is no hypotension.	I	C	-
Inotropic agents: • Dopamine	IIa	C	-
• Dobutamine (inotropic)	IIa	C	-
• Levosimendan (inotropic/vasodilator).	IIb	C	-
An aldosterone antagonist such as spironolactone or eplerenone must be used if LVEF ≤40%.	I	B	282, 314
Ultrafiltration should be considered.	IIa	B	315
Early revascularization must be considered if the patient has not been previously revascularized.	I	C	-
Treatment of cardiogenic shock (Killip class IV)			
Oxygen/mechanical respiratory support is indicated according to blood gasses.	I	C	-
Urgent echocardiography/Doppler must be performed to detect mechanical complications, assess systolic function and loading conditions.	I	C	-
High-risk patients must be transferred early to tertiary centres.	I	C	-
Emergency revascularization with either PCI or CABG in suitable patients must be considered.	I	B	100
Fibrinolysis should be considered if revascularization is unavailable.	IIa	C	-
Intra-aortic balloon pumping may be considered.	IIb	B	1, 98, 305
LV assist devices may be considered for circulatory support in patients in refractory shock.	IIb	C	-
Haemodynamic assessment with balloon floating catheter may be considered.	IIb	B	316
Inotropic/vasopressor agents should be considered: • Dopamine	IIa	C	-
• Dobutamine	IIa	C	-
• Norepinephrine (preferred over dopamine when blood pressure is low).	IIb	B	300, 317

ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker; CABG = coronary artery bypass graft; i.v. = intravenous; LV = left ventricular; LVEF = left ventricular ejection fraction; PCI = percutaneous coronary intervention.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

responding adequately to conventional measures and who exhibit evidence of ongoing ischaemia, persistent ST elevation or new LBBB, early revascularization with fibrinolysis, PCI or CABG should be considered. A strategy of early revascularization—preferably at a tertiary care centre—has demonstrated benefits in terms of improved functional status and long-term survival. The SHOCK trial demonstrated that STEMI patients with cardiogenic shock, undergoing emergency revascularization with PCI or CABG surgery, have substantially improved long-term survival, compared with patients having initial intensive medical therapy followed by no- or late in-hospital revascularization: a finding consistent with observations from registries.^{100,293} Despite longer time to treatment, transferred patients are a selected population with similar adjusted in-hospital mortality, and benefit from emergency revascularization in the same way as direct-admit patients.³⁰¹ Recognizing patients at highest risk for development of shock may facilitate the early transfer of high-risk patients before the onset of haemodynamic instability. Cardiogenic shock is the one circumstance in which it may be acceptable to embark on emergency revascularization of multivessel disease.^{100,302}

IABP counterpulsation is the most widely used mechanical support for the treatment of cardiogenic shock, based on the beneficial effect of aortic diastolic inflation and rapid systolic deflation, improving myocardial and peripheral perfusion and reducing afterload and myocardial oxygen consumption. Evidence regarding its efficacy, in the setting of acute myocardial infarction complicated by cardiogenic shock, has been reviewed recently for patients in the prefibrinolytic, the fibrinolytic and the primary PCI eras.⁹⁸ Owing to the lack of randomized trials, only registries were evaluated, and showed conflicting results for the three eras, with mortality risk differences of 29% and 18% in favour of IABP in the prethrombolytic and thrombolytic eras, and an increase of 6% in mortality with IABP in the primary PCI era. Concordantly, the *Treat angina with Aggrastat and determine Cost of Therapy with an Invasive or Conservative Strategy (TACTICS)* trial,³⁰³ which investigated the efficacy of counterpulsation in thrombolysed STEMI patients with hypotension, possible cardiogenic shock or heart failure, showed no benefit in terms of mortality from adding IABP to thrombolysis for the overall study cohort, but showed a favourable decrease in 6-month mortality for patients with more severe haemodynamic impairment allocated to IABP. Similarly, another small pilot trial in 40 patients with cardiogenic shock, who were undergoing primary PCI, showed beneficial effects on BNP for the IABP group, but no benefit in terms of the primary trial outcome [change in serial Acute Physiology And Chronic Health Evaluation II (APACHE-II) scoring].³⁰⁴ Another recent meta-analysis suggests a survival benefit from IABP in patients with cardiogenic shock.³⁰⁵ Overall, despite common use in clinical practice, there is somewhat conflicting evidence with respect to the benefit of IABP in cardiogenic shock, which is probably largely related to the difficulty of performing randomized trials in this setting.

Mechanical LV assist devices (LVADs) have been used in patients not responding to standard therapy, including inotropes, fluids and IABP, but evidence regarding their benefits is limited. A recent meta-analysis examined three randomized trials comparing a percutaneous LVAD vs. IABP in a total of 100 patients. Although the LVAD appeared safe and demonstrated better haemodynamics,

there was no improvement in 30-day mortality.³⁰⁶ Based on these results, percutaneous LVADs cannot be recommended as first-line treatment in cardiogenic shock but may be considered on an individual basis, taking into account the experience of the group, as well as patient age and co-morbidities. Similarly, in settings other than STEMI, such as transplant candidates not responding to standard therapy, surgical implantable LVADs³⁰⁷ or extracorporeal life support involving membrane oxygenators³⁰⁸ have been used as destination therapy or a bridge to transplant. Again, the evidence for benefit is still limited.

5.1.3 Arrhythmias and conduction disturbances in the acute phase

Arrhythmias and conduction disturbances are common during the early hours after myocardial infarction. According to recordings from cardiac monitors implanted within 11 ± 5 days of an acute myocardial infarction, the incidence is 28% for new-onset atrial fibrillation, 13% for non-sustained ventricular tachycardia, 10% for high-degree atrioventricular block (≤ 30 beats per minute lasting for ≥ 8 s), 7% for sinus bradycardia (≤ 30 beats per minute lasting for ≥ 8 s), 5% for sinus arrest (≥ 5 s), 3% for sustained ventricular tachycardia, and 3% for ventricular fibrillation.³¹⁸ The long-term prognostic significance of early (< 48 h) VF or sustained ventricular tachycardia (VT) in patients with acute myocardial infarction is still controversial. In patients with acute myocardial infarction, early VF/VT identified those at increased risk for 30-day mortality (22% vs. 5%) as compared to those without VF/VT.³¹⁹ ACE inhibitors/ARBs reduced the 30-day mortality in these patients. Other studies have confirmed that beta-blocker therapy, given in the first 24 h after AMI in patients with early sustained VF/VT, was associated with decreased early mortality without worsening heart failure.³²⁰ Prospective randomized studies are warranted to clarify the clinical implications of early-onset ventricular arrhythmias in this setting.

Arrhythmias after the early reperfusion period may be a manifestation of a serious underlying condition, such as continuing myocardial ischaemia, pump failure, altered autonomic tone, hypoxia, and electrolyte- (e.g. hypokalaemia) and acid-base disturbances, all of which require attention and corrective measures. High-degree atrioventricular block was a more powerful predictor of cardiac death than tachyarrhythmias in patients with left ventricular ejection fraction $< 40\%$ after myocardial infarction.³¹⁸

5.1.3.1 Supraventricular arrhythmias

Atrial fibrillation complicates some 6–28% of myocardial infarctions and is frequently associated with severe LV damage and heart failure.^{318,321} Episodes may last from minutes to hours and are often repetitive. In many cases, the arrhythmia is well tolerated and no specific treatment is required, other than anticoagulation (Table 24).²⁵⁰ In some instances, the fast ventricular rate contributes to heart failure, requiring prompt treatment. Adequate rate control is important in order to reduce myocardial oxygen demand, and can be accomplished by administration of beta-blockers or possibly calcium antagonists, either orally or intravenously (see recommendations below). In patients with extensive myocardial damage or severe LV dysfunction, rate control is more

Table 24 Management of atrial fibrillation

Recommendations	Class ^a	Level ^b	Ref ^c
Rhythm control should be considered in patients with atrial fibrillation secondary to a trigger or substrate that has been corrected (e.g. ischaemia).	IIa	C	-
Acute rate control of atrial fibrillation			
Intravenous beta-blockers or non-dihydropyridine CCB (e.g. diltiazem, verapamil) ^d are indicated if there are no clinical signs of acute heart failure.	I	A	323
Amiodarone or i.v. digitalis is indicated in case of rapid ventricular response in the presence of concomitant acute heart failure or hypotension.	I	B	324
Cardioversion			
Immediate electrical cardioversion is indicated when adequate rate control cannot be achieved promptly with pharmacological agents in patients with atrial fibrillation and on-going ischaemia, severe haemodynamic compromise or heart failure.	I	C	-
Intravenous amiodarone is indicated for conversion to sinus rhythm in stable patients with recent onset atrial fibrillation and structural heart disease.	I	A	250
Digoxin (LoE A), verapamil, sotalol, metoprolol (LoE B) and other beta-blocking agents (LoE C) are ineffective in converting recent onset atrial fibrillation to sinus rhythm and should not be used for rhythm control (although beta blockers or digoxin may be used for rate control).	III	A B C	250

Recommended doses of anti-arrhythmic agents are given in Guidelines for management of patients with atrial fibrillation.²⁵⁰

CCB = calcium-channel blocker; i.v. = intravenous; LoE A = level of evidence; LV = left ventricular.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

^dCalcium antagonists should be used cautiously or avoided in patients with heart failure because of their negative inotropic effects.

safely achieved with i.v. digoxin with or without concomitant administration of i.v. amiodarone, related to the negative inotropic effect of beta-blockers or calcium antagonists. Urgent electrical cardioversion may be considered in patients presenting with atrial fibrillation and intractable ischaemia or haemodynamic instability. Several,^{321,322} but not all,³¹⁸ studies have suggested that development of atrial fibrillation in the setting of acute myocardial infarction is an independent predictor of all-cause mortality, irrespective of the treatment given. Atrial fibrillation not only increased the risk for ischaemic stroke during the hospitalization but also during follow-up, even paroxysmal atrial fibrillation (AF) that has reversed to sinus rhythm at the time of discharge.³²¹ Patients with atrial fibrillation and risk factors for thromboembolism should therefore be adequately treated with oral anticoagulation. Because AF will generally require anticoagulation, when choosing a stent in these patients, the benefits of DES on restenosis should be weighed carefully against the substantial bleeding risks that are associated with the prolonged combination of triple antithrombotic therapy. Specific guidance with respect to selection of a rhythm- or rate control strategy, as well as on the type of stent and combination antiplatelet and anticoagulant therapy, has been given in the recent guidelines on the management of atrial fibrillation.²⁵⁰

Other supraventricular tachycardias are rare and are usually self-limited. They may respond to vagal manoeuvres. Intravenous adenosine may be considered in this setting, if atrial flutter is ruled out and the haemodynamic status is stable; the ECG should be monitored during administration. If not contraindicated, beta-blockers may be effective. Electrical cardioversion should be employed if the arrhythmia is poorly tolerated.

5.1.3.2 Ventricular arrhythmias (Table 25)

Ventricular premature beats are almost universal on the first day of the acute phase and complex arrhythmias (multiform complexes, short runs or the R-on-T phenomenon) are common. Their value as predictors of VF is questionable. No specific therapy is required.

Ventricular tachycardia should be differentiated from accelerated idioventricular rhythm—a consequence of reperfusion that is usually harmless—in which the ventricular rate is <120 beats per minute. Runs of non-sustained VT (lasting <30 s) are not reliable predictive markers for early VF and may be well tolerated, not necessarily requiring treatment. More prolonged episodes may cause hypotension and heart failure and may degenerate into VF. Since there is no evidence that suppression of asymptomatic non-sustained VT prolongs life, there is no indication to treat non-sustained VT, unless it is associated with haemodynamic instability. Sustained and/or haemodynamically unstable VT requires suppressive therapy, summarized below and outlined in the guidelines for ventricular arrhythmias.³²⁵ Electrical cardioversion (which requires sedation in conscious patients) is indicated if any VT persists and always indicated if the patient is haemodynamically unstable.³²⁶ It is the safest method for termination of sustained VT in acute STEMI. If the patient appears haemodynamically stable, i.v. amiodarone, sotalol or lidocaine (if the VT is thought to be related to ongoing myocardial ischaemia) may be initiated for its termination, but conversion rates are low. Amiodarone is the only anti-arrhythmic agent without severe pro-arrhythmic effects in patients with reduced LV function, and is therefore the drug of choice in patients with reduced left ventricular function. In a cohort of patients (in whom the majority had CAD) with stable

Table 25 Management of ventricular arrhythmias and conduction disturbances in the acute phase

Recommendations	Class ^a	Level ^b	Ref ^c
Direct current cardioversion is indicated for sustained VT and VF.	I	C	-
Sustained monomorphic VT that is recurrent or refractory to direct current cardioversion: should be considered to be treated with i.v. amiodarone. ^d	IIa	C	-
may be treated with i.v. lidocaine or sotalol. ^e	IIb	C	-
Transvenous catheter pace termination should be considered if VT is refractory to cardioversion or frequently recurrent despite antiarrhythmic medication.	IIa	C	-
Repetitive symptomatic salvoes of non-sustained monomorphic VT should be considered for either conservative management (watchful waiting) or treated with i.v. beta-blocker, ^e or sotalol, ^e or amiodarone. ^d	IIa	C	-
Polymorphic VT			
• must be treated by i.v. beta-blocker ^e	I	B	320, 336
• or i.v. amiodarone ^d	I	C	-
• urgent angiography must be performed when myocardial ischaemia cannot be excluded	I	C	-
• may be treated with i.v. lidocaine	IIb	C	330
• must prompt assessment and correction of electrolyte disturbances consider magnesium.	I	C	-
• should be treated with overdrive pacing using a temporary transvenous right ventricular lead or isoprotenerol infusion.	IIa	C	-
In cases of sinus bradycardia associated with hypotension, AV block II (Mobitz 2) or AV block III with bradycardia that causes hypotension or heart failure:			
• intravenous atropine is indicated	I	C	-
• temporary pacing is indicated in cases of failure to respond to atropine.	I	C	-
• urgent angiography with a view to revascularization is indicated if the patient has not received prior reperfusion therapy.	I	C	-
Management of ventricular arrhythmias and risk evaluation for sudden death on long term			
Specialized electrophysiological evaluation of ICD implantation for secondary prevention of sudden cardiac death is indicated in patients with significant LV dysfunction, who suffer from haemodynamically unstable sustained VT or who are resuscitated from VF occurring beyond the initial acute phase.	I	A	333
Secondary preventive ICD therapy is indicated to reduce mortality in patients with significant LV dysfunction, and haemodynamically unstable sustained VT or survived VF, not occurring within the initial acute phase.	I	A	333
Risk evaluation for sudden cardiac death should be performed to assess indication for primary preventive ICD therapy by assessing LVEF (from echocardiography) at least 40 days after the acute event in patients with LVEF ≤40%.	I	A	333

Recommended doses of anti-arrhythmic agents are given in the Guidelines for management of patients with ventricular arrhythmias and the prevention of sudden cardiac death.³²⁵ AV = atrioventricular; i.v. = intravenous; ICD = implantable cardioverter defibrillator; LVEF = left ventricular ejection fraction; VF = ventricular fibrillation; VT = ventricular tachycardia.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

^dQT-prolonging agents should not be used if baseline QT is prolonged.

^eIntravenous sotalol or other beta-blockers should not be given if ejection fraction is low.

sustained VT (but without acute myocardial infarction), both i.v. amiodarone and procainamide were relatively ineffective, with 25% and 30% conversion rates for amiodarone and procainamide, respectively. Clinically important hypotensive reactions led to cessation of amiodarone and procainamide infusion or immediate direct current cardioversion in 6% and 19% of patients, respectively.³²⁷

Ventricular fibrillation: immediate defibrillation should be performed according to recommendations outlined in the international guidelines for cardiopulmonary resuscitation and emergency cardiovascular care.^{326,328} Even though it has been demonstrated that lidocaine can reduce the incidence of VF in the acute phase of myocardial infarction, this drug increases the risk of asystole. A meta-analysis of 14 trials showed a trend towards higher mortality in lidocaine-treated patients than in controls, which is why routine prophylactic use of the drug is not justified.³²⁹ In a

retrospective analysis of STEMI patients who developed sustained VT/VF (n = 1126, 5.9%) in the GUSTO IIB and III trials, all-cause death was compared among those receiving amiodarone (n = 50, 4.4%), lidocaine (n = 664, 59.0%) or no anti-arrhythmic (n = 302, 26.8%). Among patients who survived 3 h, amiodarone was associated with increased mortality at 30 days and 6 months but lidocaine was not, an observation that reinforces the need for randomized trials in this population.³³⁰

Sustained VT or VF, developing beyond the initial acute phase (provided the ventricular tachyarrhythmia is not due to a reversible cause, such as electrolyte disturbance or transient ischaemia/reinfarction), are liable to recur and are associated with a high risk of death. Although myocardial ischaemia should always be ruled out in case of ventricular arrhythmias, it should be emphasized that revascularization is unlikely to prevent recurrent cardiac arrest in patients with markedly abnormal LV function or

sustained monomorphic VT, even if the original arrhythmia appeared to result from transient ischaemia.^{331,332} Among survivors of VF or sustained VT causing severe symptoms, ICD therapy is associated with significant mortality reduction, compared with antiarrhythmic drug therapy (mainly amiodarone).³³³ With the exception of beta-blockers, antiarrhythmic drugs have not shown to be effective as first-line management of patients with life-threatening ventricular arrhythmias and should not be used for the prevention of sudden death. An ICD is therefore recommended as secondary preventive therapy to reduce mortality in patients with significant LV dysfunction, who present with haemodynamically unstable sustained VT or who are resuscitated from VF that does not occur within the first 24–48 h.²⁹¹ Such patients should be subject to specialized electrophysiological evaluation before discharge for placement of an implantable cardioverter-defibrillator (ICD) for secondary prevention of sudden cardiac death.^{325,333}

Primary preventive ICD therapy has been shown to reduce all-cause mortality in patients with reduced left ventricular ejection fraction (EF <40%) as a result of an infarction that occurred at least 40 days earlier.^{333,334} In general, ICD implantation should be deferred until at least 40 days after the acute event. Evaluation of the need for a primary preventive ICD and implantation may, in some cases, be postponed until 3 months after revascularization procedures, to allow adequate time for recovery of LV function. Patients may be evaluated for CRT and ICD treatment whenever stunning of viable myocardium can be excluded, indications for which are outlined in guidelines.³³⁵

5.1.3.3 Sinus bradycardia and heart block

Sinus bradycardia is common in the first hours of STEMI, especially in inferior infarction. In some cases, opioids are responsible. It often requires no treatment. If accompanied by severe hypotension, sinus bradycardia should be treated by i.v. atropine, starting with a dosage of 0.25–0.5 mg, repeated up to a total of 1.5–2.0 mg. Occasionally it may be associated with hypotension at a later stage. If it then fails to respond to atropine, temporary pacing is advised.

First-degree heart block needs no treatment.

Second-degree Type I (Mobitz I or Wenckebach) atrioventricular (AV) block is usually associated with inferior infarction and seldom causes adverse haemodynamic effects. Should it do so, however, atropine should be given first. If this fails, pacing should be instituted. Agents that slow AV conduction (such as beta-blockers, digitalis, verapamil or amiodarone) should be withheld.

Second-degree Type II (Mobitz II) AV block and complete AV block may be indications for the insertion of a pacing electrode, certainly if bradycardia causes hypotension or heart failure. If the haemodynamic disturbance is severe, consideration should be given to AV sequential pacing. Revascularization should always be considered urgently in patients who have not yet received reperfusion therapy.

AV block associated with inferior wall infarction is usually supra-Hisian, i.e. located above the His bundle, and associated with transient bradycardia with a narrow QRS escape rhythm—above 40 beats per minute—and has low mortality. They usually resolve

spontaneously and rarely require intervention. AV block associated with anterior wall myocardial infarction is usually infra-Hisian, i.e. located below the AV node, associated with an unstable, wide QRS and low escape rhythm, and has a high mortality rate (up to 80%) due to the extensive myocardial necrosis. The development of a new bundle branch block or hemiblock usually indicates extensive anterior infarction. There is then a high likelihood of developing both a complete AV block and pump failure.

Asystole may follow AV block, bifascicular or trifascicular block, or electrical countershock. If a pacing electrode is in place, pacing should be attempted. Otherwise, chest compression and ventilation should be initiated, and transthoracic pacing started.

A transvenous pacing electrode should be inserted in the presence of advanced AV block with a low escape rhythm, as described above, and considered if bifascicular or trifascicular block develops. The subclavian route should be avoided following fibrinolysis or in the presence of anticoagulation. Alternative sites should be chosen in this situation. Indications for pacing are outlined in detail in the ESC Guidelines for cardiac pacing and cardiac resynchronization therapy.²⁹¹ Permanent pacing is indicated in patients with persistent third-degree AV block, in patients with persistent second-degree AV block associated with bundle branch block, and in transient Mobitz II or complete heart block associated with new onset bundle branch block.²⁹¹

5.2 Cardiac complications

Certain demographic characteristics and procedural aspects define patients at higher risk for complications, who may require extended monitoring. Advanced age, Killip II–IV symptoms, 3-vessel disease, anterior wall infarction, prolonged ischaemic time or reduced TIMI flow are frequently cited.²¹³ Several mechanical complications may occur acutely in the first days following STEMI, although their incidence has fallen with the increase in the provision of prompt and effective reperfusion therapy. They are all life-threatening and need prompt detection and management. Repeated clinical examination (at least twice daily) may pick up a new cardiac murmur, suggestive of mitral regurgitation or ventricular septal defect, which then needs to be confirmed or ruled out by immediate echocardiography. CABG should generally be performed, if appropriate, at the time of surgery in patients requiring emergency surgery for serious mechanical complications.

5.2.1 Mitral valve regurgitation

Mitral valve regurgitation may occur during the subacute phase due to LV dilatation, papillary muscle dysfunction or rupture of the tip of the papillary muscle or *chordae tendinae*. It usually presents as sudden haemodynamic deterioration with acute dyspnoea and pulmonary congestion and a new systolic murmur, which may be underappreciated in this context. The diagnosis is suspected by clinical examination and should be immediately confirmed by emergency echocardiography. Pulmonary oedema and cardiogenic shock may occur rapidly. Treatment is based on afterload reduction to reduce regurgitant volume and pulmonary congestion, if blood pressure allows. Intravenous diuretic and vasodilator/inotropic support, as well as IABP, may stabilize patients in preparation for angiography and surgery. Emergency surgical repair or valve replacement is required.³³⁷

5.2.2 Cardiac rupture

Rupture of the LV free wall may occur during the subacute phase following transmural infarction, and may present as sudden pain and cardiovascular collapse with electromechanical dissociation. The development of haemopericardium and tamponade is usually rapidly fatal. The diagnosis is confirmed by echocardiography. Subacute free wall rupture, due to sealing of the site by thrombus formation, if recognized, may permit time for pericardiocentesis and immediate surgery.

5.2.3 Ventricular septal rupture

Ventricular septal rupture usually presents as rapid-onset clinical deterioration with acute heart failure and a loud systolic murmur occurring during the subacute phase. The diagnosis is confirmed by echocardiography, which will differentiate this from acute mitral regurgitation and locate and quantify the rupture.³³⁸ The consequent left-to-right shunt may result in signs and symptoms of acute, new-onset right heart failure. An IABP may stabilize patients in preparation for angiography and surgery. Intravenous diuretics and vasodilators should be used with caution in hypotensive patients. Surgical repair is required urgently, but there is no agreement on the optimal timing for surgery.³³⁹ Early surgery is associated with a high mortality rate and a high risk of recurrent ventricular rupture, while delayed surgery allows easier septal repair in scarred tissue but carries the risk of rupture extension, tamponade and death while waiting for surgery. Mortality remains high in all patients and is higher in patients with inferobasal defects as opposed to anteroapical location.

5.2.4 Right ventricular infarction

Right ventricular infarction may occur in isolation or, far more frequently, in connection with inferior wall STEMI. It frequently presents with the triad of hypotension, clear lung fields and raised jugular venous pressure. Elevation of the ST-segment ≥ 1 mV in V_1 and V_4R is suggestive of right ventricular infarction and should routinely be sought in patients with inferior STEMI and hypotension. Doppler echocardiography typically demonstrates right ventricular dilatation, low pulmonary arterial pressure, dilated hepatic veins and varying degrees of inferior wall injury. Despite the jugular distension, fluid loading that maintains right ventricular filling pressure is a key therapy in avoiding or treating hypotension. In addition, diuretics and vasodilators should be avoided, as they may aggravate hypotension. Maintenance of sinus rhythm and atrioventricular synchrony is important and atrial fibrillation or atrioventricular block should be treated early.

5.2.5 Pericarditis

The incidence of pericarditis after STEMI has decreased with the advent of modern, effective reperfusion therapy.³⁴⁰ Pericarditis manifests as recurrent chest pain, usually characteristically sharp and, in contradistinction to recurrent ischaemia, related to posture and respiration. It may be associated with ST-segment re-elevation. However, ST-segment re-elevation is usually mild and progressive, which helps distinguish it from the abrupt ST-segment re-elevation seen in cases of coronary re-occlusion resulting from, for example, stent thrombosis. A continuous

pericardial rub may confirm the diagnosis but is often absent, especially with substantial pericardial effusion. Echocardiography will detect and quantify the size of effusion, if present, and rule out haemorrhagic effusion with tamponade. The pain usually responds to high-dose aspirin, paracetamol or colchicine. Steroids and long-term non-steroidal anti-inflammatory drugs should be avoided due to the risk of scar thinning with aneurysm development or rupture. Pericardiocentesis is rarely required, but should be performed in the presence of haemodynamic compromise with signs of tamponade. When pericardial effusion is present, anticoagulant therapy, if present (e.g. for prophylaxis of venous thrombo-embolism), should be interrupted unless absolutely indicated.

5.2.6 Left ventricular aneurysm

Patients with a large transmural infarction—especially of the anterolateral wall—may undergo infarct expansion with subsequent development of LV aneurysm. This remodelling process—of LV dilatation and aneurysm formation with volume overload—results in combined systolic and diastolic dysfunction and, frequently, mitral regurgitation. Doppler echocardiography will assess LV volume, ejection fraction, the extent and degree of wall-motion abnormalities, and detect mural thrombus necessitating anticoagulation. ACE inhibitors/ARBs and aldosterone antagonists have been shown to attenuate the remodelling process in transmural infarction and improve survival, and should be administered early following haemodynamic stabilization. Patients will frequently develop the symptoms and signs of chronic heart failure and should be treated according to guidelines for heart failure.²⁸⁴

5.2.7 Left ventricular thrombus

The frequency of mural LV thrombus has decreased, largely because of the progress made in reperfusion therapy, the widespread use of multiple antithrombotic agents in STEMI, and the limitation of myocardial infarct size produced by effective, early myocardial reperfusion.^{341,342} Although some studies suggest that up to a quarter of anterior MIs have detectable LV thrombi,³⁴³ LV thrombi are associated with poor prognosis because of their association with extensive infarcts, particularly anterior infarcts with apical involvement, and a risk of systemic embolism. Relatively old trials had shown that anticoagulation in patients with large anterior wall motion abnormalities reduced the occurrence of mural thrombi.^{344–346} Anticoagulation should therefore be considered in patients with such large anterior wall motion abnormalities, if they are at low risk of bleeding, to prevent the development of thrombi. Consensus is that mural thrombi, once diagnosed, require oral anticoagulant therapy with vitamin K antagonists for up to 6 months. However, this has not been revisited in the era of stenting and DAPT. Combining oral anticoagulation and DAPT into a triple therapy increases bleeding risks. The optimal duration of such triple antithrombotic therapy is unknown and should take into account the relative risks of bleeding and stent thrombosis. Repeated imaging of the left ventricle after 3 months of therapy may allow discontinuation of anticoagulation earlier than 6 months, if evidence of thrombus is no longer present, particularly if there is recovery of apical wall motion.

6. Gaps in the evidence and areas for future research

There remain many important areas of uncertainty in the management of STEMI that offer opportunities for future research:

- developing strategies to minimize early cardiac arrest may be associated with large gains in survival.
- improving patients' and public awareness of symptoms potentially related to STEMI and the need to call the EMS directly, preferably via a unique centralized telephone number, is an important tool for shortening patient delay.
- investigating whether pre-hospital thrombolysis still has a role in the management of patients seen early after symptom onset—and who otherwise have access to primary PCI—is an important issue currently being tested in the ongoing Strategic Reperfusion Early After Myocardial Infarction (STREAM) randomized clinical trial.
- while selected centres and geographic areas have made tremendous progress in ensuring high-quality rapid care for patients with STEMI, there remains a definite need for streamlining of pre-hospital and hospital management, in order to shorten time to diagnosis and treatment in a homogeneous fashion worldwide. Designing optimized clinical pathways for ensuring high-quality and homogeneous early STEMI diagnosis and management at a national level is important.
- reducing or minimizing myocardial injury and LV dysfunction following STEMI also remains a crucial goal. Several strategies are being tested, using a variety of pharmacological and non-pharmacological approaches.

- defining the optimal management strategy for non-culprit vessels in patients successfully treated with prior primary PCI of the culprit artery.
- there is a need to define the optimal long-term antithrombotic regimen for patients receiving stents and who have an indication for oral anticoagulants (e.g. due to high-risk atrial fibrillation, prosthetic heart valve or LV thrombus).
- new antithrombotic agents in addition to aspirin and/or ADP receptor inhibitors have reduced ischaemic events, but with increased bleeding risk. However, the optimal combination of anticoagulant and antiplatelet therapies remains to be proven.
- given the increased bleeding risk related to potent dual and triple antithrombotic therapy, it would be desirable to test simpler combinations and clarify the optimal duration of treatment for prevention of repeated ischaemic/thrombotic events.
- in patients with known diabetes or acute hyperglycaemia, the optimal glucose management strategy in the acute and post-discharge phases remains unclear, both in terms of optimal medication choice and goals of therapy.
- development of percutaneous techniques for managing ventricular septal defects may permit avoidance or delay of surgical repair, while providing potentially life-saving therapy to these very high-risk patients.
- the effectiveness and safety of cell therapy to replace myocardium, or minimize the consequences of myocardial injury, needs to be established.
- the optimal therapeutic strategy to minimize risk of sudden death in patients who develop VT or VF during or after STEMI is not entirely clear.
- more evidence is needed on effective strategies to achieve and maintain long-term effective risk factor control.



The CME text *ESC Guidelines for the management of acute myocardial infarction in patients presenting with ST-segment elevation* is accredited by the European Board for Accreditation in Cardiology (EBAC). EBAC works according to the quality standards of the European Accreditation Council for Continuing Medical Education (EACCME), which is an institution of the European Union of Medical Specialists (UEMS). In compliance with EBAC/EACCME guidelines, all authors participating in this programme have disclosed any potential conflicts of interest that might cause a bias in the article. The Organizing Committee is responsible for ensuring that all potential conflicts of interest relevant to the programme are declared to the participants prior to the CME activities.

CME questions for this article are available at: *European Heart Journal* <http://www.oxford-learning.com/eurheartj> and European Society of Cardiology <http://www.escardio.org/guidelines>.

References

1. Van de Werf F, Bax J, Betriu A, Blömlund-Lundqvist C, Crea F, Falk V, Filippatos G, Fox K, Huber K, Kastrati A, Rosengren A, Steg PG, Tubaro M, Verheugt F, Weidinger F, Weis M, Vahanian A, Camm J, De Caterina R, Dean V, Dickstein K, Funck-Brentano C, Hellems I, Kristensen SD, McGregor K, Sechtem U, Silber S, Tendera M, Widimsky P, Zamorano JL, Aguirre FV, Al-Attar N, Alegria E, Andreotti F, Benzer W, Breithardt O, Danchin N, Di Mario C, Dudek D, Gulba D, Halvorsen S, Kaufmann P, Kornowski R, Lip GY, Rutten F. Management of acute myocardial infarction in patients presenting with persistent ST-segment elevation: the Task Force on the Management of ST-Segment Elevation Acute Myocardial Infarction of the European Society of Cardiology. *Eur Heart J* 2008; **29**:2909–2945.
2. Thygesen K, Alpert JS, White HD. Universal definition of myocardial infarction. *Eur Heart J* 2007; **28**:2525–2538.
3. Hamm CW, Bassand JP, Agewall S, Bax J, Boersma E, Bueno H, Caso P, Dudek D, Gielen S, Huber K, Ohman M, Petrie MC, Sonntag F, Uva MS, Storey RF, Wijns W, Zahger D, Bax JJ, Auricchio A, Baumgartner H, Ceconi C, Dean V, Deaton C, Fagard R, Funck-Brentano C, Hasdai D, Hoes A, Knuuti J, Kolh P, McDonagh T, Moulin C, Poldermans D, Popescu BA, Reiner Z, Sechtem U, Sirnes PA, Torbicki A, Vahanian A, Windecker S, Windecker S, Achenbach S, Badimon L, Bertrand M, Botker HE, Collet JP, Crea F, Danchin N, Falk E, Goudevos J, Gulba D, Hambrecht R, Herrmann J, Kastrati A, Kjeldsen K, Kristensen SD, Lancellotti P, Mehilli J, Merkely B, Montalescot G, Neumann FJ, Neyens L, Perk J, Roffi M, Romeo F, Ruda M, Swahn E, Valgimigli M, Vrints CJ, Widimsky P. ESC Guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation: The Task Force for the management of acute coronary syndromes (ACS) in patients presenting without persistent ST-segment elevation of the European Society of Cardiology (ESC). *Eur Heart J* 2011; **32**:2999–3054.
4. Wijns W, Kolh P, Danchin N, Di Mario C, Falk V, Folliquet T, Garg S, Huber K, James S, Knuuti J, Lopez-Sendon J, Marco J, Menicanti L, Ostojic M, Piepoli MF, Pirllet C, Pomar JL, Reifart N, Ribichini FL, Schalij MJ, Sergeant P, Serruys PW, Silber S, Sousa Uva M, Taggart D, Vahanian A, Auricchio A, Bax J, Ceconi C, Dean V, Filippatos G, Funck-Brentano C, Hobbs R, Kearney P, McDonagh T, Popescu BA, Reiner Z, Sechtem U, Sirnes PA, Tendera M, Vardas PE, Widimsky P, Alfieri O, Dunning J, Elia S, Kappetein P, Lockowandt U, Sarris G, Vouhe P, von Segesser L, Agewall S, Aladashvili A, Alexopoulos D, Antunes MJ, Atalar E, Brutel de la Riviere A, Doganov A, Eha J, Fajadet J, Ferreira R, Garot J, Halcox J, Hasin Y, Janssens S, Kervinen K, Laufer G, Legrand V, Nashef SA, Neumann FJ, Niemela K, Nihoyannopoulos P, Noc M, Piek JJ, Pirk J, Rozenman Y, Sabate M, Starc R, Thielmann M, Wheatley DJ,

- Windecker S, Zembala M. Guidelines on myocardial revascularization: The Task Force on Myocardial Revascularization of the European Society of Cardiology (ESC) and the European Association for Cardio-Thoracic Surgery (EACTS). *Eur Heart J* 2010;**31**:2501–2555.
5. WHO Fact sheet N°310, updated June 2011, <http://www.who.int/mediacentre/factsheets/fs310/en/index.html>.
 6. Widimsky P, Wijns W, Fajadet J, de Belder M, Knot J, Aaberge L, Andrikopoulos G, Baz JA, Betriu A, Claeys M, Danchin N, Djambazov S, Erne P, Hartikainen J, Huber K, Kala P, Klineva M, Kristensen SD, Ludman P, Ferre JM, Merkely B, Milicic D, Morais J, Noc M, Opolski G, Ostojic M, Radovanovic D, De Servi S, Stenestrand U, Studencan M, Tubaro M, Vasiljevic Z, Weidinger F, Witkowski A, Zeymer U. Reperfusion therapy for ST elevation acute myocardial infarction in Europe: description of the current situation in 30 countries. *Eur Heart J* 2010;**31**:943–957.
 7. Widimsky P, Zelisko M, Jansky P, Tousek F, Holm F, Aschermann M. The incidence, treatment strategies, outcomes of acute coronary syndromes in the “reperfusion network” of different hospital types in the Czech Republic: results of the Czech evaluation of acute coronary syndromes in hospitalized patients (CZECH) registry. *Int J Cardiol* 2007;**119**:212–219.
 8. McManus DD, Gore J, Yarzebski J, Spencer F, Lessard D, Goldberg RJ. Recent trends in the incidence, treatment, and outcomes of patients with STEMI and NSTEMI. *Am J Med* 2011;**124**:40–47.
 9. Roger VL, Go AS, Lloyd-Jones DM, Benjamin EJ, Berry JD, Borden WB, Bravata DM, Dai S, Ford ES, Fox CS, Fullerton HJ, Gillespie C, Hailpern SM, Heit JA, Howard VJ, Kissela BM, Kittner SJ, Lackland DT, Lichtman JH, Lisabeth LD, Makuc DM, Marcus GM, Marelli A, Matchar DB, Moy CS, Mozaffarian D, Mussolino ME, Nichol G, Paynter NP, Soliman EZ, Sorlie PD, Sotodehnia N, Turan TN, Virani SS, Wong ND, Woo D, Turner MB. Executive summary: heart disease and stroke statistics—2012 update: a report from the American Heart Association. *Circulation* 2012;**125**:188–197.
 10. Mandelzweig L, Battler A, Boyko V, Bueno H, Danchin N, Filippatos G, Gitt A, Hasdai D, Hasin Y, Marrugat J, Van de Werf F, Wallentin L, Behar S. The second Euro Heart Survey on acute coronary syndromes: characteristics, treatment, and outcome of patients with ACS in Europe and the Mediterranean Basin in 2004. *Eur Heart J* 2006;**27**:2285–2293.
 11. Jernberg T, Johanson P, Held C, Svenblad B, Lindback J, Wallentin L. Association between adoption of evidence-based treatment and survival for patients with ST-elevation myocardial infarction. *J Am Med Assoc* 2011;**305**:1677–1684.
 12. Fox KA, Steg PG, Eagle KA, Goodman SG, Anderson FA Jr., Granger CB, Flather MD, Budaj A, Quill A, Gore JM. Decline in rates of death and heart failure in acute coronary syndromes, 1999–2006. *J Am Med Assoc* 2007;**297**:1892–1900.
 13. Fox KA, Dabbous OH, Goldberg RJ, Pieper KS, Eagle KA, Van de Werf F, Avezum A, Goodman SG, Flather MD, Anderson FA Jr., Granger CB. Prediction of risk of death and myocardial infarction in the six months after presentation with acute coronary syndrome: prospective multinational observational study (GRACE). *Br Med J* 2006;**333**:1091.
 14. Fox KA, Carruthers KF, Dunbar DR, Graham C, Manning JR, De Raedt H, Buyschaert I, Lambrechts D, Van de Werf F. Underestimated and under-recognized: the late consequences of acute coronary syndrome (GRACE UK–Belgian Study). *Eur Heart J* 2010;**31**:2755–2764.
 15. Tubaro M, Danchin N, Goldstein P, Filippatos G, Hasin Y, Heras M, Jansky P, Norekval TM, Swahn E, Thgesen K, Vrints C, Zahger D, Arntz HR, Bellou A, De La Coussaye JE, De Luca L, Huber K, Lambert Y, Lettino M, Lindahl B, McLean S, Nibbe L, Peacock WF, Price S, Quinn T, Spaulding C, Tatu-Chitoiu G, Van De Werf F. Pre-hospital treatment of STEMI patients. A scientific statement of the Working Group Acute Cardiac Care of the European Society of Cardiology. *Acute Card Care* 2011;**13**:56–67.
 16. Brieger D, Eagle KA, Goodman SG, Steg PG, Budaj A, White K, Montalescot G. Acute coronary syndromes without chest pain, an underdiagnosed and undertreated high-risk group: insights from the Global Registry of Acute Coronary Events. *Chest* 2004;**126**:461–469.
 17. Diercks DB, Peacock WF, Hiestand BC, Chen AY, Pollack CV Jr., Kirk JD, Smith SC Jr., Glibler WB, Ohman EM, Blomkalns AL, Newby LK, Hochman JS, Peterson ED, Roe MT. Frequency and consequences of recording an electrocardiogram >10 minutes after arrival in an emergency room in non-ST-segment elevation acute coronary syndromes (from the CRUSADE Initiative). *Am J Cardiol* 2006;**97**:437–442.
 18. Lopez-Sendon J, Coma-Canella I, Alcasena S, Seoane J, Gamallo C. Electrocardiographic findings in acute right ventricular infarction: sensitivity and specificity of electrocardiographic alterations in right precordial leads V4R, V3R, V1, V2, and V3. *J Am Coll Cardiol* 1985;**6**:1273–1279.
 19. Rokos IC, French WJ, Koenig WJ, Stratton SJ, Nighswonger B, Strunk B, Jewell J, Mahmud E, Dunford JV, Hokanson J, Smith SW, Baran KW, Swor R, Berman A, Wilson BH, Aluko AO, Gross BW, Rostkyus PS, Salvucci A, Dev V, McNally B, Manoukian SV, King SB 3rd. Integration of pre-hospital electrocardiograms and ST-elevation myocardial infarction receiving center (SRC) networks: impact on door-to-balloon times across 10 independent regions. *JACC Cardiovasc Interv* 2009;**2**:339–346.
 20. O’Doherty M, Tayler DI, Quinn E, Vincent R, Chamberlain DA. Five hundred patients with myocardial infarction monitored within one hour of symptoms. *Br Med J (Clin Res Ed)* 1983;**286**:1405–1408.
 21. Mehta RH, Starr AZ, Lopes RD, Hochman JS, Widimsky P, Pieper KS, Armstrong PW, Granger CB. Incidence of and outcomes associated with ventricular tachycardia or fibrillation in patients undergoing primary percutaneous coronary intervention. *J Am Med Assoc* 2009;**301**:1779–1789.
 22. Sgarbossa EB, Pinski SL, Barbagelata A, Underwood DA, Gates KB, Topol EJ, Califf RM, Wagner GS. Electrocardiographic diagnosis of evolving acute myocardial infarction in the presence of left bundle-branch block. GUSTO-1 (Global Utilization of Streptokinase and Tissue Plasminogen Activator for Occluded Coronary Arteries) investigators. *N Engl J Med* 1996;**334**:481–487.
 23. Shlipak MG, Lyons WL, Go AS, Chou TM, Evans GT, Browner WS. Should the electrocardiogram be used to guide therapy for patients with left bundle-branch block and suspected myocardial infarction? *JAMA* 1999;**281**:714–719.
 24. Lopes RD, Siha H, Fu Y, Mehta RH, Patel MR, Armstrong PW, Granger CB. Diagnosing acute myocardial infarction in patients with left bundle branch block. *Am J Cardiol* 2011;**108**:782–788.
 25. Widimsky P, Rohac F, Stasek J, Kala P, Rokyta R, Kuzmanov B, Jakl M, Poloczek M, Kanovsky J, Bernat I, Hlinomaz O, Belohlavek J, Kral A, Mrazek V, Grigorov V, Djambazov S, Petr R, Knot J, Bilkova D, Fischerova M, Vondrak K, Maly M, Lorencova A. Primary angioplasty in acute myocardial infarction with right bundle branch block: should new onset right bundle branch block be added to future guidelines as an indication for reperfusion therapy? *Eur Heart J* 2012;**33**:86–95.
 26. Krishnaswamy A, Lincoff AM, Menon V. Magnitude and consequences of missing the acute infarct-related circumflex artery. *Am Heart J* 2009;**158**:706–712.
 27. From AM, Best PJ, Lennon RJ, Rihal CS, Prasad A. Acute myocardial infarction due to left circumflex artery occlusion and significance of ST-segment elevation. *Am J Cardiol* 2010;**106**:1081–1085.
 28. Yan AT, Yan RT, Kannel BM, Anderson FA Jr., Budaj A, Lopez-Sendon J, Brieger D, Allegrone J, Steg G, Goodman SG. Relationship of ST elevation in lead aVR with angiographic findings and outcome in non-ST elevation acute coronary syndromes. *Am Heart J* 2007;**154**:71–78.
 29. Akashi YJ, Goldstein DS, Barbaro G, Ueyama T. Takotsubo cardiomyopathy: a new form of acute, reversible heart failure. *Circulation* 2008;**118**:2754–2762.
 30. Cabello JB, Burls A, Emparanza JJ, Bayliss S, Quinn T. Oxygen therapy for acute myocardial infarction. *Cochrane Database Syst Rev* 2010;**6**:CD007160.
 31. Spaulding CM, Joly LM, Rosenberg A, Monchi M, Weber SN, Dhainaut JF, Carli P. Immediate coronary angiography in survivors of out-of-hospital cardiac arrest. *N Engl J Med* 1997;**336**:1629–1633.
 32. Kern KB, Rahman O. Emergent percutaneous coronary intervention for resuscitated victims of out-of-hospital cardiac arrest. *Catheter Cardiovasc Interv* 2010;**75**:616–624.
 33. Garot P, Lefevre T, Eltchaninoff H, Morice MC, Tamion F, Abry B, Lesault PF, Le Tarnec JY, Pouges C, Margenet A, Monchi M, Laurent I, Dumas P, Garot J, Louvard Y. Six-month outcome of emergency percutaneous coronary intervention in resuscitated patients after cardiac arrest complicating ST-elevation myocardial infarction. *Circulation* 2007;**115**:1354–1362.
 34. Bernard SA, Gray TW, Buist MD, Jones BM, Silvester W, Gutteridge G, Smith K. Treatment of comatose survivors of out-of-hospital cardiac arrest with induced hypothermia. *N Engl J Med* 2002;**346**:557–563.
 35. Mild therapeutic hypothermia to improve the neurologic outcome after cardiac arrest. *N Engl J Med* 2002;**346**:549–556.
 36. Belliard G, Catez E, Charron C, Caille V, Aegerter P, Dubourg O, Jardin F, Vieillard-Baron A. Efficacy of therapeutic hypothermia after out-of-hospital cardiac arrest due to ventricular fibrillation. *Resuscitation* 2007;**75**:252–259.
 37. Nolan JP, Soar J, Zideman DA, Biarent D, Bossaert LL, Deakin C, Koster RW, Wyllie J, Bottiger B. European Resuscitation Council Guidelines for Resuscitation 2010 Section 1. Executive summary. *Resuscitation* 2010;**81**:1219–1276.
 38. Luepker RV, Raczynski JM, Osganian S, Goldberg RJ, Finnegan JR Jr., Hedges JR, Goff DC Jr., Eisenberg MS, Zapka JG, Feldman HA, Labarthe DR, McGovern PG, Cornell CE, Proschan MA, Simons-Morton DG. Effect of a community intervention on patient delay and emergency medical service use in acute coronary heart disease: The Rapid Early Action for Coronary Treatment (REACT) Trial. *JAMA* 2000;**284**:60–67.
 39. Terkelsen CJ, Sorensen JT, Maeng M, Jensen LO, Tilsted HH, Trautner S, Vach W, Johnsen SP, Thuesen L, Lassen JF. System delay and mortality among patients with STEMI treated with primary percutaneous coronary intervention. *JAMA* 2010;**304**:763–771.
 40. Steg PG, Bonnefoy E, Chabaud S, Lapostolle F, Dubien PY, Cristofini P, Leizorovicz A, Touboul P. Impact of time to treatment on mortality after pre-

- hospital fibrinolysis or primary angioplasty: data from the CAPTIM randomized clinical trial. *Circulation* 2003;**108**:2851–2856.
41. Pinto DS, Kirtane AJ, Nallamothu BK, Murphy SA, Cohen DJ, Laham RJ, Cutlip DE, Bates ER, Frederick PD, Miller DP, Carrozza JP Jr., Antman EM, Cannon CP, Gibson CM. Hospital delays in reperfusion for ST-elevation myocardial infarction: implications when selecting a reperfusion strategy. *Circulation* 2006;**114**:2019–2025.
 42. Huber K, De Caterina R, Kristensen SD, Verheugt FW, Montalescot G, Maestri LB, Van de Werf F. Pre-hospital reperfusion therapy: a strategy to improve therapeutic outcome in patients with ST-elevation myocardial infarction. *Eur Heart J* 2005;**26**:2063–2074.
 43. Welsh RC, Chang W, Goldstein P, Adgey J, Granger CB, Verheugt FW, Wallentin L, Van de Werf F, Armstrong PW. Time to treatment and the impact of a physician on pre-hospital management of acute ST elevation myocardial infarction: insights from the ASSENT-3 PLUS trial. *Heart* 2005;**91**:1400–1406.
 44. Danchin N, Coste P, Ferrieres J, Steg PG, Cottin Y, Blanchard D, Belle L, Ritz B, Kirkorian G, Angioi M, Sans P, Charbonnier B, Eltchaninoff H, Gueret P, Khalife K, Asseman P, Puel J, Goldstein P, Cambou JP, Simon T. Comparison of thrombolysis followed by broad use of percutaneous coronary intervention with primary percutaneous coronary intervention for ST-segment-elevation acute myocardial infarction: data from the French registry on acute ST-elevation myocardial infarction (FAST-MI). *Circulation* 2008;**118**:268–276.
 45. Bonnefoy E, Steg PG, Boutitie F, Dubien PY, Lapostolle F, Roncalli J, Dissait F, Vanzetto G, Leizorowicz A, Kirkorian G, Mercier C, McFadden EP, Touboul P. Comparison of primary angioplasty and pre-hospital fibrinolysis in acute myocardial infarction (CAPTIM) trial: a 5-year follow-up. *Eur Heart J* 2009;**30**:1598–1606.
 46. McMullan JT, Hinckley W, Bentley J, Davis T, Fermann GJ, Gunderman M, Hart KW, Knight WA, Lindsell CJ, Miller C, Shackelford A, Brian Gibler W. Ground emergency medical services requests for helicopter transfer of ST-segment elevation myocardial infarction patients decrease medical contact to balloon times in rural and suburban settings. *Acad Emerg Med* 2012;**19**:153–160.
 47. Kalla K, Christ G, Karnik R, Malzer R, Norman G, Pracher H, Schreiber W, Unger G, Glogar HD, Kaff A, Lagner AN, Maurer G, Mlczech J, Slany J, Weber HS, Huber K. Implementation of guidelines improves the standard of care: the Viennese registry on reperfusion strategies in ST-elevation myocardial infarction (Vienna STEMI registry). *Circulation* 2006;**113**:2398–2405.
 48. Henry TD, Sharkey SW, Burke MN, Chavez IJ, Graham KJ, Henry CR, Lips DL, Madison JD, Menssen KM, Mooney MR, Newell MC, Pedersen WR, Poulouse AK, Traverse JH, Unger BT, Wang YL, Larson DM. A regional system to provide timely access to percutaneous coronary intervention for ST-elevation myocardial infarction. *Circulation* 2007;**116**:721–728.
 49. Le May MR, So DY, Dionne R, Glover CA, Froeschl MP, Wells GA, Davies RF, Sherrard HL, Maloney J, Marquis JF, O'Brien ER, Trickett J, Poirier P, Ryan SC, Ha A, Joseph PG, Labinaz M. A citywide protocol for primary PCI in ST-segment elevation myocardial infarction. *N Engl J Med* 2008;**358**:231–240.
 50. Bradley EH, Herrin J, Wang Y, Barton BA, Webster TR, Mattera JA, Roumanis SA, Curtis JP, Nallamothu BK, Magid DJ, McNamara RL, Parkosewich J, Loeb JM, Krumholz HM. Strategies for reducing the door-to-balloon time in acute myocardial infarction. *N Engl J Med* 2006;**355**:2308–2320.
 51. Widimsky P, Fajadet J, Danchin N, Wijns W. "Stent 4 Life" targeting PCI at all who will benefit the most. A joint project between EAPCI, Euro-PCR, EUCOMED and the ESC Working Group on Acute Cardiac Care. *EuroIntervention* 2009;**4**:555, 557.
 52. Knot J, Widimsky P, Wijns W, Stenestrand U, Kristensen SD, Van THA, Weidinger F, Janzon M, Norgaard BL, Soerensen JT, van de Wetering H, Thygesen K, Bergsten PA, Digerfeldt C, Potgieter A, Tomer N, Fajadet J. How to set up an effective national primary angioplasty network: lessons learned from five European countries. *EuroIntervention* 2009;**5**:299, 301–309.
 53. Ting HH, Krumholz HM, Bradley EH, Cone DC, Curtis JP, Drew BJ, Field JM, French WJ, Gibler WB, Goff DC, Jacobs AK, Nallamothu BK, O'Connor RE, Schuur JD. Implementation and integration of pre-hospital ECGs into systems of care for acute coronary syndrome: a scientific statement from the American Heart Association Interdisciplinary Council on Quality of Care and Outcomes Research, Emergency Cardiovascular Care Committee, Council on Cardiovascular Nursing, and Council on Clinical Cardiology. *Circulation* 2008;**118**:1066–1079.
 54. Amit G, Cafri C, Gilutz H, Ilia R, Zaher D. Benefit of direct ambulance to coronary care unit admission of acute myocardial infarction patients undergoing primary percutaneous intervention. *Int J Cardiol* 2007;**119**:355–358.
 55. Nallamothu BK, Krumholz HM, Ko DT, LaBresh KA, Rathore S, Roe MT, Schwamm L. Development of systems of care for ST-elevation myocardial infarction patients: gaps, barriers, and implications. *Circulation* 2007;**116**:e68–e72.
 56. Rathore SS, Curtis JP, Chen J, Wang Y, Nallamothu BK, Epstein AJ, Krumholz HM. Association of door-to-balloon time and mortality in patients admitted to hospital with ST elevation myocardial infarction: national cohort study. *Br Med J* 2009;**338**:b1807.
 57. Nielsen PH, Terkelsen CJ, Nielsen TT, Thuesen L, Kruse LR, Thayssen P, Kelbaek H, Abildgaard U, Villadsen AB, Andersen HR, Maeng M. System delay and timing of intervention in acute myocardial infarction (from the Danish Acute Myocardial Infarction-2 [DANAMI-2] trial). *Am J Cardiol* 2011;**108**:776–781.
 58. Steg PG, Cambou JP, Goldstein P, Durand E, Sauval P, Kadri Z, Blanchard D, Lablanche JM, Gueret P, Cottin Y, Juliard JM, Hanania G, Vaur L, Danchin N. Bypassing the emergency room reduces delays and mortality in ST elevation myocardial infarction: theUSIC 2000 registry. *Heart* 2006;**92**:1378–1383.
 59. Hackett D, Davies G, Chierchia S, Maseri A. Intermittent coronary occlusion in acute myocardial infarction. Value of combined thrombolytic and vasodilator therapy. *N Engl J Med* 1987;**317**:1055–1059.
 60. Schomig A, Mehilli J, Antoniucci D, Ndrepepa G, Markwardt C, Di Pede F, Nekolla SG, Schlotterbeck K, Schuhen H, Pache J, Seyfarth M, Martinoff S, Benzer W, Schmitt C, Dirschinger J, Schwaiger M, Kastrati A. Mechanical reperfusion in patients with acute myocardial infarction presenting more than 12 hours from symptom onset: a randomized controlled trial. *JAMA* 2005;**293**:2865–2872.
 61. Ndrepepa G, Kastrati A, Mehilli J, Antoniucci D, Schomig A. Mechanical reperfusion and long-term mortality in patients with acute myocardial infarction presenting 12 to 48 hours from onset of symptoms. *JAMA* 2009;**301**:487–488.
 62. Hochman JS, Lamas GA, Buller CE, Dzavik V, Reynolds HR, Abramsky SJ, Forman S, Ruzyllo W, Maggioni AP, White H, Sadowski Z, Carvalho AC, Rankin JM, Renkin JP, Steg PG, Mascette AM, Sopko G, Pfisterer ME, Leor J, Fridrich V, Mark DB, Knatterud GL. Coronary intervention for persistent occlusion after myocardial infarction. *N Engl J Med* 2006;**355**:2395–2407.
 63. Ioannidis JP, Katritsis DG. Percutaneous coronary intervention for late reperfusion after myocardial infarction in stable patients. *Am Heart J* 2007;**154**:1065–1071.
 64. Menon V, Pearte CA, Buller CE, Steg PG, Forman SA, White HD, Marino PN, Katritsis DG, Caramori P, Lasevitch R, Loboz-Grudzien K, Zurawski A, Lamas GA, Hochman JS. Lack of benefit from percutaneous intervention of persistently occluded infarct arteries after the acute phase of myocardial infarction is time independent: insights from Occluded Artery Trial. *Eur Heart J* 2009;**30**:183–191.
 65. Boersma E, Maas AC, Deckers JW, Simoons ML. Early thrombolytic treatment in acute myocardial infarction: reappraisal of the golden hour. *Lancet* 1996;**348**:771–775.
 66. Boersma E. Does time matter? A pooled analysis of randomized clinical trials comparing primary percutaneous coronary intervention and in-hospital fibrinolysis in acute myocardial infarction patients. *Eur Heart J* 2006;**27**:779–788.
 67. Gierlotka M, Gasior M, Wilczek K, Hawranek M, Szkodziński J, Paczek P, Lekston A, Kalarus Z, Zembala M, Polonski L. Reperfusion by primary percutaneous coronary intervention in patients with ST-segment elevation myocardial infarction within 12 to 24 hours of the onset of symptoms (from a prospective national observational study [PL-ACS]). *Am J Cardiol* 2011;**107**:501–508.
 68. Zijlstra F, Hoorntje JC, de Boer MJ, Reijnders K, Miedema K, Otervanger JP, van't Hof AW, Suryapranata H. Long-term benefit of primary angioplasty as compared with thrombolytic therapy for acute myocardial infarction. *N Engl J Med* 1999;**341**:1413–1419.
 69. Keeley EC, Boura JA, Grines CL. Primary angioplasty vs. intravenous thrombolytic therapy for acute myocardial infarction: a quantitative review of 23 randomised trials. *Lancet* 2003;**361**:13–20.
 70. Widimsky P, Budesinsky T, Vorac D, Groch L, Zelizko M, Aschermann M, Branny M, Stasek J, Formanek P. Long distance transport for primary angioplasty vs. immediate thrombolysis in acute myocardial infarction. Final results of the randomized national multicentre trial—PRAGUE-2. *Eur Heart J* 2003;**24**:94–104.
 71. Andersen HR, Nielsen TT, Rasmussen K, Thuesen L, Kelbaek H, Thayssen P, Abildgaard U, Pedersen F, Madsen JK, Grande P, Villadsen AB, Kruse LR, Hagfelt L, Lomholt P, Husted SE, Vigholt E, Kjaergard HK, Mortensen LS. A comparison of coronary angioplasty with fibrinolytic therapy in acute myocardial infarction. *N Engl J Med* 2003;**349**:733–742.
 72. Morrison LJ, Verbeek PR, McDonald AC, Sawadsky BV, Cook DJ. Mortality and pre-hospital thrombolysis for acute myocardial infarction: A meta-analysis. *JAMA* 2000;**283**:2686–2692.
 73. Bonnefoy E, Lapostolle F, Leizorowicz A, Steg G, McFadden EP, Dubien PY, Cattani S, Boulenger E, Machecourt J, Lacroix JM, Cassagnes J, Dissait F,

- Touboul P. Primary angioplasty vs. pre-hospital fibrinolysis in acute myocardial infarction: a randomised study. *Lancet* 2002;**360**:825–829.
74. Pinto DS, Frederick PD, Chakraborti AK, Kirtane AJ, Ullman E, Dejam A, Miller DP, Henry TD, Gibson CM. Benefit of transferring ST-segment-elevation myocardial infarction patients for percutaneous coronary intervention compared with administration of onsite fibrinolytic declines as delays increase. *Circulation* 2011;**124**:2512–2521.
 75. Widimsky P, Holmes DR Jr. How to treat patients with ST-elevation acute myocardial infarction and multi-vessel disease? *Eur Heart J* 2011;**32**:396–403.
 76. Cavender MA, Milford-Beland S, Roe MT, Peterson ED, Weintraub WS, Rao SV. Prevalence, predictors, and in-hospital outcomes of non-infarct artery intervention during primary percutaneous coronary intervention for ST-segment elevation myocardial infarction (from the National Cardiovascular Data Registry). *Am J Cardiol* 2009;**104**:507–513.
 77. Kornowski R, Mehran R, Dangas G, Nikolsky E, Assali A, Claessen BE, Gersh BJ, Wong SC, Witzentichler B, Guagliumi G, Dudek D, Fahy M, Lansky AJ, Stone GW. Prognostic impact of staged vs. “one-time” multivessel percutaneous intervention in acute myocardial infarction: analysis from the HORIZONS-AMI (Harmonizing Outcomes with Revascularization and Stents in Acute Myocardial Infarction) trial. *J Am Coll Cardiol* 2011;**58**:704–711.
 78. Jolly SS, Yusuf S, Cairns J, Niemela K, Xavier D, Widimsky P, Budaj A, Niemela M, Valentin V, Lewis BS, Avezum A, Steg PG, Rao SV, Gao P, Afzal R, Joyner CD, Chrolavicius S, Mehta SR. Radial vs. femoral access for coronary angiography and intervention in patients with acute coronary syndromes (RIVAL): a randomised, parallel group, multicentre trial. *Lancet* 2011;**377**:1409–1420.
 79. Ragnoli E, Biondi-Zoccai G, Sciahbasi A, Politi L, Rigattieri S, Pendenza G, Di Russo C, Summari F, Patrizi R, Moretti C, Agostoni P, Loschiavo P, Lioy E, Sheiban I, Sangiorgi GM. Radial vs. femoral randomized investigation in ST elevation acute coronary syndromes: The RIFLE STEACS study. *J Am Coll Cardiol* 2012; in press.
 80. Kastrati A, Dibra A, Spaulding C, Laarman GJ, Menichelli M, Valgimigli M, Di Lorenzo E, Kaiser C, Tieraal I, Mehilli J, Seyfarth M, Varenne O, Dirksen MT, Percoco G, Varricchio A, Pittl U, Syvanne M, Suttrop MJ, Violini R, Schomig A. Meta-analysis of randomized trials on drug-eluting stents vs. bare-metal stents in patients with acute myocardial infarction. *Eur Heart J* 2007;**28**:2706–2713.
 81. Piccolo R, Cassese S, Galasso G, De Rosa R, D’Anna C, Piscione F. Long-term safety and efficacy of drug-eluting stents in patients with acute myocardial infarction: a meta-analysis of randomized trials. *Atherosclerosis* 2011;**217**:149–157.
 82. Stone GW, Witzentichler B, Guagliumi G, Peruga JZ, Brodie BR, Dudek D, Kornowski R, Hartmann F, Gersh BJ, Pocock SJ, Dangas G, Wong SC, Fahy M, Parise H, Mehran R. Heparin plus a glycoprotein IIb/IIIa inhibitor vs. bivalirudin monotherapy and paclitaxel-eluting stents vs. bare-metal stents in acute myocardial infarction (HORIZONS-AMI): final 3-year results from a multicentre, randomised controlled trial. *Lancet* 2011;**377**:2193–2204.
 83. Svilaas T, Vlaar PJ, van der Horst IC, Diercks GF, de Smet BJ, van den Heuvel AF, Anthonio RL, Jessurun GA, Tan ES, Suurmeijer AJ, Zijlstra F. Thrombus aspiration during primary percutaneous coronary intervention. *N Engl J Med* 2008;**358**:557–567.
 84. Vlaar PJ, Svilaas T, van der Horst IC, Diercks GF, Fokkema ML, de Smet BJ, van den Heuvel AF, Anthonio RL, Jessurun GA, Tan ES, Suurmeijer AJ, Zijlstra F. Cardiac death and reinfarction after 1 year in the Thrombus Aspiration during Percutaneous coronary intervention in Acute myocardial infarction Study (TAPAS): a 1-year follow-up study. *Lancet* 2008;**371**:1915–1920.
 85. Burzotta F, De Vita M, Gu YL, Isshiki T, Lefevre T, Kaltoft A, Dudek D, Sardella G, Orrego PS, Antoniucci D, De Luca L, Biondi-Zoccai GG, Crea F, Zijlstra F. Clinical impact of thrombectomy in acute ST-elevation myocardial infarction: an individual patient-data pooled analysis of 11 trials. *Eur Heart J* 2009;**30**:2193–2203.
 86. Bavry AA, Kumbhani DJ, Bhatt DL. Role of adjunctive thrombectomy and embolic protection devices in acute myocardial infarction: a comprehensive meta-analysis of randomized trials. *Eur Heart J* 2008;**29**:2989–3001.
 87. Stone GW, Maehara A, Witzentichler B, Godlewski J, Parise H, Dambrink JH, Ochala A, Carlton TW, Cristea E, Wolff SD, Brenner SJ, Chowdhary S, El-Omar M, Neunteufl T, Metzger DC, Karwoski T, Dixon JM, Mehran R, Gibson CM. Intracoronary abiximab and aspiration thrombectomy in patients with large anterior myocardial infarction: the INFUSE-AMI randomized trial. *JAMA* 2012;**307**:1817–1826.
 88. Frobert O, Lagerqvist B, Gudnason T, Thuesen L, Svensson R, Olivecrona GK, James SK. Thrombus Aspiration in ST-Elevation myocardial infarction in Scandinavia (TASTE trial). A multicenter, prospective, randomized, controlled clinical registry trial based on the Swedish angiography and angioplasty registry (SCAAR) platform. Study design and rationale. *Am Heart J* 2010;**160**:1042–1048.
 89. A trial of routine aspiration thrombectomy with percutaneous coronary intervention (PCI) vs. PCI alone in patients with ST-segment elevation myocardial infarction (STEMI) undergoing primary PCI (TOTAL). www.clinicaltrials.gov/ct2/show/NCT01149044.
 90. Botker HE, Kharbanda R, Schmidt MR, Bottcher M, Kaltoft AK, Terkelsen CJ, Munk K, Andersen NH, Hansen TM, Trautner S, Lassen JF, Christiansen EH, Krusell LR, Kristensen SD, Thuesen L, Nielsen SS, Rehling M, Sorensen HT, Redington AN, Nielsen TT. Remote ischaemic conditioning before hospital admission, as a complement to angioplasty, and effect on myocardial salvage in patients with acute myocardial infarction: a randomised trial. *Lancet* 2010;**375**:727–734.
 91. Staat P, Rioufol G, Piot C, Cottin Y, Cung TT, L’Huillier I, Aupetit JF, Bonnefoy E, Finet G, Andre-Fouet X, Ovize M. Postconditioning the human heart. *Circulation* 2005;**112**:2143–2148.
 92. Thibault H, Piot C, Staat P, Bontemps L, Sportouch C, Rioufol G, Cung TT, Bonnefoy E, Angoulvant D, Aupetit JF, Finet G, Andre-Fouet X, Macia JC, Racza F, Rossi R, Itti R, Kirkorian G, Derumeaux G, Ovize M. Long-term benefit of postconditioning. *Circulation* 2008;**117**:1037–1044.
 93. Piot C, Croisille P, Staat P, Thibault H, Rioufol G, Mewton N, Elbelghiti R, Cung TT, Bonnefoy E, Angoulvant D, Macia C, Racza F, Sportouch C, Gahide G, Finet G, Andre-Fouet X, Revel D, Kirkorian G, Monassier JP, Derumeaux G, Ovize M. Effect of cyclosporine on reperfusion injury in acute myocardial infarction. *N Engl J Med* 2008;**359**:473–481.
 94. Sorensson P, Saleh N, Bouvier F, Bohm F, Settergren M, Caidahl K, Tornvall P, Arheden H, Ryden L, Pernow I. Effect of postconditioning on infarct size in patients with ST elevation myocardial infarction. *Heart* 2010;**96**(21):1710–1715.
 95. Freixa X, Bellera N, Ortiz-Perez JT, Jimenez M, Pare C, Bosch X, De Caralt TM, Betriu A, Masotti M. Ischaemic postconditioning revisited: lack of effects on infarct size following primary percutaneous coronary intervention. *Eur Heart J* 2012;**33**:103–112.
 96. Lonborg J, Kelbaek H, Vejstrup N, Jorgensen E, Helqvist S, Saunamaki K, Clemmensen P, Holmvang L, Treiman M, Jensen JS, Engstrom T. Cardioprotective effects of ischemic postconditioning in patients treated with primary percutaneous coronary intervention, evaluated by magnetic resonance. *Circ Cardiovasc Interv* 2010;**3**:34–41.
 97. Patel MR, Smalling RW, Thiele H, Barnhart HX, Zhou Y, Chandra P, Chew D, Cohen M, French J, Perera D, Ohman EM. Intra-aortic balloon counterpulsation and infarct size in patients with acute anterior myocardial infarction without shock: the CRISP AMI randomized trial. *JAMA* 2011;**306**:1329–1337.
 98. Sjaauw KD, Engstrom AE, Vis MM, van der Schaaf RJ, Baan J Jr., Koch KT, de Winter RJ, Piek JJ, Tijssen JG, Henriques JP. A systematic review and meta-analysis of intra-aortic balloon pump therapy in ST-elevation myocardial infarction: should we change the guidelines? *Eur Heart J* 2009;**30**:459–468.
 99. Cucherat M, Bonnefoy E, Treneau G. Primary angioplasty vs. intravenous thrombolysis for acute myocardial infarction. *Cochrane Database Syst Rev* 2003;(3):CD001560.
 100. Hochman JS, Sleeper LA, Webb JG, Sanborn TA, White HD, Talley JD, Buller CE, Jacobs AK, Slater JN, Col J, McKinlay SM, Lejemtel TH. Early revascularization in acute myocardial infarction complicated by cardiogenic shock. SHOCK Investigators. Should We Emergently Revascularize Occluded Coronaries for Cardiogenic Shock. *N Engl J Med* 1999;**341**:625–634.
 101. Nordmann AJ, Hengstler P, Harr T, Young J, Bucher HC. Clinical outcomes of primary stenting versus balloon angioplasty in patients with myocardial infarction: a meta-analysis of randomized controlled trials. *Am J Med* 2004;**116**:253–262.
 102. Stone GW, Brodie BR, Griffin JJ, Costantini C, Morice MC, St Goar FG, Overlie PA, Popma JJ, McDonnell J, Jones D, O’Neill WW, Grines CL. Clinical and angiographic follow-up after primary stenting in acute myocardial infarction: the Primary Angioplasty in Myocardial Infarction (PAMI) stent pilot trial. *Circulation* 1999;**99**:1548–1554.
 103. Hannan EL, Samadashvili Z, Walford G, Holmes DR Jr., Jacobs AK, Stamato NJ, Venditti FJ, Sharma S, King SB 3rd. Culprit vessel percutaneous coronary intervention versus multivessel and staged percutaneous coronary intervention for ST-segment elevation myocardial infarction patients with multivessel disease. *JACC Cardiovasc Interv* 2010;**3**:22–31.
 104. Toma M, Buller CE, Westerhout CM, Fu Y, O’Neill WW, Holmes DR Jr., Hamm CW, Granger CB, Armstrong PW. Non-culprit coronary artery percutaneous coronary intervention during acute ST-segment elevation myocardial infarction: insights from the APEX-AMI trial. *Eur Heart J* 2010;**31**:1701–1707.
 105. Vlaar PJ, Mahmoud KD, Holmes DR Jr., van Valkenhoef G, Hillege HL, van der Horst IC, Zijlstra F, de Smet BJ. Culprit vessel only versus multivessel and staged percutaneous coronary intervention for multivessel disease in patients presenting with ST-segment elevation myocardial infarction: a pairwise and network meta-analysis. *J Am Coll Cardiol* 2011;**58**:692–703.
 106. Wijnbergen I, Helmes H, Tijssen J, Brueren G, Peels K, van Dantzig JM, Van’t Veer M, Koolen JJ, Pijls NH, Michels R. Comparison of drug-eluting and bare-metal stents for primary percutaneous coronary intervention with or without

- abciximab in ST-segment elevation myocardial infarction: DEBATER: the Eindhoven reperfusion study. *JACC Cardiovasc Interv* 2012;**5**:313–322.
107. De Luca G, Dirksen MT, Spaulding C, Kelbaek H, Schalij M, Thuesen L, van der Hoeven B, Vink MA, Kaiser C, Musto C, Chechi T, Spaziani G, Diaz de la Llera LS, Pasceri V, Di Lorenzo E, Violini R, Cortese G, Suryapranata H, Stone GW. Drug-eluting vs bare-metal stents in primary angioplasty: a pooled patient-level meta-analysis of randomized trials. *Arch Intern Med* 2012;**172**: 611–621.
 108. Kelbaek H, Terkelsen CJ, Helqvist S, Lassen JF, Clemmensen P, Klovgaard L, Kaltoft A, Engstrom T, Botker HE, Saunamaki K, Krusell LR, Jorgensen E, Hansen HH, Christiansen EH, Ravkilde J, Kober L, Kofoed KF, Thuesen L. Randomized comparison of distal protection versus conventional treatment in primary percutaneous coronary intervention: the drug elution and distal protection in ST-elevation myocardial infarction (DEDICATION) trial. *J Am Coll Cardiol* 2008;**51**:899–905.
 109. Wiviott SD, Braunwald E, McCabe CH, Montalescot G, Ruzyllo W, Gottlieb S, Neumann FJ, Ardissino D, De Servi S, Murphy SA, Riesmeyer J, Weerakkody G, Gibson CM, Antman EM. Prasugrel versus clopidogrel in patients with acute coronary syndromes. *N Engl J Med* 2007;**357**:2001–2015.
 110. Wallentin L, Becker RC, Budaj A, Cannon CP, Emanuelsson H, Held C, Horrow J, Husted S, James S, Katus H, Mahaffey KW, Scirica BM, Skene A, Steg PG, Storey RF, Harrington RA, Freij A, Thorsen M. Ticagrelor versus clopidogrel in patients with acute coronary syndromes. *N Engl J Med* 2009;**361**: 1045–1057.
 111. Montalescot G, Wiviott SD, Braunwald E, Murphy SA, Gibson CM, McCabe CH, Antman EM. Prasugrel compared with clopidogrel in patients undergoing percutaneous coronary intervention for ST-elevation myocardial infarction (TRITON-TIMI 38): double-blind, randomised controlled trial. *Lancet* 2009;**373**:723–731.
 112. http://www.ema.europa.eu/docs/en_GB/document_library/EPAR_-_Product_Information/human/000984/WC500021971.pdf.
 113. Steg PG, James S, Harrington RA, Ardissino D, Becker RC, Cannon CP, Emanuelsson H, Finkelstein A, Husted S, Katus H, Kilhamn J, Olofsson S, Storey RF, Weaver WD, Wallentin L. Ticagrelor versus clopidogrel in patients with ST-elevation acute coronary syndromes intended for reperfusion with primary percutaneous coronary intervention: A Platelet Inhibition and Patient Outcomes (PLATO) trial subgroup analysis. *Circulation* 2010;**122**:2131–2141.
 114. Storey RF, Becker RC, Harrington RA, Husted S, James SK, Cools F, Steg PG, Khurmi NS, Emanuelsson H, Cooper A, Cairns R, Cannon CP, Wallentin L. Characterization of dyspnoea in PLATO study patients treated with ticagrelor or clopidogrel and its association with clinical outcomes. *Eur Heart J* 2011;**32**: 2945–2953.
 115. Mehta SR, Tanguay JF, Eikelboom JW, Jolly SS, Joyner CD, Granger CB, Faxon DP, Rupprecht HJ, Budaj A, Avezum A, Widimsky P, Steg PG, Bassand JP, Montalescot G, Macaya C, Di Pasquale G, Niemela K, Ajani AE, White HD, Chrolavicius S, Gao P, Fox KA, Yusuf S. Double-dose versus standard-dose clopidogrel and high-dose versus low-dose aspirin in individuals undergoing percutaneous coronary intervention for acute coronary syndromes (CURRENT-OASIS 7): a randomised factorial trial. *Lancet* 2010;**376**:1233–1243.
 116. Koul S, Smith JG, Schersten F, James S, Lagerqvist B, Erlinge D. Effect of upstream clopidogrel treatment in patients with ST-segment elevation myocardial infarction undergoing primary percutaneous coronary intervention. *Eur Heart J* 2011;**32**:2989–2997.
 117. Dorler J, Edlinger M, Alber HF, Altenberger J, Benzer W, Grimm G, Huber K, Pachinger O, Schuchlenz H, Siostroznek P, Zenker G, Weidinger F. Clopidogrel pre-treatment is associated with reduced in-hospital mortality in primary percutaneous coronary intervention for acute ST-elevation myocardial infarction. *Eur Heart J* 2011;**32**:2954–2961.
 118. Yusuf S, Mehta SR, Chrolavicius S, Afzal R, Pogue J, Granger CB, Budaj A, Peters RJ, Bassand JP, Wallentin L, Joyner C, Fox KA. Effects of fondaparinux on mortality and reinfarction in patients with acute ST-segment elevation myocardial infarction: the OASIS-6 randomized trial. *JAMA* 2006;**295**:1519–1530.
 119. Silvain J, Beygui F, Barthelémy O, Pollack C Jr., Cohen M, Zeymer U, Huber K, Goldstein P, Cayla G, Collet JP, Vicaut E, Montalescot G. Efficacy and safety of enoxaparin versus unfractionated heparin during percutaneous coronary intervention: systematic review and meta-analysis. *Br Med J* 2012;**344**:e553.
 120. Montalescot G, Ellis SG, de Belder MA, Janssens L, Katz O, Pluta W, Ecollan P, Tendera M, van Boven AJ, Widimsky P, Andersen HR, Betriu A, Armstrong P, Brodie BR, Herrmann HC, Neumann FJ, Efron MB, Lu J, Barnathan ES, Topol EJ. Enoxaparin in primary and facilitated percutaneous coronary intervention A formal prospective nonrandomized substudy of the FINESSE trial (Facilitated Intervention with Enhanced Reperfusion Speed to Stop Events). *JACC Cardiovasc Interv* 2010;**3**:203–212.
 121. Navarese EP, De Luca G, Castriota F, Kozinski M, Gurbel PA, Gibson CM, Andreotti F, Buffon A, Siller-Matula JM, Sukiennik A, De Servi S, Kubica J. Low-molecular-weight heparins vs. unfractionated heparin in the setting of percutaneous coronary intervention for ST-elevation myocardial infarction: a meta-analysis. *J Thromb Haemost* 2011;**9**:1902–1915.
 122. Montalescot G, Zeymer U, Silvain J, Boulanger B, Cohen M, Goldstein P, Ecollan P, Combes X, Huber K, Pollack C Jr., Benezet JF, Stibbe O, Filippi E, Teiger E, Cayla G, Elhadad S, Adnet F, Chouhied T, Gallula S, Greffet A, Aout M, Collet JP, Vicaut E. Intravenous enoxaparin or unfractionated heparin in primary percutaneous coronary intervention for ST-elevation myocardial infarction: the international randomised open-label ATOLL trial. *Lancet* 2011;**378**:693–703.
 123. Mehran R, Lansky AJ, Witzencbichler B, Guagliumi G, Peruga JZ, Brodie BR, Dudek D, Kornowski R, Hartmann F, Gersh BJ, Pocock SJ, Wong SC, Nikolsky E, Gambone L, Vandertie L, Parise H, Dangas GD, Stone GW. Bivalirudin in patients undergoing primary angioplasty for acute myocardial infarction (HORIZONS-AMI): 1-year results of a randomised controlled trial. *Lancet* 2009;**374**:1149–1159.
 124. Stone GW, Witzencbichler B, Guagliumi G, Peruga JZ, Brodie BR, Dudek D, Kornowski R, Hartmann F, Gersh BJ, Pocock SJ, Dangas G, Wong SC, Kirtane AJ, Parise H, Mehran R. Bivalirudin during primary PCI in acute myocardial infarction. *N Engl J Med* 2008;**358**:2218–2230.
 125. Dangas GD, Caixeta A, Mehran R, Parise H, Lansky AJ, Cristea E, Brodie BR, Witzencbichler B, Guagliumi G, Peruga JZ, Dudek D, Moeckel M, Stone GW. Frequency and predictors of stent thrombosis after percutaneous coronary intervention in acute myocardial infarction. *Circulation* 2011;**123**:1745–1756.
 126. De Luca G, Navarese E, Marino P. Risk profile and benefits from Gp IIb/IIIa inhibitors among patients with ST-segment elevation myocardial infarction treated with primary angioplasty: a meta-regression analysis of randomized trials. *Eur Heart J* 2009;**30**:2705–2713.
 127. Ellis SG, Tendera M, de Belder MA, van Boven AJ, Widimsky P, Janssens L, Andersen HR, Betriu A, Savonitto S, Adamus J, Peruga JZ, Kosmider M, Katz O, Neunteufl T, Jorgova J, Dorobantu M, Grinfeld L, Armstrong P, Brodie BR, Herrmann HC, Montalescot G, Neumann FJ, Efron MB, Barnathan ES, Topol EJ. Facilitated PCI in patients with ST-elevation myocardial infarction. *N Engl J Med* 2008;**358**:2205–2217.
 128. Herrmann HC, Lu J, Brodie BR, Armstrong PW, Montalescot G, Betriu A, Neuman FJ, Efron MB, Barnathan ES, Topol EJ, Ellis SG. Benefit of facilitated percutaneous coronary intervention in high-risk ST-segment elevation myocardial infarction patients presenting to nonpercutaneous coronary intervention hospitals. *JACC Cardiovasc Interv* 2009;**2**:917–924.
 129. en Berg JM, van 't Hof AW, Dill T, Heestermans T, van Werkum JW, Mosterd A, van Houwelingen G, Koopmans PC, Stella PR, Boersma E, Hamm C. Effect of early, pre-hospital initiation of high bolus dose tirofiban in patients with ST-segment elevation myocardial infarction on short- and long-term clinical outcome. *J Am Coll Cardiol* 2010;**55**:2446–2455.
 130. Mehili J, Kastrati A, Schulz S, Frungel S, Nekolla SG, Moshage W, Dotzer F, Huber K, Pache J, Dirschinger J, Seyfarth M, Martinoff S, Schwaiger M, Schomig A. Abciximab in patients with acute ST-segment-elevation myocardial infarction undergoing primary percutaneous coronary intervention after clopidogrel loading: a randomized double-blind trial. *Circulation* 2009;**119**:1933–1940.
 131. Friedland S, Eisenberg MJ, Shimony A. Meta-analysis of randomized controlled trials of intracoronary versus intravenous administration of glycoprotein IIb/IIIa inhibitors during percutaneous coronary intervention for acute coronary syndrome. *Am J Cardiol* 2011;**108**:1244–1251.
 132. Thiele H, Wöhrle J, Hambrecht R, Rittger H, Birkemeyer R, Lauer B, Neuhaus P, Brosteanu O, Sick P, Wiemer M, Kerber S, Kleinertz K, Eitel I, Desch S, Schuler G. Intracoronary versus intravenous bolus abciximab during primary percutaneous coronary intervention in patients with acute ST-elevation myocardial infarction: a randomised trial. *Lancet* 2012;**379**:923–931.
 133. Randomised trial of intravenous streptokinase, oral aspirin, both, or neither among 17,187 cases of suspected acute myocardial infarction: ISIS-2. (Second International Study of Infarct Survival) Collaborative Group. *Lancet* 1988;**2**:349–360.
 134. Patrono C, Andreotti F, Arnesen H, Badimon L, Baigent C, Collet JP, De Caterina R, Gulba D, Huber K, Husted S, Kristensen SD, Morais J, Neumann FJ, Rasmussen LH, Siegbahn A, Steg PG, Storey RF, Van de Werf F, Verheugt F. Antiplatelet agents for the treatment and prevention of atherothrombosis. *Eur Heart J* 2011;**32**:2922–2932.
 135. Leon MB, Baim DS, Popma JJ, Gordon PC, Cutlip DE, Ho KK, Giambartolomei A, Diver DJ, Lasorda DM, Williams DO, Pocock SJ, Kuntz RE. A clinical trial comparing three antithrombotic-drug regimens after coronary-artery stenting. Stent Anticoagulation Restenosis Study Investigators. *N Engl J Med* 1998;**339**: 1665–1671.
 136. Schomig A, Neumann FJ, Kastrati A, Schühlen H, Blasini R, Hadamitzky M, Walter H, Zitzmann-Roth EM, Richardt G, Alt E, Schmitt C, Ulm K. A randomized comparison of antiplatelet and anticoagulant therapy after the placement of coronary-artery stents. *N Engl J Med* 1996;**334**(17):1084–1089.

137. De Luca G, Suryapranata H, Stone GW, Antoniucci D, Tcheng JE, Neumann FJ, Van de Werf F, Antman EM, Topol EJ. Abciximab as adjunctive therapy to reperfusion in acute ST-segment elevation myocardial infarction: a meta-analysis of randomized trials. *JAMA* 2005;**293**:1759–1765.
138. Zeymer U, Margenet A, Haude M, Bode C, Lablanche JM, Heuer H, Schroder R, Kropff S, Bourkaib R, Banik N, Zahn R, Teiger E. Randomized comparison of eptifibatide versus abciximab in primary percutaneous coronary intervention in patients with acute ST-segment elevation myocardial infarction: results of the EVA-AMI Trial. *J Am Coll Cardiol* 2010;**56**:463–469.
139. Akerblom A, James SK, Koutouzis M, Lagerqvist B, Stenestrand U, Svennblad B, Oldgren J. Eptifibatide is noninferior to abciximab in primary percutaneous coronary intervention: results from the SCAAR (Swedish Coronary Angiography and Angioplasty Registry). *J Am Coll Cardiol* 2010;**56**:470–475.
140. Van 't Hof AW, Ten Berg J, Heestermans T, Dill T, Funck RC, van Werkum W, Dambrink JH, Suryapranata H, van Houwelingen G, Ottervanger JP, Stella P, Giannitsis E, Hamm C. Pre-hospital initiation of tirofiban in patients with ST-elevation myocardial infarction undergoing primary angioplasty (On-TIME 2): a multicentre, double-blind, randomised controlled trial. *Lancet* 2008;**372**:537–546.
141. Valgimigli M, Biondi-Zoccai G, Tebaldi M, van 't Hof AW, Campo G, Hamm C, ten Berg J, Bolognese L, Saia F, Danzi GB, Briguori C, Okmen E, King SB, Moliterno DJ, Topol EJ. Tirofiban as adjunctive therapy for acute coronary syndromes and percutaneous coronary intervention: a meta-analysis of randomized trials. *Eur Heart J* 2010;**31**:35–49.
142. De Luca G, Bellandi F, Huber K, Noc M, Petronio AS, Arntz HR, Maioli M, Gabriel HM, Zorman S, M DEC, Rakowski T, Gyongyosi M, Dudek D. Early glycoprotein IIb/IIIa inhibitors in primary angioplasty-abciximab long-term results (EGYPT-ALT) cooperation: individual patient's data meta-analysis. *J Thromb Haemost* 2011;**9**:2361–2370.
143. Primary versus tenecteplase-facilitated percutaneous coronary intervention in patients with ST-segment elevation acute myocardial infarction (ASSENT-4 PCI): randomised trial. *Lancet* 2006;**367**:569–578.
144. Sorajja P, Gersh BJ, Costantini C, McLaughlin MG, Zimetbaum P, Cox DA, Garcia E, Tcheng JE, Mehran R, Lansky AJ, Kandzari DE, Grines CL, Stone GW. Combined prognostic utility of ST-segment recovery and myocardial blush after primary percutaneous coronary intervention in acute myocardial infarction. *Eur Heart J* 2005;**26**:667–674.
145. Fokkema ML, Vlaar PJ, Svilaas T, Vogelzang M, Amo D, Diercks GF, Suurmeijer AJ, Zijlstra F. Incidence and clinical consequences of distal embolization on the coronary angiogram after percutaneous coronary intervention for ST-elevation myocardial infarction. *Eur Heart J* 2009;**30**:908–915.
146. Indications for fibrinolytic therapy in suspected acute myocardial infarction: collaborative overview of early mortality and major morbidity results from all randomised trials of more than 1000 patients. Fibrinolytic Therapy Trialists' (FTT) Collaborative Group. *Lancet* 1994;**343**:311–322.
147. White HD. Thrombolytic therapy in the elderly. *Lancet* 2000;**356**(9247):2028–2030.
148. Armstrong PW, Gershlick A, Goldstein P, Wilcox R, Danays T, Bluhmki E, Van de Werf F. The Strategic Reperfusion Early After Myocardial Infarction (STREAM) study. *Am Heart J* 2010;**160**:30–35 e31.
149. Van de Werf F, Barron HV, Armstrong PW, Granger CB, Berioli S, Barbash G, Pehrsson K, Verheugt FW, Meyer J, Betriu A, Califf RM, Li X, Fox NL. Incidence and predictors of bleeding events after fibrinolytic therapy with fibrin-specific agents: a comparison of TNK-tPA and rt-PA. *Eur Heart J* 2001;**22**:2253–2261.
150. Van De Werf F, Adgey J, Ardissino D, Armstrong PW, Aylward P, Barbash G, Betriu A, Binbrek AS, Califf R, Diaz R, Fanebust R, Fox K, Granger C, Heikkila J, Husted S, Jansky P, Langer A, Lupi E, Maseri A, Meyer J, Mlczoch J, Moccetti D, Myburgh D, Oto A, Paolasso E, Pehrsson K, Seabra-Gomes R, Soares-Piegas L, Sugrue D, Tendera M, Topol E, Toutouzas P, Vahanian A, Verheugt F, Wallentin L, White H. Single-bolus tenecteplase compared with front-loaded alteplase in acute myocardial infarction: the ASSENT-2 double-blind randomised trial. *Lancet* 1999;**354**:716–722.
151. A comparison of reteplase with alteplase for acute myocardial infarction. The Global Use of Strategies to Open Occluded Coronary Arteries (GUSTO III) Investigators. *N Engl J Med* 1997;**337**:1118–1123.
152. Berkowitz SD, Granger CB, Pieper KS, Lee KL, Gore JM, Simoons M, Armstrong PW, Topol EJ, Califf RM. Incidence and predictors of bleeding after contemporary thrombolytic therapy for myocardial infarction. The Global Utilization of Streptokinase and Tissue Plasminogen activator for Occluded coronary arteries (GUSTO) I Investigators. *Circulation* 1997;**95**:2508–2516.
153. An international randomized trial comparing four thrombolytic strategies for acute myocardial infarction. The GUSTO investigators. *N Engl J Med* 1993;**329**:673–682.
154. Bottiger BW, Arntz HR, Chamberlain DA, Bluhmki E, Belmans A, Danays T, Carli PA, Adgey JA, Bode C, Wenzel V. Thrombolysis during resuscitation for out-of-hospital cardiac arrest. *N Engl J Med* 2008;**359**:2651–2662.
155. Bjorklund E, Stenestrand U, Lindback J, Svensson L, Wallentin L, Lindahl B. Pre-hospital thrombolysis delivered by paramedics is associated with reduced time delay and mortality in ambulance-transported real-life patients with ST-elevation myocardial infarction. *Eur Heart J* 2006;**27**:1146–1152.
156. Chen ZM, Jiang LX, Chen YP, Xie JX, Pan HC, Peto R, Collins R, Liu LS. Addition of clopidogrel to aspirin in 45,852 patients with acute myocardial infarction: randomised placebo-controlled trial. *Lancet* 2005;**366**:1607–1621.
157. Sabatine MS, Cannon CP, Gibson CM, Lopez-Sendon JL, Montalescot G, Theroux P, Claeys MJ, Cools F, Hill KA, Skene AM, McCabe CH, Braunwald E. Addition of clopidogrel to aspirin and fibrinolytic therapy for myocardial infarction with ST-segment elevation. *N Engl J Med* 2005;**353**:1179–1189.
158. Efficacy and safety of tenecteplase in combination with enoxaparin, abciximab, or unfractionated heparin: the ASSENT-3 randomised trial in acute myocardial infarction. *Lancet* 2001;**358**:605–613.
159. Wallentin L, Goldstein P, Armstrong PW, Granger CB, Adgey AA, Arntz HR, Bogaerts K, Danays T, Lindahl B, Makijarvi M, Verheugt F, Van de Werf F. Efficacy and safety of tenecteplase in combination with the low-molecular-weight heparin enoxaparin or unfractionated heparin in the pre-hospital setting: the Assessment of the Safety and Efficacy of a New Thrombolytic Regimen (ASSENT)-3 PLUS randomized trial in acute myocardial infarction. *Circulation* 2003;**108**:135–142.
160. Giraldez RR, Nicolau JC, Corbalan R, Gurfinkel EP, Juarez U, Lopez-Sendon J, Parkhomenko A, Molhoek P, Mohanavelu S, Morrow DA, Antman EM. Enoxaparin is superior to unfractionated heparin in patients with ST elevation myocardial infarction undergoing fibrinolysis regardless of the choice of lytic: an ExTRACT-TIMI 25 analysis. *Eur Heart J* 2007;**28**:1566–1573.
161. White HD, Braunwald E, Murphy SA, Jacob AJ, Gotcheva N, Polonetsky L, Antman EM. Enoxaparin vs. unfractionated heparin with fibrinolysis for ST-elevation myocardial infarction in elderly and younger patients: results from ExTRACT-TIMI 25. *Eur Heart J* 2007;**28**:1066–1071.
162. Ross AM, Molhoek P, Lundergan C, Knudtson M, Draoui Y, Regalado L, Le Louer V, Bigonzi F, Schwartz W, de Jong E, Coyne K. Randomized comparison of enoxaparin, a low-molecular-weight heparin, with unfractionated heparin adjunctive to recombinant tissue plasminogen activator thrombolysis and aspirin: second trial of Heparin and Aspirin Reperfusion Therapy (HART II). *Circulation* 2001;**104**:648–652.
163. Antman EM, Louchenburg HW, Baars HF, Wesdorp JC, Hamer B, Bassand JP, Bigonzi F, Pisapia G, Gibson CM, Heidbuchel H, Braunwald E, Van de Werf F. Enoxaparin as adjunctive antithrombin therapy for ST-elevation myocardial infarction: results of the ENTIRE-Thrombolysis in Myocardial Infarction (TIMI) 23 Trial. *Circulation* 2002;**105**:1642–1649.
164. Peters RJ, Joyner C, Bassand JP, Afzal R, Chrolavicius S, Mehta SR, Oldgren J, Wallentin L, Budaj A, Fox KA, Yusuf S. The role of fondaparinux as an adjunct to thrombolytic therapy in acute myocardial infarction: a subgroup analysis of the OASIS-6 trial. *Eur Heart J* 2008;**29**:324–331.
165. Gershlick AH, Stephens-Lloyd A, Hughes S, Abrams KR, Stevens SE, Uren NG, de Belder A, Davis J, Pitt M, Banning A, Baumbach A, Shiu MF, Schofield P, Dawkins KD, Henderson RA, Oldroyd KG, Wilcox R. Rescue angioplasty after failed thrombolytic therapy for acute myocardial infarction. *N Engl J Med* 2005;**353**:2758–2768.
166. Ellis SG, da Silva ER, Heyndrickx G, Talley JD, Cernigliaro C, Steg G, Spaulding C, Nobuyoshi M, Erbel R, Vassanelli C et al. Randomized comparison of rescue angioplasty with conservative management of patients with early failure of thrombolysis for acute anterior myocardial infarction. *Circulation* 1994;**90**:2280–2284.
167. Hochman JS, Sleeper LA, White HD, Dzavik V, Wong SC, Menon V, Webb JG, Steingart R, Picard MH, Menegus MA, Boland J, Sanborn T, Buller CE, Modur S, Forman R, Desvigne-Nickens P, Jacobs AK, Slater JN, Lejemtel TH. One-year survival following early revascularization for cardiogenic shock. *JAMA* 2001;**285**:190–192.
168. Cantor WJ, Fitchett D, Borgundvaag B, Ducas J, Heffernan M, Cohen EA, Morrison LJ, Langer A, Dzavik V, Mehta SR, Lazzam C, Schwartz B, Casanova A, Goodman SG. Routine early angioplasty after fibrinolysis for acute myocardial infarction. *N Engl J Med* 2009;**360**:2705–2718.
169. Di Mario C, Dudek D, Piscione F, Mielecki W, Savonitto S, Murena E, Dimopoulos K, Manari A, Gasparone A, Ochala A, Zmudka K, Bolognese L, Steg PG, Flather M. Immediate angioplasty versus standard therapy with rescue angioplasty after thrombolysis in the Combined Abciximab Reteplase Stent Study in Acute Myocardial Infarction (CARESS-in-AMI): an open, prospective, randomised, multicentre trial. *Lancet* 2008;**371**:559–568.
170. Bohmer E, Hoffmann P, Abdelnoor M, Arnesen H, Halvorsen S. Efficacy and safety of immediate angioplasty versus ischemia-guided management after

- thrombolysis in acute myocardial infarction in areas with very long transfer distances results of the NORDISTEMI (NORwegian study on District treatment of ST-elevation myocardial infarction). *J Am Coll Cardiol* 2010;**55**:102–110.
171. Fernandez-Aviles F, Alonso JJ, Castro-Beiras A, Vazquez N, Blanco J, Alonso-Briales J, Lopez-Mesa J, Fernandez-Vazquez F, Calvo I, Martinez-Elbal L, San Roman JA, Ramos B. Routine invasive strategy within 24 hours of thrombolysis versus ischaemia-guided conservative approach for acute myocardial infarction with ST-segment elevation (GRACIA-1): a randomised controlled trial. *Lancet* 2004;**364**:1045–1053.
 172. Borgja F, Goodman SG, Halvorsen S, Cantor WJ, Piscione F, Le May MR, Fernandez-Aviles F, Sanchez PL, Dimopoulos K, Scheller B, Armstrong PW, Di Mario C. Early routine percutaneous coronary intervention after fibrinolysis vs. standard therapy in ST-segment elevation myocardial infarction: a meta-analysis. *Eur Heart J* 2010;**31**:2156–2169.
 173. Sanchez PL, Gimeno F, Ancillo P, Sanz JJ, Alonso-Briales JH, Bosa F, Santos I, Sanchis J, Bethencourt A, Lopez-Mesa J, de Prado AP, Alonso JJ, San Roman JA, Fernandez-Aviles F. Role of the paclitaxel-eluting stent and tirofiban in patients with ST-elevation myocardial infarction undergoing postfibrinolysis angioplasty: the GRACIA-3 randomized clinical trial. *Circ Cardiovasc Interv* 2010;**3**:297–307.
 174. de Bono DP, Simoons ML, Tijssen J, Arnold AE, Betriu A, Burgersdijk C, Lopez Bescos L, Mueller E, Pfisterer M, Van de Werf F, Zijlstra F, Verstraete M. Effect of early intravenous heparin on coronary patency, infarct size, and bleeding complications after alteplase thrombolysis: results of a randomised double blind European Cooperative Study Group trial. *Br Heart J* 1992;**67**:122–128.
 175. The effects of tissue plasminogen activator, streptokinase, or both on coronary-artery patency, ventricular function, and survival after acute myocardial infarction. The GUSTO Angiographic Investigators. *N Engl J Med* 1993;**329**:1615–1622.
 176. White H. Thrombin-specific anticoagulation with bivalirudin versus heparin in patients receiving fibrinolytic therapy for acute myocardial infarction: the HERO-2 randomised trial. *Lancet* 2001;**358**:1855–1863.
 177. Fernandez-Aviles F, Alonso JJ, Pena G, Blanco J, Alonso-Briales J, Lopez-Mesa J, Fernandez-Vazquez F, Moreu J, Hernandez RA, Castro-Beiras A, Gabriel R, Gibson CM, Sanchez PL. Primary angioplasty vs. early routine post-fibrinolysis angioplasty for acute myocardial infarction with ST-segment elevation: the GRACIA-2 non-inferiority, randomized, controlled trial. *Eur Heart J* 2007;**28**:949–960.
 178. Scheller B, Hennen B, Hammer B, Walle J, Hofer C, Hilpert V, Winter H, Nickenig G, Bohm M. Beneficial effects of immediate stenting after thrombolysis in acute myocardial infarction. *J Am Coll Cardiol* 2003;**42**:634–641.
 179. Le May MR, Wells GA, Labinaz M, Davies RF, Turek M, Leddy D, Maloney J, McKibbin T, Quinn B, Beanlands RS, Glover C, Marquis JF, O'Brien ER, Williams WL, Higginson LA. Combined angioplasty and pharmacological intervention versus thrombolysis alone in acute myocardial infarction (CAPITAL AMI study). *J Am Coll Cardiol* 2005;**46**:417–424.
 180. D'Souza SP, Mamas MA, Fraser DG, Fath-Ordoubadi F. Routine early coronary angioplasty versus ischaemia-guided angioplasty after thrombolysis in acute ST-elevation myocardial infarction: a meta-analysis. *Eur Heart J* 2011;**32**:972–982.
 181. Madsen JK, Grande P, Saunamaki K, Thayssen P, Kassis E, Eriksen U, Rasmussen K, Haunso S, Nielsen TT, Haghfelt T, Fritz-Hansen P, Hjelms E, Paulsen PK, Alstrup P, Arendrup H, Niebuhr-Jorgensen U, Andersen LI. Danish multicenter randomized study of invasive versus conservative treatment in patients with inducible ischemia after thrombolysis in acute myocardial infarction (DANAMI). DANish trial in Acute Myocardial Infarction. *Circulation* 1997;**96**:748–755.
 182. Savonitto S, D'Urbano M, Caracciolo M, Barlocco F, Mariani G, Nichelatti M, Klugmann S, De Servi S. Urgent surgery in patients with a recently implanted coronary drug-eluting stent: a phase II study of 'bridging' antiplatelet therapy with tirofiban during temporary withdrawal of clopidogrel. *Br J Anaesth* 2010;**104**:285–291.
 183. Angiolillo DJ, Firstenberg MS, Price MJ, Tummala PE, Hutrya M, Welsby IJ, Voeltz MD, Chandna H, Ramaiah C, Brtko M, Cannon L, Dyke C, Liu T, Montalescot G, Manoukian SV, Prats J, Topol EJ. Bridging antiplatelet therapy with cangrelor in patients undergoing cardiac surgery: a randomized controlled trial. *JAMA* 2012;**307**:265–274.
 184. Cohen M, Gensini GF, Maritz F, Gurfinkel EP, Huber K, Timerman A, Krzeminska-Pakula M, Danchin N, White HD, Santopinto J, Bigonzi F, Hecquet C, Vittori L. The safety and efficacy of subcutaneous enoxaparin versus intravenous unfractionated heparin and tirofiban versus placebo in the treatment of acute ST-segment elevation myocardial infarction patients ineligible for reperfusion (TETAMI): a randomized trial. *J Am Coll Cardiol* 2003;**42**:1348–1356.
 185. Oldgren J, Wallentin L, Afzal R, Bassand JP, Budaj A, Chrolavicius S, Fox KA, Granger CB, Mehta SR, Pais P, Peters RJ, Xavier D, Zhu J, Yusuf S. Effects of fondaparinux in patients with ST-segment elevation acute myocardial infarction not receiving reperfusion treatment. *Eur Heart J* 2008;**29**:315–323.
 186. Steg PG, Jolly SS, Mehta SR, Afzal R, Xavier D, Rupprecht HJ, Lopez-Sendon JL, Budaj A, Diaz R, Avezum A, Widimsky P, Rao SV, Chrolavicius S, Meeks B, Joyner C, Pogue J, Yusuf S. Low-dose vs standard-dose unfractionated heparin for percutaneous coronary intervention in acute coronary syndromes treated with fondaparinux: the FUTURA/OASIS-8 randomized trial. *JAMA* 2010;**304**:1339–1349.
 187. Yusuf S, Zhao F, Mehta SR, Chrolavicius S, Tognoni G, Fox KK. Effects of clopidogrel in addition to aspirin in patients with acute coronary syndromes without ST-segment elevation. *N Engl J Med* 2001;**345**:494–502.
 188. James SK, Roe MT, Cannon CP, Cornel JH, Horrow J, Husted S, Katus H, Morais J, Steg PG, Storey RF, Stevens S, Wallentin L, Harrington RA. Ticagrelor versus clopidogrel in patients with acute coronary syndromes intended for non-invasive management: substudy from prospective randomised PLATElet inhibition and patient Outcomes (PLATO) trial. *Br Med J* 2011;**342**:d3527.
 189. Brieger D, Eagle KA, Goodman SG, Steg PG, Budaj A, White K, Montalescot G. Acute coronary syndromes without chest pain, an underdiagnosed and under-treated high-risk group: insights from the Global Registry of Acute Coronary Events. 2004. *Chest* 2009;**136**(5 Suppl):e30.
 190. Alexander KP, Chen AY, Roe MT, Newby LK, Gibson CM, Allen-LaPointe NM, Pollack C, Glibler WB, Ohman EM, Peterson ED. Excess dosing of antiplatelet and antithrombin agents in the treatment of non-ST-segment elevation acute coronary syndromes. *JAMA* 2005;**294**:3108–3116.
 191. Bairey Merz N, Bonow RO, Sopko G, Balaban RS, Cannon RO 3rd, Gordon D, Hand MM, Hayes SN, Lewis JF, Long T, Manolio TA, Maseri A, Nabel EG, Desvigne Nickens P, Pepine CJ, Redberg RF, Rossouw JE, Selker HP, Shaw LJ, Waters DD. Women's Ischemic Syndrome Evaluation: current status and future research directions: report of the National Heart, Lung and Blood Institute workshop: October 2–4, 2002: executive summary. *Circulation* 2004;**109**:805–807.
 192. Milcent C, Dormont B, Durand-Zaleski I, Steg PG. Gender differences in hospital mortality and use of percutaneous coronary intervention in acute myocardial infarction: microsimulation analysis of the 1999 nationwide French hospitals database. *Circulation* 2007;**115**:833–839.
 193. Kang SH, Suh JW, Yoon CH, Cho MC, Kim YJ, Chae SC, Yoon JH, Gwon HC, Han KR, Kim JH, Ahn YK, Jeong MH, Kim HS, Choi DJ. Sex Differences in Management and Mortality of Patients With ST-Elevation Myocardial Infarction (from the Korean Acute Myocardial Infarction National Registry). *Am J Cardiol* 2012;**109**:787–793.
 194. Zhang Z, Fang J, Gillespie C, Wang G, Hong Y, Yoon PW. Age-Specific Gender Differences in In-Hospital Mortality by Type of Acute Myocardial Infarction. *Am J Cardiol* 2012;**109**:1097–1103.
 195. Mehilii J, Kastrati A, Dirschinger J, Pache J, Seyfarth M, Blasini R, Hall D, Neumann FJ, Schomig A. Sex-based analysis of outcome in patients with acute myocardial infarction treated predominantly with percutaneous coronary intervention. *JAMA* 2002;**287**:210–215.
 196. Szumner K, Lundman P, Jacobson SH, Schon S, Lindback J, Stenestrand U, Wallentin L, Jernberg T. Relation between renal function, presentation, use of therapies and in-hospital complications in acute coronary syndrome: data from the SWEDEHEART register. *J Intern Med* 2010;**268**:40–49.
 197. James S, Budaj A, Aylward P, Buck KK, Cannon CP, Cornel JH, Harrington RA, Horrow J, Katus H, Keltai M, Lewis BS, Parikh K, Storey RF, Szumner K, Wojdyla D, Wallentin L. Ticagrelor versus clopidogrel in acute coronary syndromes in relation to renal function: results from the Platelet Inhibition and Patient Outcomes (PLATO) trial. *Circulation* 2010;**122**:1056–1067.
 198. Wiviott SD, Braunwald E, Angiolillo DJ, Meisel S, Dalby AJ, Verheugt FW, Goodman SG, Corbalan R, Purdy DA, Murphy SA, McCabe CH, Antman EM. Greater clinical benefit of more intensive oral antiplatelet therapy with prasugrel in patients with diabetes mellitus in the trial to assess improvement in therapeutic outcomes by optimizing platelet inhibition with prasugrel-Thrombolysis in Myocardial Infarction 38. *Circulation* 2008;**118**:1626–1636.
 199. James S, Angiolillo DJ, Cornel JH, Erlinge D, Husted S, Kontny F, Maya J, Nicolau JC, Spinar J, Storey RF, Stevens SR, Wallentin L. Ticagrelor vs. clopidogrel in patients with acute coronary syndromes and diabetes: a substudy from the PLATElet inhibition and patient Outcomes (PLATO) trial. *Eur Heart J* 2010;**31**:3006–3016.
 200. Bartnik M, Ryden L, Ferrari R, Malmberg K, Pyorala K, Simoons M, Standl E, Soler-Soler J, Ohrvik J. The prevalence of abnormal glucose regulation in patients with coronary artery disease across Europe. The Euro Heart Survey on diabetes and the heart. *Eur Heart J* 2004;**25**:1880–1890.
 201. Timmer JR, Hoekstra M, Nijsten MW, van der Horst IC, Ottervanger JP, Slingerland RJ, Dambrink JH, Bilo HJ, Zijlstra F, van 't Hof AW. Prognostic

- value of admission glycosylated hemoglobin and glucose in nondiabetic patients with ST-segment-elevation myocardial infarction treated with percutaneous coronary intervention. *Circulation* 2011;**124**:704–711.
202. De Caterina R, Madonna R, Sourij H, Wascher T. Glycaemic control in acute coronary syndromes: prognostic value and therapeutic options. *Eur Heart J* 2010;**31**:1557–1564.
 203. Diaz R, Goyal A, Mehta SR, Afzal R, Xavier D, Pais P, Chrolavicius S, Zhu J, Kazmi K, Liu L, Budaj A, Zubaid M, Avezum A, Ruda M, Yusuf S. Glucose–insulin–potassium therapy in patients with ST-segment elevation myocardial infarction. *JAMA* 2007;**298**:2399–2405.
 204. Finfer S, Chittock DR, Su SY, Blair D, Foster D, Dhingra V, Bellomo R, Cook D, Dodek P, Henderson WR, Hebert PC, Heritier S, Heyland DK, McArthur C, McDonald E, Mitchell I, Myburgh JA, Norton R, Potter J, Robinson BG, Ronco JJ. Intensive versus conventional glucose control in critically ill patients. *N Engl J Med* 2009;**360**:1283–1297.
 205. Ryden L, Standl E, Bartnik M, Van den Bergh G, Betteridge J, de Boer MJ, Cosentino F, Jonsson B, Laakso M, Malmberg K, Priori S, Ostergren J, Tuomilehto J, Thraainsdottir I, Vanhorebeek I, Stramba-Badiale M, Lindgren P, Qiao Q, Priori SG, Blanc JJ, Budaj A, Camm J, Dean V, Deckers J, Dickstein K, Lekakis J, McGregor K, Metra M, Morais J, Osterspey A, Tamargo J, Zamorano JL, Deckers JW, Bertrand M, Charbonnel B, Erdmann E, Ferrannini E, Flyvbjerg A, Gohlke H, Juanatey JR, Graham I, Monteiro PF, Parhofer K, Pyorala K, Raz I, Schemthaner G, Volpe M, Wood D. Guidelines on diabetes, pre-diabetes, and cardiovascular diseases: executive summary. The Task Force on Diabetes and Cardiovascular Diseases of the European Society of Cardiology (ESC) and of the European Association for the Study of Diabetes (EASD). *Eur Heart J* 2007;**28**:88–136.
 206. Senthinathan A, Kelly V, Dzingina M, Jones D, Baker M, Longson D. Hyperglycaemia in acute coronary syndromes: summary of NICE guidance. *Br Med J* 2011;**343**:d6646.
 207. Kosiborod M, McGuire DK. Glucose-lowering targets for patients with cardiovascular disease: focus on inpatient management of patients with acute coronary syndromes. *Circulation* 2010;**122**:2736–2744.
 208. Bartnik M, Malmberg K, Norhammar A, Tenerz A, Ohrvik J, Ryden L. Newly detected abnormal glucose tolerance: an important predictor of long-term outcome after myocardial infarction. *Eur Heart J* 2004;**25**:1990–1997.
 209. Hasin Y, Danchin N, Filippatos GS, Heras M, Janssens U, Leor J, Nahir M, Parkhomenko A, Thygesen K, Tubaro M, Wallentin LC, Zakke I. Recommendations for the structure, organization, and operation of intensive cardiac care units. *Eur Heart J* 2005;**26**:1676–1682.
 210. Spencer FA, Lessard D, Gore JM, Yarzebski J, Goldberg RJ. Declining length of hospital stay for acute myocardial infarction and postdischarge outcomes: a community-wide perspective. *Arch Intern Med* 2004;**164**:733–740.
 211. Berger AK, Duval S, Jacobs DR Jr., Barber C, Vazquez G, Lee S, Luepker RV. Relation of length of hospital stay in acute myocardial infarction to postdischarge mortality. *Am J Cardiol* 2008;**101**:428–434.
 212. Grines CL, Marsalese DL, Brodie B, Griffin J, Donohue B, Costantini CR, Balestrini C, Stone G, Wharton T, Esente P, Spain M, Moses J, Nobuyoshi M, Ayres M, Jones D, Mason D, Sachs D, Grines LL, O'Neill W. Safety and cost-effectiveness of early discharge after primary angioplasty in low risk patients with acute myocardial infarction. PAMI-II Investigators. Primary Angioplasty in Myocardial Infarction. *J Am Coll Cardiol* 1998;**31**:967–972.
 213. De Luca G, Suryapranata H, van 't Hof AW, de Boer MJ, Hoorntje JC, Dambink JH, Gosselink AT, Ottervanger JP, Zijlstra F. Prognostic assessment of patients with acute myocardial infarction treated with primary angioplasty: implications for early discharge. *Circulation* 2004;**109**:2737–2743.
 214. Estevez-Loureiro R, Calvino-Santos R, Vazquez JM, Barge-Caballero E, Salgado-Fernandez J, Pineiro M, Freire-Tellado M, Varela-Portas J, Martinez L, Gomez S, Rodriguez JA, Vazquez N, Castro-Beiras A. Safety and feasibility of returning patients early to their originating centers after transfer for primary percutaneous coronary intervention. *Rev Esp Cardiol* 2009;**62**:1356–1364.
 215. Newby LK, Hasselblad V, Armstrong PW, Van de Werf F, Mark DB, White HD, Topol EJ, Califf RM. Time-based risk assessment after myocardial infarction. Implications for timing of discharge and applications to medical decision-making. *Eur Heart J* 2003;**24**:182–189.
 216. Kotowycz MA, Cosman TL, Tartaglia C, Afzal R, Syal RP, Natarajan MK. Safety and feasibility of early hospital discharge in ST-segment elevation myocardial infarction—a prospective and randomized trial in low-risk primary percutaneous coronary intervention patients (the Safe-Depart Trial). *Am Heart J* 2010;**159**:117.e111–e116.
 217. Morrow DA, Antman EM, Charlesworth A, Cairns R, Murphy SA, de Lemos JA, Giugliano RP, McCabe CH, Braunwald E. TIMI risk score for ST-elevation myocardial infarction: A convenient, bedside, clinical score for risk assessment at presentation: An intravenous nPA for treatment of infarcting myocardium early II trial substudy. *Circulation* 2000;**102**:2031–2037.
 218. Fox KA, Anderson FA Jr., Dabbous OH, Steg PG, Lopez-Sendon J, Van de Werf F, Budaj A, Gurfinkel EP, Goodman SG, Brieger D. Intervention in acute coronary syndromes: do patients undergo intervention on the basis of their risk characteristics? The Global Registry of Acute Coronary Events (GRACE). *Heart* 2007;**93**:177–182.
 219. Lee KL, Woodlief LH, Topol EJ, Weaver WD, Betriu A, Col J, Simoons M, Aylward P, Van de Werf F, Califf RM. Predictors of 30-day mortality in the era of reperfusion for acute myocardial infarction. Results from an international trial of 41,021 patients. GUSTO-I Investigators. *Circulation* 1995;**91**:1659–1668.
 220. Allman KC, Shaw LJ, Hachamovitch R, Udellson JE. Myocardial viability testing and impact of revascularization on prognosis in patients with coronary artery disease and left ventricular dysfunction: a meta-analysis. *J Am Coll Cardiol* 2002;**39**:1151–1158.
 221. St John Sutton M, Pfeffer MA, Plappert T, Rouleau JL, Moye LA, Dagenais GR, Lamas GA, Klein M, Sussex B, Goldman S et al. Quantitative two-dimensional echocardiographic measurements are major predictors of adverse cardiovascular events after acute myocardial infarction. The protective effects of captopril. *Circulation* 1994;**89**:68–75.
 222. Beanlands RS, Nichol G, Huszti E, Humen D, Racine N, Freeman M, Gulenchyn KY, Garrard L, deKemp R, Guo A, Ruddy TD, Benard F, Lamy A, Iwanochko RM. F-18-fluorodeoxyglucose positron emission tomography imaging-assisted management of patients with severe left ventricular dysfunction and suspected coronary disease: a randomized, controlled trial (PARR-2). *J Am Coll Cardiol* 2007;**50**:2002–2012.
 223. La Canna G, Rahimtoola SH, Visioli O, Giubbini R, Alfieri O, Zognio M, Milan E, Ceconi C, Gargano M, Lo Russo R, Ferrari R. Sensitivity, specificity, and predictive accuracies of non-invasive tests, singly and in combination, for diagnosis of hibernating myocardium. *Eur Heart J* 2000;**21**:1358–1367.
 224. Perk J, De Backer G, Gohlke H, Graham I, Reiner Z, Verschuren WMM, Albus C, Benlian P, Boysen G, Cifkova R, Deaton C, Ebrahim S, Fisher M, Germano G, Hobbs R, Hoes A, Karadeniz S, Mezzani A, Prescott E, Ryden L, Schermer M, Syvanne M, Scholte Op Reimer WJ, Vrints C, Wood D, Zamorano JL, Zannad F, Cooney MT, Bax J, Baumgartner H, Ceconi C, Dean V, Deaton C, Fagard R, Funck-Brentano C, Hasdai D, Hoes A, Kirchhof P, Knuuti J, Kolh P, McDonagh T, Moulin C, Popescu BA, Reiner Z, Sechtem U, Tendera M, Torbicki A, Vahanian A, Windecker S, Sirnes PA, Aboyans V, Esquerre EA, Baigent C, Brotons C, Burell G, Ceriello A, De Sutter J, Deckers J, Del Prato S, Diener HC, Fitzsimons D, Fras Z, Hambrecht R, Jankowski P, Keil U, Kirby M, Larsen ML, Mancia G, Manolis AJ, McMurray J, Pajak A, Parkhomenko A, Rallidis L, Rigo F, Rocha E, Ruilope LM, van der Velde E, Vanuzzo D, Viigimaa M, Volpe M, Wiklund O, Wolpert C. The Fifth Joint Task Force of the European Society of Cardiology and Other Societies on Cardiovascular Disease Prevention in Clinical Practice (constituted by representatives of nine societies and by invited experts) * Developed with the special contribution of the European Association for Cardiovascular Prevention & Rehabilitation (EACPR). *Eur Heart J* 2012;**33**:1635–1701.
 225. Thomson CC, Rigotti NA. Hospital- and clinic-based smoking cessation interventions for smokers with cardiovascular disease. *Prog Cardiovasc Dis* 2003;**45**:459–479.
 226. Meine TJ, Patel MR, Washam JB, Pappas PA, Jollis JG. Safety and effectiveness of transdermal nicotine patch in smokers admitted with acute coronary syndromes. *Am J Cardiol* 2005;**95**:976–978.
 227. Taylor CB, Houston-Miller N, Killen JD, DeBusk RF. Smoking cessation after acute myocardial infarction: effects of a nurse-managed intervention. *Ann Intern Med* 1990;**113**:118–123.
 228. Taylor RS, Brown A, Ebrahim S, Jolliffe J, Noorani H, Rees K, Skidmore B, Stone JA, Thompson DR, Oldridge N. Exercise-based rehabilitation for patients with coronary heart disease: systematic review and meta-analysis of randomized controlled trials. *Am J Med* 2004;**116**:682–692.
 229. Myers J, Prakash M, Froelicher V, Do D, Partington S, Atwood JE. Exercise capacity and mortality among men referred for exercise testing. *N Engl J Med* 2002;**346**:793–801.
 230. Bangalore S, Qin J, Sloan S, Murphy SA, Cannon CP. What is the optimal blood pressure in patients after acute coronary syndromes? Relationship of blood pressure and cardiovascular events in the Pravastatin OR atorvastatin Evaluation and Infection Therapy-Thrombolysis In Myocardial Infarction (PROVE IT-TIMI) 22 trial. *Circulation* 2010;**122**:2142–2151.
 231. Gulliksson M, Burell G, Vessby B, Lundin L, Toss H, Svardsudd K. Randomized controlled trial of cognitive behavioral therapy vs standard treatment to prevent recurrent cardiovascular events in patients with coronary heart disease: Secondary Prevention in Uppsala Primary Health Care project (SUPRIM). *Arch Intern Med* 2011;**171**:134–140.
 232. Lawler PR, Filion KB, Eisenberg MJ. Efficacy of exercise-based cardiac rehabilitation post-myocardial infarction: a systematic review and meta-analysis of randomized controlled trials. *Am Heart J* 2011;**162**(4):571–584.e572.

233. Heran BS, Chen JM, Ebrahim S, Moxham T, Oldridge N, Rees K, Thompson DR, Taylor RS. Exercise-based cardiac rehabilitation for coronary heart disease. *Cochrane Database Syst Rev* 2011;(7):CD001800.
234. West RR, Jones DA, Henderson AH. Rehabilitation after myocardial infarction trial (RAMIT): multi-centre randomised controlled trial of comprehensive cardiac rehabilitation in patients following acute myocardial infarction. *Heart* 2011;**98**:637–644.
235. Giannuzzi P, Temporelli PL, Marchioli R, Maggioni AP, Balestroni G, Ceci V, Chieffo C, Gattone M, Griffo R, Schweiger C, Tavazzi L, Urbinati S, Valagussa F, Vanuzzo D. Global secondary prevention strategies to limit event recurrence after myocardial infarction: results of the GOSPEL study, a multicenter, randomized controlled trial from the Italian Cardiac Rehabilitation Network. *Arch Intern Med* 2008;**168**:2194–2204.
236. Dalal HM, Zawada A, Jolly K, Moxham T, Taylor RS. Home based versus centre based cardiac rehabilitation: Cochrane systematic review and meta-analysis. *Br Med J* 2010;**340**:b5631.
237. Baigent C, Blackwell L, Collins R, Emberson J, Godwin J, Peto R, Buring J, Hennekens C, Kearney P, Meade T, Patrono C, Roncagliani MC, Zanchetti A. Aspirin in the primary and secondary prevention of vascular disease: collaborative meta-analysis of individual participant data from randomised trials. *Lancet* 2009;**373**:1849–1860.
238. Capodanno D, Patel A, Dharmashankar K, Ferreiro JL, Ueno M, Kodali M, Tomasello SD, Capranzano P, Seecheran N, Darlington A, Tello-Montoliu A, Desai B, Bass TA, Angiolillo DJ. Pharmacodynamic effects of different aspirin dosing regimens in type 2 diabetes mellitus patients with coronary artery disease. *Circ Cardiovasc Interv* 2011;**4**:180–187.
239. Henry P, Vermeil A, Boval B, Guyetand C, Petroni T, Dillinger JG, Sideris G, Sollier CB, Drouet L. 24-hour time-dependent aspirin efficacy in patients with stable coronary artery disease. *Thromb Haemost* 2011;**105**:336–344.
240. Silberman S, Neukirch-Stoop C, Steg PG. Rapid desensitization procedure for patients with aspirin hypersensitivity undergoing coronary stenting. *Am J Cardiol* 2005;**95**:509–510.
241. Page NA, Schroeder WS. Rapid desensitization protocols for patients with cardiovascular disease and aspirin hypersensitivity in an era of dual antiplatelet therapy. *Ann Pharmacother* 2007;**41**:61–67.
242. Rossini R, Angiolillo DJ, Musumeci G, Scuri P, Invernizzi P, Bass TA, Mihalcsik L, Gavazzi A. Aspirin desensitization in patients undergoing percutaneous coronary interventions with stent implantation. *Am J Cardiol* 2008;**101**:786–789.
243. A randomised, blinded, trial of clopidogrel versus aspirin in patients at risk of ischaemic events (CAPRIE). CAPRIE Steering Committee. *Lancet* 1996;**348**:1329–1339.
244. Grines CL, Bonow RO, Casey DE Jr., Gardner TJ, Lockhart PB, Moliterno DJ, O'Gara P, Whitlow P. Prevention of premature discontinuation of dual antiplatelet therapy in patients with coronary artery stents: a science advisory from the American Heart Association, American College of Cardiology, Society for Cardiovascular Angiography and Interventions, American College of Surgeons, and American Dental Association, with representation from the American College of Physicians. *Circulation* 2007;**115**:813–818.
245. Gwon HC, Hahn JY, Park KW, Song YB, Chae IH, Lim DS, Han KR, Choi JH, Choi SH, Kang HJ, Koo BK, Ahn T, Yoon JH, Jeong MH, Hong TJ, Chung WY, Choi YJ, Hur SH, Kwon HM, Jeon DW, Kim BO, Park SH, Lee NH, Jeon HK, Jang Y, Kim HS. Six-Month versus Twelve-Month Dual Antiplatelet Therapy after Implantation of Drug-Eluting Stents: 'EXCELLENT' Randomized, Multicenter Study. *Circulation* 2012;**125**:505–513.
246. Park SJ, Park DW, Kim YH, Kang SJ, Lee SW, Lee CW, Han KH, Park SW, Yun SC, Lee SG, Rha SW, Seong IW, Jeong MH, Hur SH, Lee NH, Yoon J, Yang JY, Lee BK, Choi YJ, Chung WS, Lim DS, Cheong SS, Kim KS, Chae JK, Nah DY, Jeon DS, Seung KB, Jang JS, Park HS, Lee K. Duration of dual antiplatelet therapy after implantation of drug-eluting stents. *N Engl J Med* 2010;**362**:1374–1382.
247. Valgimigli M, Campo G, Percoco G, Monti M, Ferrari F, Tumscitz C, Zuffi A, Colombo F, Kubbaiah M, Cavazza C, Cangiano E, Tebaldi M, Minarelli M, Arcozzi C, Scalone A, Frangione A, Borghesi M, Marchesini J, Parrinello G, Ferrari R. Randomized comparison of 6- versus 24-month clopidogrel therapy after balancing anti-intimal hyperplasia stent potency in all-comer patients undergoing percutaneous coronary intervention Design and rationale for the PROlonging Dual-antiplatelet treatment after Grading stent-induced Intimal hyperplasia study (PRODIGY). *Am Heart J* 2010;**160**:804–811.
248. Mauri L, Kereiakes DJ, Normand SL, Wiviott SD, Cohen DJ, Holmes DR, Bangalore S, Cutlip DE, Pencina M, Massaro JM. Rationale and design of the dual antiplatelet therapy study, a prospective, multicenter, randomized, double-blind trial to assess the effectiveness and safety of 12 versus 30 months of dual antiplatelet therapy in subjects undergoing percutaneous coronary intervention with either drug-eluting stent or bare metal stent placement for the treatment of coronary artery lesions. *Am Heart J* 2010;**160**:1035–1041, e1031.
249. Mehta SR, Yusuf S, Peters RJ, Bertrand ME, Lewis BS, Natarajan MK, Malmberg K, Rupprecht H, Zhao F, Chrolavicius S, Copland I, Fox KA. Effects of pretreatment with clopidogrel and aspirin followed by long-term therapy in patients undergoing percutaneous coronary intervention: the PCI-CURE study. *Lancet* 2001;**358**:527–533.
250. Camm AJ, Kirchhof P, Lip GY, Schotten U, Savelieva I, Ernst S, Van Gelder IC, Al-Attar N, Hindricks G, Prendergast B, Heidbuchel H, Alfieri O, Angelini A, Atar D, Colonna P, De Caterina R, De Sutter J, Goette A, Gorennek B, Heldal M, Hohloser SH, Kolh P, Le Heuzey JY, Ponikowski P, Rutten FH. Guidelines for the management of atrial fibrillation: the Task Force for the Management of Atrial Fibrillation of the European Society of Cardiology (ESC). *Eur Heart J* 2010;**31**:2369–2429.
251. Lip GY. Anticoagulation therapy and the risk of stroke in patients with atrial fibrillation at 'moderate risk' [CHADS2 score=1]: simplifying stroke risk assessment and thromboprophylaxis in real-life clinical practice. *Thromb Haemost* 2010;**103**:683–685.
252. Hansen ML, Sørensen R, Clausen MT, Fog-Petersen ML, J. R, Gadsbøll N, Gislason GH, Folke F, Andersen SS, Schramm TK, Abildstrøm SZ, Poulsen HE, Køber L, Torp-Pedersen C. Risk of bleeding with single, dual, or triple therapy with warfarin, aspirin, and clopidogrel in patients with atrial fibrillation. *Arch Intern Med* 2010;**170**:1433–1441.
253. Lip GY, Huber K, Andreotti F, Arnesen H, Airaksinen JK, Cuisset T, Kirchhof P, Marin F. Antithrombotic management of atrial fibrillation patients presenting with acute coronary syndrome and/or undergoing coronary stenting: executive summary—a Consensus Document of the European Society of Cardiology Working Group on Thrombosis, endorsed by the European Heart Rhythm Association (EHRA) and the European Association of Percutaneous Cardiovascular Interventions (EAPCI). *Eur Heart J* 2010;**31**:1311–1318.
254. Faxon DP, Eikelboom JW, Berger PB, Holmes DR Jr., Bhatt DL, Moliterno DJ, Becker RC, Angiolillo DJ. Antithrombotic therapy in patients with atrial fibrillation undergoing coronary stenting: a North American perspective: executive summary. *Circ Cardiovasc Interv* 2011;**4**:522–534.
255. Rubboli A, Halperin JL, Airaksinen KE, Buerke M, Eeckhout E, Freedman SB, Gershlick AH, Schlitt A, Tse HF, Verheugt FW, Lip GY. Antithrombotic therapy in patients treated with oral anticoagulation undergoing coronary artery stenting. An expert consensus document with focus on atrial fibrillation. *Ann Med* 2008;**40**:428–436.
256. Abraham NS, Hlatky MA, Antman EM, Bhatt DL, Bjorkman DJ, Clark CB, Furberg CD, Johnson DA, Kahi CJ, Laine L, Mahaffey KW, Quigley EM, Scheiman J, Sperling LS, Tomaselli GF. ACCF/ACG/AHA 2010 expert consensus document on the concomitant use of proton pump inhibitors and thienopyridines: a focussed update of the ACCF/ACG/AHA 2008 expert consensus document on reducing the gastrointestinal risks of antiplatelet therapy and NSAID use. A Report of the American College of Cardiology Foundation Task Force on Expert Consensus Documents. *J Am Coll Cardiol* 2010;**56**:2051–2066.
257. Gilard M, Arnaud B, Cornily JC, Le Gal G, Lacut K, Le Calvez G, Mansourati J, Mottier D, Abgrall JF, Bosch J. Influence of omeprazole on the antiplatelet action of clopidogrel associated with aspirin: the randomized, double-blind OCLA (Omeprazole CLopidogrel Aspirin) study. *J Am Coll Cardiol* 2008;**51**:256–260.
258. Goodman SG, Clare R, Pieper KS, Nicolau JC, Storey RF, Cantor WJ, Mahaffey KW, Angiolillo DJ, Husted S, Cannon CP, James SK, Kilhamn J, Steg PG, Harrington RA, Wallentin L. Association of Proton Pump Inhibitor Use on Cardiovascular Outcomes with Clopidogrel and Ticagrelor: Insights from PLATO. *Circulation* 2012.
259. O'Donoghue ML, Braunwald E, Antman EM, Murphy SA, Bates ER, Rozenman Y, Michelson AD, Hautvast RW, Ver Lee PN, Close SL, Shen L, Mega JL, Sabatine MS, Wiviott SD. Pharmacodynamic effect and clinical efficacy of clopidogrel and prasugrel with or without a proton-pump inhibitor: an analysis of two randomised trials. *Lancet* 2009;**374**:989–997.
260. Bhatt DL, Cryer BL, Contant CF, Cohen M, Lanas A, Schnitzer TJ, Shook TL, Lapuerta P, Goldsmith MA, Laine L, Scirica BM, Murphy SA, Cannon CP. Clopidogrel with or without Omeprazole in Coronary Artery Disease. *N Engl J Med* 2010;**363**(20):1909–1917.
261. Simon T, Steg PG, Gilard M, Blanchard D, Bonello L, Hanssen M, Lardoux H, Coste P, Lefevre T, Drouet E, Mulak G, Bataille V, Ferrieres J, Verstuyft C, Danchin N. Clinical events as a function of proton pump inhibitor use, clopidogrel use, and cytochrome P450 2C19 genotype in a large nationwide cohort of acute myocardial infarction: results from the French Registry of Acute ST-Elevation and Non-ST-Elevation Myocardial Infarction (FAST-MI) registry. *Circulation* 2010;**123**:474–482.
262. Mega JL, Braunwald E, Wiviott SD, Bassand JP, Bhatt DL, Bode C, Burton P, Cohen M, Cook-Bruns N, Fox KA, Goto S, Murphy SA, Plotnikov AN, Schneider D, Sun X, Verheugt FW, Gibson CM. Rivaroxaban in Patients with a Recent Acute Coronary Syndrome. *N Engl J Med* 2012;**366**:9–19.

263. Alexander JH, Lopes RD, James S, Kilaru R, He Y, Mohan P, Bhatt DL, Goodman S, Verheugt FW, Flather M, Huber K, Liaw D, Husted SE, Lopez-Sendon J, De Caterina R, Jansky P, Darius H, Vinereanu D, Cornel JH, Cools F, Atar D, Leiva-Pons JL, Keltai M, Ogawa H, Pais P, Parkhomenko A, Ruzyllo W, Diaz R, White H, Ruda M, Geraldes M, Lawrence J, Harrington RA, Wallentin L. Apixaban with antiplatelet therapy after acute coronary syndrome. *N Engl J Med* 2011;**365**(8):699–708.
264. Steg PG, Mehta SR, Jukema JW, Lip GY, Gibson CM, Kovar F, Kala P, Garcia-Hernandez A, Renfurm RW, Granger CB. RUBY-1: a randomized, double-blind, placebo-controlled trial of the safety and tolerability of the novel oral factor Xa inhibitor darexaban (YM150) following acute coronary syndrome. *Eur Heart J* 2011;**32**:2541–2554.
265. Oldgren J, Budaj A, Granger CB, Khder Y, Roberts J, Siegbahn A, Tijssen JG, Van de Werf F, Wallentin L. Dabigatran vs. placebo in patients with acute coronary syndromes on dual antiplatelet therapy: a randomized, double-blind, phase II trial. *Eur Heart J* 2011;**32**:2781–2789.
266. Chen ZM, Pan HC, Chen YP, Peto R, Collins R, Jiang LX, Xie JX, Liu LS. Early intravenous then oral metoprolol in 45,852 patients with acute myocardial infarction: randomised placebo-controlled trial. *Lancet* 2005;**366**:1622–1632.
267. Baigent C, Keech A, Kearney PM, Blackwell L, Buck G, Pollicino C, Kirby A, Sourjina T, Peto R, Collins R, Simes R. Efficacy and safety of cholesterol-lowering treatment: prospective meta-analysis of data from 90,056 participants in 14 randomised trials of statins. *Lancet* 2005;**366**:1267–1278.
268. Schwartz GG, Olsson AG, Ezekowitz MD, Ganz P, Oliver MF, Waters D, Zeiher A, Chaitman BR. Atorvastatin for acute coronary syndromes. *JAMA* 2001;**286**:533–535.
269. Cannon CP, Braunwald E, McCabe CH, Rader DJ, Rouleau JL, Belder R, Joyal SV, Hill KA, Pfeffer MA, Skene AM. Intensive versus moderate lipid lowering with statins after acute coronary syndromes. *N Engl J Med* 2004;**350**:1495–1504.
270. Reiner Z, Catapano AL, De Backer G, Graham I, Taskinen MR, Wiklund O, Agewall S, Alegria E, Chapman MJ, Durrington P, Erdine S, Halcov J, Hobbs R, Kjekshus J, Filardi PP, Riccardi G, Storey RF, Wood D, Bax J, Vahanian A, Auricchio A, Baumgartner H, Ceconi C, Dean V, Deaton C, Fagard R, Filippatos G, Hasdai D, Hobbs R, Hoes A, Kearney P, Knuuti J, Kolh P, McDonagh T, Moulin C, Poldermans D, Popescu BA, Reiner Z, Sechtem U, Simons PA, Tendera M, Torbicki A, Vardas P, Widimsky P, Windecker S, Funck-Brentano C, Berkenboom G, De Graaf J, Descamps O, Gotcheva N, Griffith K, Guida GF, Gulec S, Henkin Y, Huber K, Kesaniemi YA, Lekakis J, Manolis AJ, Marques-Vidal P, Masana L, McMurray J, Mendes M, Pagava Z, Pedersen T, Prescott E, Rato Q, Rosano G, Sans S, Stalenhoef A, Tokgozoglu L, Viigimaa M, Wittekoek ME, Zamorano JL. ESC/EAS Guidelines for the management of dyslipidaemias: the Task Force for the management of dyslipidaemias of the European Society of Cardiology (ESC) and the European Atherosclerosis Society (EAS). *Eur Heart J* 2011;**32**:1769–1818.
271. <http://www.fda.gov/Drugs/DrugSafety/ucm256581.htm>.
272. Dietary supplementation with n-3 polyunsaturated fatty acids and vitamin E after myocardial infarction: results of the GISSI-Prevenzione trial. Gruppo Italiano per lo Studio della Sopravvivenza nell'Infarto miocardico. *Lancet* 1999;**354**:447–455.
273. Kromhout D, Giltay EJ, Geleijnse JM. n-3 fatty acids cardiovascular events after myocardial infarction. *N Engl J Med* 2010;**363**:2015–2026.
274. Rauch B, Schiele R, Schneider S, Diller F, Victor N, Gohlke H, Gottwik M, Steinbeck G, Del Castillo U, Sack R, Worth H, Katus H, Spitzer W, Sabin G, Senges J. OMEGA, a randomized, placebo-controlled trial to test the effect of highly purified omega-3 fatty acids on top of modern guideline-adjusted therapy after myocardial infarction. *Circulation* 2010;**122**:2152–2159.
275. Yusuf S, Held P, Furberg C. Update of effects of calcium antagonists in myocardial infarction or angina in light of the second Danish Verapamil Infarction Trial (DAVIT-II) and other recent studies. *Am J Cardiol* 1991;**67**:1295–1297.
276. Secondary prevention with verapamil after myocardial infarction. The Danish Study Group on Verapamil in Myocardial Infarction. *Am J Cardiol* 1990;**66**:p331–401.
277. Effect of verapamil on mortality and major events after acute myocardial infarction (the Danish Verapamil Infarction Trial II–DAVIT II). *Am J Cardiol* 1990;**66**:779–785.
278. Poole-Wilson PA, Lubben J, Kirwan BA, van Dalen FJ, Wagener G, Danchin N, Just H, Fox KA, Pocock SJ, Clayton TC, Motro M, Parker JD, Bourassa MG, Dart AM, Hildebrandt P, Hjalmarson A, Kragten JA, Molhoek GP, Otterstad JE, Seabra-Gomes R, Soler-Soler J, Weber S. Effect of long-acting nifedipine on mortality and cardiovascular morbidity in patients with stable angina requiring treatment (ACTION trial): randomised controlled trial. *Lancet* 2004;**364**:849–857.
279. ISIS-4: a randomised factorial trial assessing early oral captopril, oral mononitrate, and intravenous magnesium sulphate in 58,050 patients with suspected acute myocardial infarction. ISIS-4 (Fourth International Study of Infarct Survival) Collaborative Group. *Lancet* 1995;**345**:669–685.
280. Dickstein K, Kjekshus J. Effects of losartan captopril on mortality morbidity in high-risk patients after acute myocardial infarction: the OPTIMAAL randomised trial. Optimal Trial in Myocardial Infarction with Angiotensin II Antagonist Losartan. *Lancet* 2002;**360**:752–760.
281. Pfeffer MA, McMurray JJ, Velazquez EJ, Rouleau JL, Kober L, Maggioni AP, Solomon SD, Swedberg K, Van de Werf F, White H, Leimberger JD, Henis M, Edwards S, Zelenkofske S, Sellers MA, Califf RM. Valsartan, captopril, or both in myocardial infarction complicated by heart failure, left ventricular dysfunction, or both. *N Engl J Med* 2003;**349**:1893–1906.
282. Pitt B, Remme W, Zannad F, Neaton J, Martinez F, Roniker B, Bittman R, Hurley S, Kleiman J, Gatlin M. Eplerenone, a selective aldosterone blocker, in patients with left ventricular dysfunction after myocardial infarction. *N Engl J Med* 2003;**348**:1309–1321.
283. Steinhubl SR, Berger PB, Mann JT 3rd, Fry ET, DeLago A, Wilmer C, Topol EJ. Early and sustained dual oral antiplatelet therapy following percutaneous coronary intervention: a randomized controlled trial. *JAMA* 2002;**288**:2411–2420.
284. Dickstein K, Cohen-Solal A, Filippatos G, McMurray JJ, Ponikowski P, Poole-Wilson PA, Stromberg A, van Veldhuisen DJ, Atar D, Hoes AW, Keren A, Mebazaa A, Nieminen M, Priori SG, Swedberg K. ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure 2008: the Task Force for the Diagnosis and Treatment of Acute and Chronic Heart Failure 2008 of the European Society of Cardiology. Developed in collaboration with the Heart Failure Association of the ESC (HFA) and endorsed by the European Society of Intensive Care Medicine (ESICM). *Eur Heart J* 2008;**29**:2388–2442.
285. Dargie HJ. Effect of carvedilol on outcome after myocardial infarction in patients with left-ventricular dysfunction: the CAPRICORN randomised trial. *Lancet* 2001;**357**:1385–1390.
286. Packer M, Coats AJ, Fowler MB, Katus HA, Krum H, Mohacs P, Rouleau JL, Tendera M, Castaigne A, Roecker EB, Schultz MK, DeMets DL. Effect of carvedilol on survival in severe chronic heart failure. *N Engl J Med* 2001;**344**:1651–1658.
287. The Cardiac Insufficiency Bisoprolol Study II (CIBIS-II): a randomised trial. *Lancet* 1999;**353**:9–13.
288. Poole-Wilson PA, Swedberg K, Cleland JG, Di Lenarda A, Hanrath P, Komajda M, Lubben J, Lutiger B, Metra M, Remme WJ, Torp-Pedersen C, Scherhag A, Skene A. Comparison of carvedilol and metoprolol on clinical outcomes in patients with chronic heart failure in the Carvedilol Or Metoprolol European Trial (COMET): randomised controlled trial. *Lancet* 2003;**362**:7–13.
289. Fox KM. Efficacy of perindopril in reduction of cardiovascular events among patients with stable coronary artery disease: randomised, double-blind, placebo-controlled, multicentre trial (the EUROPA study). *Lancet* 2003;**362**:782–788.
290. Yusuf S, Sleight P, Pogue J, Bosch J, Davies R, Dagenais G. Effects of an angiotensin-converting-enzyme inhibitor, ramipril, on cardiovascular events in high-risk patients. The Heart Outcomes Prevention Evaluation Study Investigators. *N Engl J Med* 2000;**342**:145–153.
291. Vardas PE, Auricchio A, Blanc JJ, Daubert JC, Drexler H, Ector H, Gasparini M, Linde C, Morgado FB, Oto A, Sutton R, Trusz-Gluza M. Guidelines for cardiac pacing and cardiac resynchronization therapy: The Task Force for Cardiac Pacing and Cardiac Resynchronization Therapy of the European Society of Cardiology. Developed in collaboration with the European Heart Rhythm Association. *Eur Heart J* 2007;**28**:2256–2295.
292. Goldberg RJ, Spencer FA, Gore JM, Lessard D, Yarzebski J. Thirty-year trends (1975 to 2005) in the magnitude of, management of, and hospital death rates associated with cardiogenic shock in patients with acute myocardial infarction: a population-based perspective. *Circulation* 2009;**119**:1211–1219.
293. Dzavik V, Sleeper LA, Cocke TP, Moscucci M, Saucedo J, Hosat S, Jiang X, Slater J, Lejemtel T, Hochman JS. Early revascularization is associated with improved survival in elderly patients with acute myocardial infarction complicated by cardiogenic shock: a report from the SHOCK Trial Registry. *Eur Heart J* 2003;**24**:828–837.
294. Menon V, White H, Lejemtel T, Webb JG, Sleeper LA, Hochman JS. The clinical profile of patients with suspected cardiogenic shock due to predominant left ventricular failure: a report from the SHOCK Trial Registry. Should we emergently revascularize Occluded Coronaries in cardiogenic shock? *J Am Coll Cardiol* 2000;**36**(Suppl A):1071–1076.
295. Picard MH, Davidoff R, Sleeper LA, Mendes LA, Thompson CR, Dzavik V, Steingart R, Gin K, White HD, Hochman JS. Echocardiographic predictors of survival and response to early revascularization in cardiogenic shock. *Circulation* 2003;**107**:279–284.
296. Engstrom AE, Vis MM, Bouma BJ, van den Brink RB, Baan J Jr., Claessen BE, Kikkert WJ, Sjaauw KD, Meuwissen M, Koch KT, de Winter RJ, Tijssen JG, Piek JJ, Henriques JP. Right ventricular dysfunction is an independent predictor for mortality in ST-elevation myocardial infarction patients presenting with cardiogenic shock on admission. *Eur J Heart Fail* 2010;**12**:276–282.

297. Jeger RV, Lowe AM, Buller CE, Pfisterer ME, Dzavik V, Webb JG, Hochman JS, Jorde UP. Hemodynamic parameters are prognostically important in cardiogenic shock but similar following early revascularization or initial medical stabilization: a report from the SHOCK Trial. *Chest* 2007;**132**:1794–1803.
298. Alexander JH, Reynolds HR, Stebbins AL, Dzavik V, Harrington RA, Van de Werf F, Hochman JS. Effect of tilarginine acetate in patients with acute myocardial infarction and cardiogenic shock: the TRIUMPH randomized controlled trial. *JAMA* 2007;**297**:1657–1666.
299. Thackray S, Easthaugh J, Freemantle N, Cleland JG. The effectiveness and relative effectiveness of intravenous inotropic drugs acting through the adrenergic pathway in patients with heart failure—a meta-regression analysis. *Eur J Heart Fail* 2002;**4**:515–529.
300. De Backer D, Biston P, Devriendt J, Madl C, Chochrad D, Aldecoa C, Brasseur A, Defrance P, Gottignies P, Vincent JL. Comparison of dopamine and norepinephrine in the treatment of shock. *N Engl J Med* 2010;**362**:779–789.
301. Jeger RV, Tseng CH, Hochman JS, Bates ER. Interhospital transfer for early revascularization in patients with ST-elevation myocardial infarction complicated by cardiogenic shock—a report from the SHould we revascularize Occluded Coronaries for cardiogenic shock? (SHOCK) trial and registry. *Am Heart J* 2006;**152**:686–692.
302. Hussain F, Philipp RK, Ducas RA, Elliott J, Dzavik V, Jassal DS, Tam JW, Roberts D, Garber PJ, Ducas J. The ability to achieve complete revascularization is associated with improved in-hospital survival in cardiogenic shock due to myocardial infarction: Manitoba cardiogenic SHOCK Registry investigators. *Catheter Cardiovasc Interv* 2011;**78**:540–548.
303. Ohman EM, Nanas J, Stomel RJ, Leeser MA, Nielsen DW, O'Dea D, Rogers FJ, Harber D, Hudson MP, Fraulo E, Shaw LK, Lee KL. Thrombolysis and counterpulsation to improve survival in myocardial infarction complicated by hypotension and suspected cardiogenic shock or heart failure: results of the TACTICS Trial. *J Thromb Thrombolysis* 2005;**19**:33–39.
304. Prondzinsky R, Lemm H, Swyter M, Wegener N, Unverzagt S, Carter JM, Russ M, Schlitt A, Buerke U, Christoph A, Schmidt H, Winkler M, Thiery J, Werdan K, Buerke M. Intra-aortic balloon counterpulsation in patients with acute myocardial infarction complicated by cardiogenic shock: the prospective, randomized IABP SHOCK Trial for attenuation of multiorgan dysfunction syndrome. *Crit Care Med* 2010;**38**:152–160.
305. Bahekar A, Singh M, Singh S, Bhuriya R, Ahmad K, Khosla S, Arora R. Cardiovascular outcomes using intra-aortic balloon pump in high-risk acute myocardial infarction with or without cardiogenic shock: a meta-analysis. *J Cardiovasc Pharmacol Ther* 2011;**17**:44–56.
306. Cheng JM, den Uil CA, Hoeks SE, van der Ent M, Jewbali LS, van Domburg RT, Serruys PW. Percutaneous left ventricular assist devices vs. intra-aortic balloon pump counterpulsation for treatment of cardiogenic shock: a meta-analysis of controlled trials. *Eur Heart J* 2009;**30**:2102–2108.
307. Starling RC, Naka Y, Boyle AJ, Gonzalez-Stawinski G, John R, Jorde U, Russell SD, Conte JV, Aaronson KD, McGee EC Jr., Cotts WG, DeNofrio D, Pham DT, Farrar DJ, Pagani FD. Results of the post-U.S. Food and Drug Administration-approval study with a continuous flow left ventricular assist device as a bridge to heart transplantation: a prospective study using the INTERMACS (Interagency Registry for Mechanically Assisted Circulatory Support). *J Am Coll Cardiol* 2011;**57**:1890–1898.
308. Sheu JJ, Tsai TH, Lee FY, Fang HY, Sun CK, Leu S, Yang CH, Chen SM, Hang CL, Hsieh YK, Chen CJ, Wu CJ, Yip HK. Early extracorporeal membrane oxygenator-assisted primary percutaneous coronary intervention improved 30-day clinical outcomes in patients with ST-segment elevation myocardial infarction complicated with profound cardiogenic shock. *Crit Care Med* 2010;**38**:1810–1817.
309. McAlister FA, Stewart S, Ferrua S, McMurray JJ. Multidisciplinary strategies for the management of heart failure patients at high risk for admission: a systematic review of randomized trials. *J Am Coll Cardiol* 2004;**44**:810–819.
310. Effect of ramipril on mortality and morbidity of survivors of acute myocardial infarction with clinical evidence of heart failure. The Acute Infarction Ramipril Efficacy (AIRE) Study Investigators. *Lancet* 1993;**342**:821–828.
311. Pfeffer MA, Braunwald E, Moye LA, Basta L, Brown EJ Jr., Cuddy TE, Davis BR, Geltman EM, Goldman S, Flaker GC et al. Effect of captopril on mortality and morbidity in patients with left ventricular dysfunction after myocardial infarction. Results of the survival and ventricular enlargement trial. The SAVE Investigators. *N Engl J Med* 1992;**327**:669–677.
312. Kober L, Torp-Pedersen C, Carlsen JE, Bagger H, Eliassen P, Lyngborg K, Videbaek J, Cole DS, Auclert L, Pauly NC. A clinical trial of the angiotensin-converting-enzyme inhibitor trandolapril in patients with left ventricular dysfunction after myocardial infarction. Trandolapril Cardiac Evaluation (TRACE) Study Group. *N Engl J Med* 1995;**333**:1670–1676.
313. Taylor AL, Ziesche S, Yancy C, Carson P, D'Agostino R Jr., Ferdinand K, Taylor M, Adams K, Sabolinski M, Worcel M, Cohn JN. Combination of isosorbide dinitrate and hydralazine in blacks with heart failure. *N Engl J Med* 2004;**351**:2049–2057.
314. Pitt B, Zannad F, Remme WJ, Cody R, Castaigne A, Perez A, Palensky J, Wittes J. The effect of spironolactone on morbidity and mortality in patients with severe heart failure. Randomized Aldactone Evaluation Study Investigators. *N Engl J Med* 1999;**341**:709–717.
315. Costanzo MR, Guglin ME, Saltzberg MT, Jessup ML, Bart BA, Teerlink JR, Jaski BE, Fang JC, Feller ED, Haas GJ, Anderson AS, Schollmeyer MP, Sobotka PA. Ultrafiltration versus intravenous diuretics for patients hospitalized for acute decompensated heart failure. *J Am Coll Cardiol* 2007;**49**:675–683.
316. Shah MR, Hasselblad V, Stevenson LW, Binanay C, O'Connor CM, Sopko G, Califf RM. Impact of the pulmonary artery catheter in critically ill patients: meta-analysis of randomized clinical trials. *JAMA* 2005;**294**:1664–1670.
317. Levy B, Perez P, Perny J, Thivillier C, Gerard A. Comparison of norepinephrine-dobutamine to epinephrine for hemodynamics, lactate metabolism, and organ function variables in cardiogenic shock. A prospective, randomized pilot study. *Crit Care Med* 2011;**39**:450–455.
318. Bloch Thomsen PE, Jons C, Raatikainen MJ, Moerch Joergensen R, Hartikainen J, Virtanen V, Boland J, Anttonen O, Gang UJ, Hoest N, Boersma LV, Platou ES, Becker D, Messier MD, Huikuri HV. Long-term recording of cardiac arrhythmias with an implantable cardiac monitor in patients with reduced ejection fraction after acute myocardial infarction: the Cardiac Arrhythmias and Risk Stratification After Acute Myocardial Infarction (CARISMA) study. *Circulation* 2010;**122**:1258–1264.
319. Askari AT, Shishehbor MH, Kaminski MA, Riley MJ, Hsu A, Lincoff AM. The association between early ventricular arrhythmias, renin-angiotensin-aldosterone system antagonism, and mortality in patients with ST-segment-elevation myocardial infarction: Insights from Global Use of Strategies to Open coronary arteries (GUSTO) V. *Am Heart J* 2009;**158**:238–243.
320. Piccini JP, Hranitzky PM, Kilaru R, Rouleau JL, White HD, Aylward PE, Van de Werf F, Solomon SD, Califf RM, Velazquez EJ. Relation of mortality to failure to prescribe beta blockers acutely in patients with sustained ventricular tachycardia and ventricular fibrillation following acute myocardial infarction (from the VALsartan In Acute myocardial iNfarcTion trial [VALIANT] Registry). *Am J Cardiol* 2008;**102**:1427–1432.
321. Schmitt J, Duray G, Gersh BJ, Hohnloser SH. Atrial fibrillation in acute myocardial infarction: a systematic review of the incidence, clinical features and prognostic implications. *Eur Heart J* 2009;**30**:1038–1045.
322. Jabre P, Roger VL, Murad MH, Chamberlain AM, Prokop L, Adnet F, Jouven X. Mortality associated with atrial fibrillation in patients with myocardial infarction: a systematic review and meta-analysis. *Circulation* 2011;**123**:1587–1593.
323. Segal JB, McNamara RL, Miller MR, Kim N, Goodman SN, Powe NR, Robinson K, Yu D, Bass EB. The evidence regarding the drugs used for ventricular rate control. *J Fam Pract* 2000;**49**:47–59.
324. Hou ZY, Chang MS, Chen CY, Tu MS, Lin SL, Chiang HT, Woosley RL. Acute treatment of recent-onset atrial fibrillation and flutter with a tailored dosing regimen of intravenous amiodarone. A randomized, digoxin-controlled study. *Eur Heart J* 1995;**16**:521–528.
325. Zipes DP, Camm AJ, Borggrefe M, Buxton AE, Chaitman B, Fromer M, Gregoratos G, Klein G, Moss AJ, Myerburg RJ, Priori SG, Quinones MA, Roden DM, Silka MJ, Tracy C, Blanc JJ, Budaj A, Dean V, Deckers JW, Despres C, Dickstein K, Lekakis J, McGregor K, Metra M, Morais J, Osterspey A, Tamargo JL, Zamorano JL, Smith SC Jr., Jacobs AK, Adams CD, Antman EM, Anderson JL, Hunt SA, Halperin JL, Nishimura R, Ornato JP, Page RL, Riegel B. ACC/AHA/ESC 2006 guidelines for management of patients with ventricular arrhythmias and the prevention of sudden cardiac death—executive summary: A report of the American College of Cardiology/American Heart Association Task Force and the European Society of Cardiology Committee for Practice Guidelines (Writing Committee to Develop Guidelines for Management of Patients with Ventricular Arrhythmias and the Prevention of Sudden Cardiac Death) Developed in collaboration with the European Heart Rhythm Association and the Heart Rhythm Society. *Eur Heart J* 2006;**27**:2099–2140.
326. Hazinski MF, Nolan JP, Billi JE, Bottiger BW, Bossaert L, de Caen AR, Deakin CD, Drajer S, Eigel B, Hickey RW, Jacobs I, Kleinman ME, Kloeck W, Koster RW, Lim SH, Mancini ME, Montgomery WH, Morley PT, Morrison LJ, Nadkarni VM, O'Connor RE, Okada K, Perlman JM, Sayre MR, Shuster M, Soar J, Sunde K, Travers AH, Wyllie J, Zideman D. Part 1: Executive summary: 010 International Consensus on Cardiopulmonary Resuscitation and Emergency Cardiovascular Care Science With Treatment Recommendations. *Circulation* 2010;**122**(Suppl 2):S250–275.
327. Marill KA, deSouza IS, Nishijima DK, Senecal EL, Setnik GS, Stair TO, Ruskin JN, Ellinor PT. Amiodarone or procainamide for the termination of sustained stable ventricular tachycardia: an historical multicenter comparison. *Acad Emerg Med* 2010;**17**:297–306.

328. Link MS, Atkins DL, Passman RS, Halperin HR, Samson RA, White RD, Cudnik MT, Berg MD, Kudenchuk PJ, Kerber RE. Part 6: electrical therapies: automated external defibrillators, defibrillation, cardioversion, and pacing: 2010 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. *Circulation* 2010;**122**(Suppl 3):S706–719.
329. Hine LK, Laird N, Hewitt P, Chalmers TC. Meta-analytic evidence against prophylactic use of lidocaine in acute myocardial infarction. *Arch Intern Med* 1989;**149**:2694–2698.
330. Piccini JP, Schulte PJ, Pieper KS, Mehta RH, White HD, Van de Werf F, Ardissino D, Califf RM, Granger CB, Ohman EM, Alexander JH. Antiarrhythmic drug therapy for sustained ventricular arrhythmias complicating acute myocardial infarction. *Crit Care Med* 2011;**39**:78–83.
331. Brugada J, Aguinaga L, Mont L, Betriu A, Mulet J, Sanz G. Coronary artery revascularization in patients with sustained ventricular arrhythmias in the chronic phase of a myocardial infarction: effects on the electrophysiologic substrate and outcome. *J Am Coll Cardiol* 2001;**37**(2):529–533.
332. Natale A, Sra J, Axtell K, Maglio C, Dhala A, Blanck Z, Deshpande S, Jazayeri M, Akhtar M. Ventricular fibrillation and polymorphic ventricular tachycardia with critical coronary artery stenosis: does bypass surgery suffice? *J Cardiovasc Electro-physiol* 1994;**5**:988–994.
333. Lee DS, Green LD, Liu PP, Dorian P, Newman DM, Grant FC, Tu JV, Alter DA. Effectiveness of implantable defibrillators for preventing arrhythmic events and death: a meta-analysis. *J Am Coll Cardiol* 2003;**41**:1573–1582.
334. Bardy GH, Lee KL, Mark DB, Poole JE, Packer DL, Boineau R, Domanski M, Troutman C, Anderson J, Johnson G, McNulty SE, Clapp-Channing N, Davidson-Ray LD, Fraulo ES, Fishbein DP, Luceri RM, Ip JH. Amiodarone or an implantable cardioverter-defibrillator for congestive heart failure. *N Engl J Med* 2005;**352**:225–237.
335. Dickstein K, Vardas PE, Auricchio A, Daubert JC, Linde C, McMurray J, Ponikowski P, Priori SG, Sutton R, van Veldhuisen DJ, Vahanian A, Auricchio A, Bax J, Ceconi C, Dean V, Filippatos G, Funck-Brentano C, Hobbs R, Kearney P, McDonagh T, Popescu BA, Reiner Z, Sechtem U, Sirnes PA, Tendera M, Vardas P, Widimsky P, Tendera M, Anker SD, Blanc JJ, Gasparini M, Hoes AW, Israel CW, Kalarus Z, Merkely B, Swedberg K, Camm AJ. 2010 Focussed Update of ESC Guidelines on device therapy in heart failure: an update of the 2008 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure and the 2007 ESC guidelines for cardiac and resynchronization therapy. Developed with the special contribution of the Heart Failure Association and the European Heart Rhythm Association. *Eur Heart J* 2010;**31**:2677–2687.
336. Huikuri HV, Cox M, Interian A Jr., Kessler KM, Glicksman F, Castellanos A, Myerburg RJ. Efficacy of intravenous propranolol for suppression of inducibility of ventricular tachyarrhythmias with different electrophysiologic characteristics in coronary artery disease. *Am J Cardiol* 1989;**64**:1305–1309.
337. Chevalier P, Burri H, Fahrat F, Cucherat M, Jegaden O, Obadia JF, Kirkorian G, Touboul P. Perioperative outcome and long-term survival of surgery for acute post-infarction mitral regurgitation. *Eur J Cardiothorac Surg* 2004;**26**:330–335.
338. Topaz O, Taylor AL. Interventricular septal rupture complicating acute myocardial infarction: from pathophysiologic features to the role of invasive and non-invasive diagnostic modalities in current management. *Am J Med* 1992;**93**:683–688.
339. Lemery R, Smith HC, Giuliani ER, Gersh BJ. Prognosis in rupture of the ventricular septum after acute myocardial infarction and role of early surgical intervention. *Am J Cardiol* 1992;**70**:147–151.
340. Imazio M, Negro A, Belli R, Beqaraj F, Forno D, Giammaria M, Trincherio R, Adler Y, Spodick D. Frequency and prognostic significance of pericarditis following acute myocardial infarction treated by primary percutaneous coronary intervention. *Am J Cardiol* 2009;**103**:1525–1529.
341. Osheroov AB, Borovik-Raz M, Aronson D, Agmon Y, Kapeliovich M, Kerner A, Grenadier E, Hammerman H, Nikolsky E, Roguin A. Incidence of early left ventricular thrombus after acute anterior wall myocardial infarction in the primary coronary intervention era. *Am Heart J* 2009;**157**:1074–1080.
342. Solheim S, Seljeflot I, Lunde K, Bjornerheim R, Aakhus S, Forfang K, Arnesen H. Frequency of left ventricular thrombus in patients with anterior wall acute myocardial infarction treated with percutaneous coronary intervention and dual antiplatelet therapy. *Am J Cardiol* 2010;**106**:1197–1200.
343. Porter A, Kandalkar H, Iakobishvili Z, Sagie A, Imbar S, Battler A, Hasdai D. Left ventricular mural thrombus after anterior ST-segment-elevation acute myocardial infarction in the era of aggressive reperfusion therapy—still a frequent complication. *Coron Artery Dis* 2005;**16**:275–279.
344. Reeder GS, Lengyel M, Tajik AJ, Seward JB, Smith HC, Danielson GK. Mural thrombus in left ventricular aneurysm: incidence, role of angiography, and relation between anticoagulation and embolization. *Mayo Clin Proc* 1981;**56**:77–81.
345. Keeley EC, Hillis LD. Left ventricular mural thrombus after acute myocardial infarction. *Clin Cardiol* 1996;**19**:83–86.
346. Turpie AG, Robinson JG, Doyle DJ, Mulji AS, Mishkel GJ, Sealey BJ, Cairns JA, Skingley L, Hirsh J, Gent M. Comparison of high-dose with low-dose subcutaneous heparin to prevent left ventricular mural thrombosis in patients with acute transmural anterior myocardial infarction. *N Engl J Med* 1989;**320**:352–357.